Mathematics 4024
Study Guide
GRADE 12



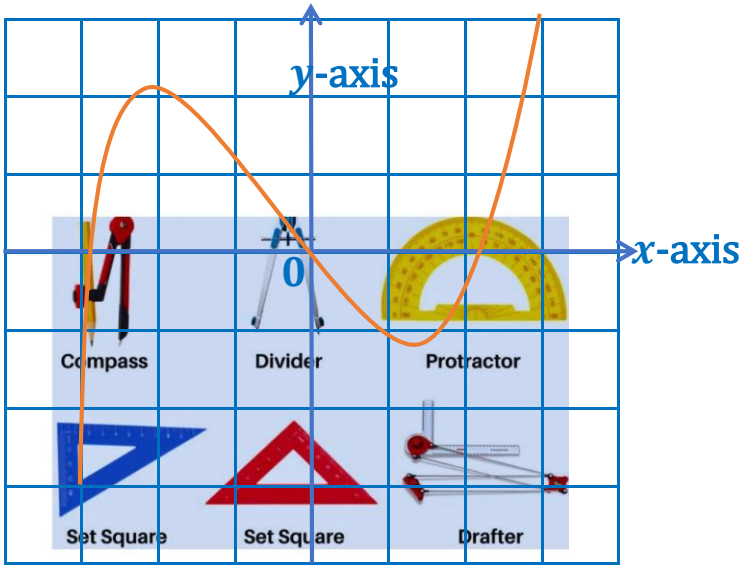
Complete Study Guide with Past Exam Questions and Teaching Notes.

Mastering Grade 12 Mathematics: A Comprehensive Guide for Grades 12

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = -\frac{b}{2a}$$

$$y = \frac{4ac-b^2}{4a}$$



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VECTORS

Objectives

By the end of this topic, learners should

1. Define what a vector is and provide examples of physical quantities that can be represented by vectors.
2. Understand how to represent vectors graphically and algebraically.
3. Learn how to perform basic vector operations, including addition, subtraction, and scalar multiplication.
4. Understand how to find the magnitude (length) and direction of a vector.
5. Apply vectors to solve problems in physics, engineering, and other fields.
6. Understand the concept of unit vectors and how to use them to simplify vector calculations.
7. Learn how to resolve vectors into their component parts.
8. Understand how to use vectors to represent displacement, velocity, acceleration, and force.

Definition

- A vector is a mathematical object that has both magnitude (length) and direction.
- It is typically represented as an arrow, where the length of the arrow indicates the magnitude, and the direction of the arrow indicates the direction of the vector.

Vector Representation

- Graphical representation: Arrow notation.
- Component form: (a, b) or $ai + bj$.
- Magnitude (length): $|v| = \sqrt{a^2 + b^2}$.
- Direction: $\theta = \tan^{-1}\left(\frac{b}{a}\right)$.

The magnitude of a vector

- The magnitude of a vector is the length of a directed line segment.
- The magnitude of the vector is length or modulus of a given vector.
- If A and B are two points on the XOY – plane, then vector AB is written as $\overrightarrow{AB}$ and its magnitude is represented by the notation $\left|\overrightarrow{AB}\right|$ and the magnitude of the vector $\underline{a}$ is written as $\left|\underline{a}\right|$.
- The length of a line can be found using Pythagoras theorem.
- For example,
  $\left|\overrightarrow{AB}\right| = \sqrt{x^2 + y^2}$ or $\left|\overrightarrow{AB}\right| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

The Midpoint

- The midpoint of a line segment is the point that divides the segment into two equal parts.
- It is equidistant from the endpoints of the segment.
- The midpoint of a line segment with endpoints (x_1, y_1) and (x_2, y_2) is given by M.P $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

2018.P1.Oct

Z is a point (-1, 8) and M is a point (2, 12). Find the magnitude of $\overrightarrow{ZM}$.

Solution

Z(-1, 8) and M(2, 12).

$$\overrightarrow{OZ} = \begin{pmatrix} -1 \\ 8 \end{pmatrix} \text{ and } \overrightarrow{OM} = \begin{pmatrix} 2 \\ 12 \end{pmatrix}$$

$$\overrightarrow{ZM} = \overrightarrow{ZO} + \overrightarrow{OM}$$

$$= \begin{pmatrix} 1 \\ -8 \end{pmatrix} + \begin{pmatrix} 2 \\ 12 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\left|\overrightarrow{ZM}\right| = \sqrt{x^2 + y^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5 \text{ units}$$

2019.P1.Nov

M is the point (0, 5) and $\overrightarrow{MN} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$. Find $\overrightarrow{ON}$.

Solution

$$\overrightarrow{OM} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} \text{ and } \overrightarrow{MN} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

$$\overrightarrow{MN} = \overrightarrow{MO} + \overrightarrow{ON}$$

$$\begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \end{pmatrix} + \overrightarrow{ON}$$

$$\overrightarrow{ON} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ -5 \end{pmatrix}$$

$$\overrightarrow{ON} = \begin{pmatrix} -3 \\ 9 \end{pmatrix}$$

2020.P1.Nov

The vector $\overrightarrow{RS} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. Given that R is a point (-2, 1), find the coordinates of point S.

Solution

$$\overrightarrow{RS} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \text{ and } \overrightarrow{0R} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\overrightarrow{RS} = \overrightarrow{RO} + \overrightarrow{OS}$$

$$\begin{pmatrix} -4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + \overrightarrow{OS}$$

$$\overrightarrow{OS} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OS} = \begin{pmatrix} -6 \\ 4 \end{pmatrix}$$

$$\therefore S(-6, 4)$$

2021.P1.Nov

A is the point (1, 2) and B is the point (-2, 5). Find AB as a column vector.

Solution

A(1, 2) and B(-2, 5)

$$\overrightarrow{OA} = \begin{pmatrix}1\\2\end{pmatrix} \text{ and } \overrightarrow{OB} = \begin{pmatrix}-2\\5\end{pmatrix}$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

$$= \begin{pmatrix}-1\\-2\end{pmatrix} + \begin{pmatrix}-2\\5\end{pmatrix}$$

$$= \begin{pmatrix}-3\\3\end{pmatrix}$$

2022.P1.Nov

The vector $\overrightarrow{RS} = \begin{pmatrix}-4\\5\end{pmatrix}$. Given that the coordinates of the point S are (1, 2), find the coordinates of point R.

Solution

$$\overrightarrow{OS} = \begin{pmatrix}1\\2\end{pmatrix} \text{ and } \overrightarrow{RS} = \begin{pmatrix}-4\\5\end{pmatrix}$$

$$\overrightarrow{RS} = \overrightarrow{RO} + \overrightarrow{OS}$$

$$\begin{pmatrix}-4\\5\end{pmatrix} = \overrightarrow{RO} + \begin{pmatrix}1\\2\end{pmatrix}$$

$$\overrightarrow{RO} = \begin{pmatrix}-4\\5\end{pmatrix} - \begin{pmatrix}1\\2\end{pmatrix}$$

$$\overrightarrow{RO} = \begin{pmatrix}-5\\3\end{pmatrix}$$

$$\overrightarrow{OR} = \begin{pmatrix}5\\-3\end{pmatrix}$$

$$R(5, -3)$$

2023.P1.Nov

Given that $\overrightarrow{OP} = \begin{pmatrix} 4 \\ 13 \end{pmatrix}$ and the point Q is (16, 8), find $|\overrightarrow{PQ}|$.

Solution

$$\overrightarrow{OP} = \begin{pmatrix} 4 \\ 13 \end{pmatrix} \text{ and } \overrightarrow{OQ} = \begin{pmatrix} 16 \\ 8 \end{pmatrix}$$

$$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ}$$

$$= \begin{pmatrix} -4 \\ -13 \end{pmatrix} + \begin{pmatrix} 16 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} 12 \\ -5 \end{pmatrix}$$

$$|\overrightarrow{PQ}| = \sqrt{x^2 + y^2}$$

$$= \sqrt{12^2 + (-5)^2}$$

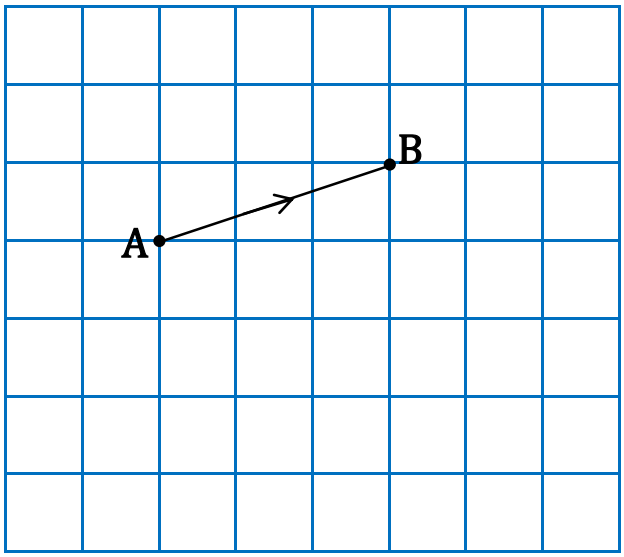
$$= \sqrt{144 + 25}$$

$$= \sqrt{169}$$

$$= 13 \text{ units}$$

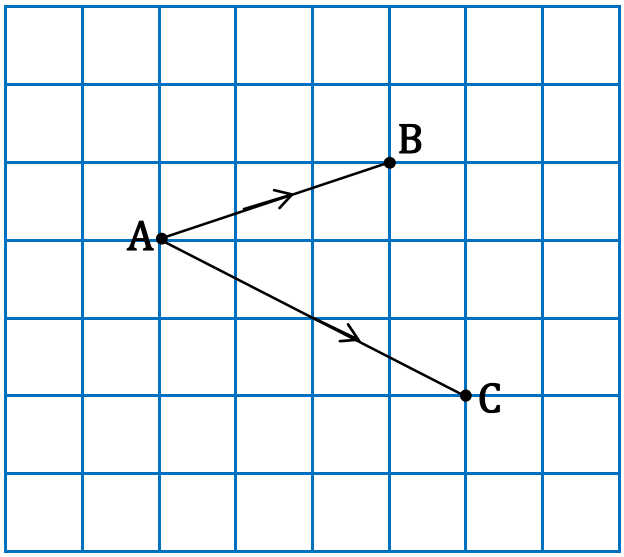
2024.P1.GCE

In the answer space below, draw $\overrightarrow{AC}$, given that $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$.



Solution

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$
$$= \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$
$$= \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$



Collinear points

- Collinear points are three or more points that lie on the same straight line.
- To determine if three points are collinear, you can use the following methods:

1 Visual Inspection
- Plot the points on a coordinate plane and check if they lie on the same straight line.

2 Slope (Gradient) Formula
- Calculate the slope of the line passing through two pairs of points. If the slopes are equal, the points are collinear.

3 Vector Method
- If the vectors formed by the points are parallel, the points are collinear.

Example

Determine if the points A(2, 3), B(4, 6), and C(6, 9) are collinear.

Solution

$\overrightarrow{\mathbf{OA}} = \begin{pmatrix}\mathbf{2}\\\mathbf{3}\end{pmatrix}, \overrightarrow{\mathbf{OB}} = \begin{pmatrix}\mathbf{4}\\\mathbf{6}\end{pmatrix}$ and $\overrightarrow{\mathbf{OC}} = \begin{pmatrix}\mathbf{6}\\\mathbf{9}\end{pmatrix}$

$\overrightarrow{\mathbf{AB}} = \overrightarrow{\mathbf{AO}} + \overrightarrow{\mathbf{OB}}$

$\overrightarrow{\mathbf{AB}} = \begin{pmatrix}\mathbf{-2}\\\mathbf{-3}\end{pmatrix} + \begin{pmatrix}\mathbf{4}\\\mathbf{6}\end{pmatrix}$

$\overrightarrow{\mathbf{AB}} = \begin{pmatrix}\mathbf{2}\\\mathbf{3}\end{pmatrix}$

$\overrightarrow{\mathbf{AB}} = \mathbf{1}\begin{pmatrix}\mathbf{2}\\\mathbf{3}\end{pmatrix}$

$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$

$\overrightarrow{BC} = \begin{pmatrix} -4 \\ -6 \end{pmatrix} + \begin{pmatrix} 6 \\ 9 \end{pmatrix}$

$\overrightarrow{BC} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$\overrightarrow{BC} = 1\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC}$

$\overrightarrow{AC} = \begin{pmatrix} -2 \\ -3 \end{pmatrix} + \begin{pmatrix} 6 \\ 9 \end{pmatrix}$

$\overrightarrow{AC} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

$\overrightarrow{AC} = 2\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

∴ The points A, B and C are collinear since $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$ gives a common factor of $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Exercise

1 Determine if the points (1, 2), (3, 4), and (5, 6) are collinear.
2 Show that the points (2, 3), (4, 5), and (6, 7) are collinear using the slope formula.
3 Determine if the points (-1, 2), (1, 4), and (3, 6) are collinear using the vector method.

2018.P2.GCE

Show that the points L(-2, -10), M(2, 2) and N(5, 11) are collinear.

Solution

$\overrightarrow{OL} = \begin{pmatrix} -2 \\ -10 \end{pmatrix}, \overrightarrow{OM} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ and $\overrightarrow{ON} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}$

$\overrightarrow{LM} = \overrightarrow{LO} + \overrightarrow{OM}$

$= \begin{pmatrix} 2 \\ 10 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 4 \\ 12 \end{pmatrix}$

$= 4\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$$\overrightarrow{MN} = \overrightarrow{MO} + \overrightarrow{ON}$$

$$= \begin{pmatrix} -2 \\ -2 \end{pmatrix} + \begin{pmatrix} 5 \\ 11 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 9 \end{pmatrix}$$

$$= 3\begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\overrightarrow{LN} = \overrightarrow{LO} + \overrightarrow{ON}$$

$$= \begin{pmatrix} 2 \\ 10 \end{pmatrix} + \begin{pmatrix} 5 \\ 11 \end{pmatrix}$$

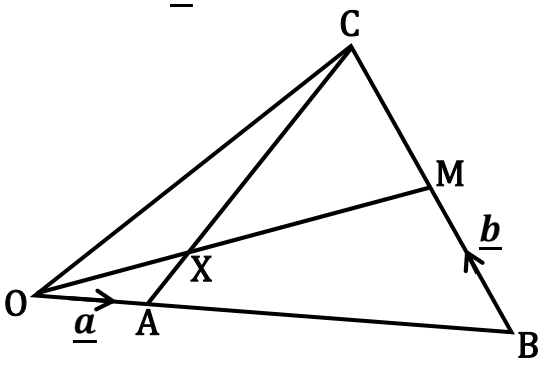
$$= \begin{pmatrix} 7 \\ 21 \end{pmatrix}$$

$$= 7\begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$\therefore$ The points L, M and N are collinear since the vectors $\overrightarrow{LM}$, $\overrightarrow{MN}$ and $\overrightarrow{LN}$ gives a common factor of $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

2019.P2.Nov

In the diagram below, $\overrightarrow{OB} = 4\overrightarrow{OA}$ and $\overrightarrow{AC} = 5\overrightarrow{AX}$. M is the midpoint of BC, $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{BM} = \underline{b}$.



(i) Express in terms of $\underline{a}$ and/or $\underline{b}$

(a) $\overrightarrow{AB}$, [1]

(b) $\overrightarrow{AC}$, [1]

(c) $\overrightarrow{OM}$. [1]

(ii) Show that $\overrightarrow{OX} = \frac{2}{5}(4\underline{a} + \underline{b})$. [2]

Solution

(ii) (a) $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$
$= 4\underline{a} - \underline{a}$
$= 3\underline{a}$

(b) $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$
$= 3\underline{a} + 2\underline{b}$

(c) $\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BM}$
$= 4\underline{a} + \underline{b}$

(i) $\overrightarrow{OX} = \frac{2}{5}(4\underline{a} + \underline{b})$

$\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AX}$

$= \underline{a} + \frac{1}{5}\overrightarrow{AC}$

$= \underline{a} + \frac{1}{5}(3\underline{a} + 2\underline{b})$

$= \underline{a} + \frac{3}{5}\underline{a} + \frac{2}{5}\underline{b}$

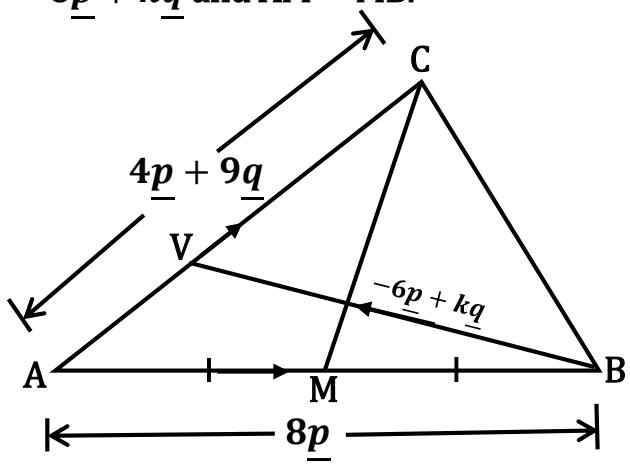
$= \frac{8}{5}\underline{a} + \frac{2}{5}\underline{b}$

$= \frac{2}{5}(4\underline{a} + \underline{b})$

Hence, shown.

2020.P2.Nov

In the diagram below, AVC is a straight line, $\overrightarrow{AB} = 8\underline{p}$, $\overrightarrow{AC} = 4\underline{p} + 9\underline{q}$, $\overrightarrow{BV} = -6\underline{p} + k\underline{q}$ and AM = MB.



(i) Express in terms $\underline{p}$, $\underline{q}$ and/or k

(a) $\overrightarrow{AM}$, [1]

(b) $\overrightarrow{AV}$. [1]

(ii) Given that $\overrightarrow{AV} = h\overrightarrow{AC}$, by forming an equation involving $\underline{p}$, $\underline{q}$, h and k or otherwise, find the numerical values of h and k. [3]

Solution

(i) (a) $\overrightarrow{AM} = \frac{1}{2}\overrightarrow{AB}$

$= \frac{1}{2}(8\underline{p})$

$= 4\underline{p}$

(b) $\overrightarrow{AV} = \overrightarrow{AB} + \overrightarrow{BV}$

$= 8\underline{p} - 6\underline{p} + k\underline{q}$

$= 2\underline{p} + k\underline{q}$

(ii) $\overrightarrow{AV} = h\overrightarrow{AC}$

$= h(4\underline{p} + 9\underline{q})$

$= 4h\underline{p} + 9h\underline{q}$

$2\underline{p} + k\underline{q} = 4h\underline{p} + 9h\underline{q}$

$2\underline{p} = 4h\underline{p}$ and $k\underline{q} = 9h\underline{q}$

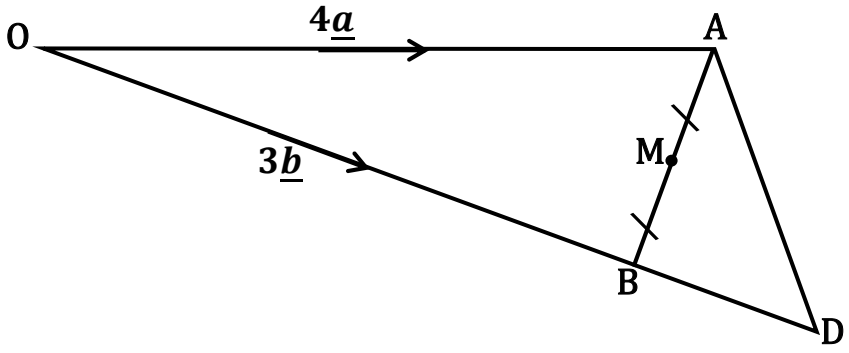
$2h = 1$ and $k = 9h$

$h = \frac{1}{2}$ and $k = \frac{9}{2}$

$k = 4\frac{1}{2}$

2021.P2.Nov

In the diagram below, $\overrightarrow{OA} = 4\underline{a}$, $\overrightarrow{OB} = 3\underline{b}$, M is the midpoint of AB and OB = 3BD.



Express in terms of $\underline{a}$ and/or $\underline{b}$

(i) $\overrightarrow{AB}$, [1]

(ii) $\overrightarrow{OM}$ [1]

(iii) $\overrightarrow{AD}$, [1]

(iv) $\overrightarrow{MD}$. [2]

Solution

(i) $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$

$= -4\underline{a} + 3\underline{b}$

$= 3\underline{b} - 4\underline{a}$

(ii) $\overrightarrow{OM} = \overrightarrow{OA} + \overrightarrow{AM}$

$= 4\underline{a} + \frac{1}{2}\overrightarrow{AB}$

$= 4\underline{a} + \frac{1}{2}(3\underline{b} - 4\underline{a})$

$= 4\underline{a} + \frac{3}{2}\underline{b} - 2\underline{a}$

$= 4\underline{a} - 2\underline{a} + \frac{3}{2}\underline{b}$

$= 2\underline{a} + \frac{3}{2}\underline{b}$

$= \frac{4\underline{a}+3\underline{b}}{2}$

(iii) $\overrightarrow{AD} = \overrightarrow{AO} + \overrightarrow{OD}$

$= -4\underline{a} + 4\underline{b}$

$= 4(\underline{b} - \underline{a})$

(iv) $\overrightarrow{MD} = \overrightarrow{MO} + \overrightarrow{OD}$

$= -2\underline{a} - \frac{3}{2}\underline{b} + 4\underline{b}$

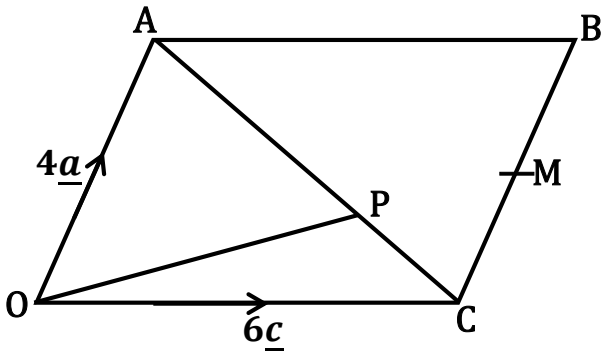
$= -2\underline{a} + \frac{5}{2}\underline{b}$

$= \frac{5}{2}\underline{b} - 2\underline{a}$

$= \frac{5\underline{b}-4\underline{a}}{2}$

2022.P2.Nov

In the parallelogram OABC, $\overrightarrow{OA} = 4\underline{a}$, $\overrightarrow{OC} = 6\underline{c}$, $\overrightarrow{AP} = \frac{2}{3}\overrightarrow{AC}$ and M is the midpoint of BC.



(i) Find in terms of $\underline{a}$ and/or $\underline{c}$

(a) $\overrightarrow{AC}$, [1]

(b) $\overrightarrow{OP}$, [1]

(c) $\overrightarrow{OM}$. [1]

(ii) Hence or otherwise, show that the points O, P and M are collinear. [2]

Solution

(i) (a) $\overrightarrow{AC} = \overrightarrow{AO} - \overrightarrow{OC}$

$= -4\underline{q} + 6\underline{c}$

$= 6\underline{c} - 4\underline{q}$

$= 2(3\underline{c} - 2\underline{q})$

(b) $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$

$= 4\underline{q} + \frac{2}{3}\overrightarrow{AC}$

$= 4\underline{q} + \frac{2}{3}(6\underline{c} - 4\underline{q})$

$= 4\underline{q} + 4\underline{c} - \frac{8}{3}\underline{q}$

$= 4\underline{q} - \frac{8}{3}\underline{q} + 4\underline{c}$

$= \frac{4}{3}\underline{q} + 4\underline{c}$

$= \frac{4}{3}(\underline{q} + 3\underline{c})$

(c) $\overrightarrow{OM} = \overrightarrow{OC} + \overrightarrow{CM}$

$= 6\underline{c} + \frac{1}{2}\overrightarrow{CB}$

$= 6\underline{c} + \frac{1}{2}(4\underline{q})$

$= 6\underline{c} + 2\underline{q}$

$= 2(3\underline{c} + \underline{q})$

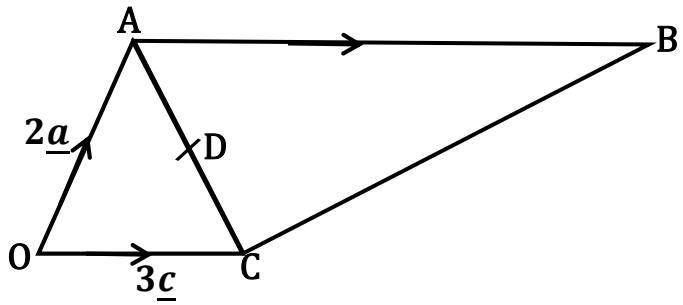
(ii) $\overrightarrow{OP} = \frac{4}{3}(3\underline{c} + \underline{q})$

$\overrightarrow{OM} = 2(3\underline{c} + \underline{q})$

∴The points O, P and Q are collinear since $\overrightarrow{OP}$ and $\overrightarrow{OM}$ gives a common factor of $3\underline{c} + \underline{q}$. Hence, shown.

2023.P2.Nov

The diagram below shows a trapezium OABC in which $\overrightarrow{OA} = 2\underline{a}$, $\overrightarrow{OC} = 3\underline{c}$, $\overrightarrow{AB} = 3\overrightarrow{OC}$ and D is the midpoint of AC.



Express in terms of $\underline{a}$ and/or $\underline{c}$

(i) $\overrightarrow{AC}$, [1]

(ii) $\overrightarrow{OD}$ [1]

(iii) $\overrightarrow{OB}$, [1]

(iv) $\overrightarrow{BD}$. [2]

Solution

(i) $\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC}$

$= -2\underline{a} + 3\underline{C}$

$= 3\underline{c} - 2\underline{a}$

(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$

$= 2\underline{a} + \frac{1}{2}\overrightarrow{AC}$

$= 2\underline{a} + \frac{1}{2}(3\underline{c} - 2\underline{a})$

$= 2\underline{a} + \frac{3}{2}\underline{c} - \underline{a}$

$= 2\underline{a} - \underline{a} + \frac{3}{2}\underline{c}$

$= \underline{a} + \frac{3}{2}\underline{c}$

$= \frac{2\underline{a}+3\underline{c}}{2}$

(iii) $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$

$= 2\underline{a} + 9\underline{c}$

(iv) $\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$

$= -9\underline{c} + \frac{1}{2}\overrightarrow{AC}$

$= -9\underline{c} + \frac{1}{2}(3\underline{c} - 2\underline{a})$

$= -9\underline{c} + \frac{3}{2}\underline{c} - \underline{a}$

$= \frac{3}{2}\underline{c} - 9\underline{c} - \underline{a}$

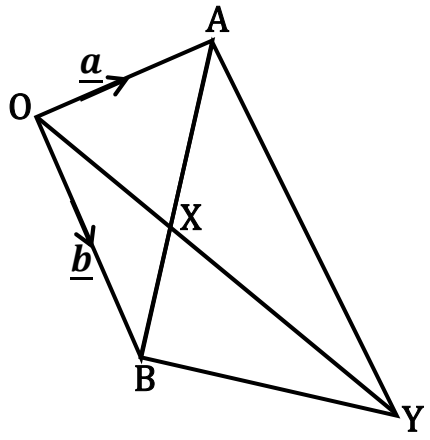
$= -\frac{15}{2}\underline{c} - \underline{a}$

$= \frac{-15\underline{c}-2\underline{a}}{2}$

$= \frac{-(15\underline{c}+2\underline{a})}{2}$

2024.P2.GCE

In the diagram below, $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and AX : AB = 2 : 3.



(i) Express in terms of $\underline{a}$ and/or $\underline{b}$

(a) $\overrightarrow{AB}$, [1]

(b) $\overrightarrow{XB}$, [1]

(c) $\overrightarrow{XO}$. [1]

(ii) Given that $\overrightarrow{OY} = 3\overrightarrow{OX}$, show that $\overrightarrow{BY} = \underline{a} + \underline{b}$. [2]

Solution

(i) (a) $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$

$= -\underline{a} + \underline{b}$

$= \underline{b} - \underline{a}$

(b) $\overrightarrow{XB} = \frac{1}{3}\overrightarrow{AB}$

$= \frac{1}{3}(\underline{b} - \underline{a})$

$= \frac{1}{3}\underline{b} - \frac{1}{3}\underline{a}$

$= \frac{\underline{b}-\underline{a}}{3}$

(c) $\overrightarrow{XO} = \overrightarrow{XB} + \overrightarrow{BO}$

$= \frac{1}{3}\underline{b} - \frac{1}{3}\underline{a} - \underline{b}$

$= \frac{1}{3}\underline{b} - \underline{b} - \frac{1}{3}\underline{a}$

$= -\frac{2}{3}\underline{b} - \frac{1}{3}\underline{a}$

$= -\frac{1}{3}(2\underline{b} + \underline{a})$

(ii) $\overrightarrow{OY} = 3\overrightarrow{OX}$

$= 3\left(\frac{2}{3}\underline{b} + \frac{1}{3}\underline{a}\right)$

$= \frac{6}{3}\underline{b} + \frac{3}{3}\underline{a}$

$= 2\underline{b} + \underline{a}$

$\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY}$

$= -\underline{b} + 2\underline{b} + \underline{a}$

$= \underline{b} + \underline{a}$

Hence, shown.

LINEAR PROGRAMMING

Objectives

By the end of this topic, learners should

1. Define linear programming and its importance in real-life decision-making.
2. Identify the key components of a linear programming problem, including variables, objective function, and constraints.
3. Formulate a linear programming problem using mathematical equations and inequalities.
4. Graph a linear programming problem on a coordinate plane.
5. Use graphical methods to find the feasible region and optimal solution.
6. Apply linear programming to solve real-world problems in fields such as business, economics, and engineering.

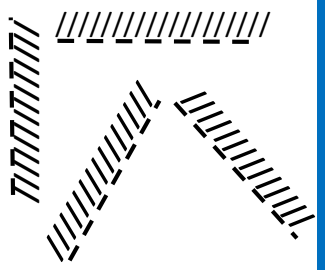
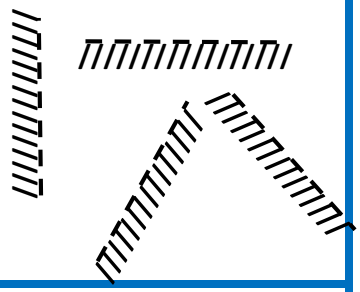
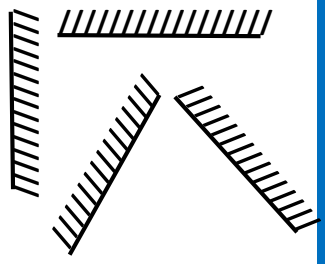
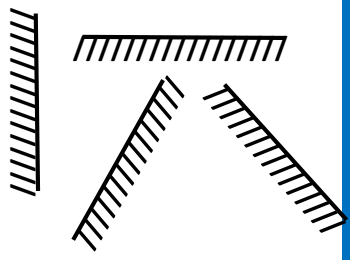
Definition

- Linear programming is a mathematical method used to optimize a linear objective function, subject to a set of linear constraints.
- It is a powerful tool used to make decisions in a wide range of fields, including business, economics, engineering, and computer science.

Key Elements

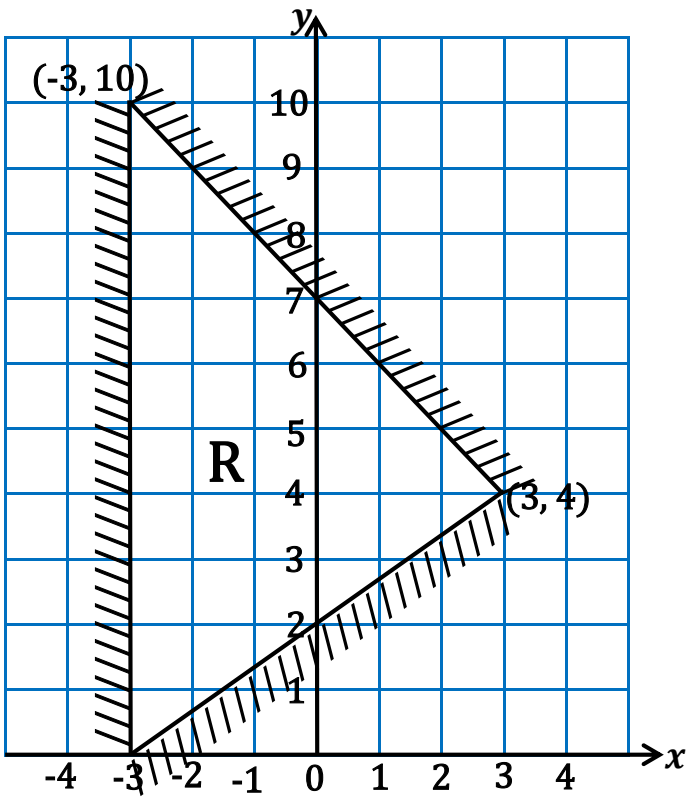
1. Linear Objective Function: A mathematical function that describes the objective of the problem.
2. Linear Constraints: Limitations or restrictions on the variables.
3. Decision Variables: The variables that are adjusted to achieve the objective.
4. Optimal Solution: The solution that maximizes or minimizes the objective function, subject to the constraints.

TERMINOLOGIES

INEQUALITY SYMBOL	TERMINOLOGIES	REGION (R)
$<$	Less than	
$>$	Greater than	
$\leq$	Less than or equal to At most Not more than Minimum Not exceeding Up to Not greater than	
$\geq$	Greater than or equal to At least More than Maximum Exceeding Not less than	
$+$	Addition Altogether Total	

2018.P1.oct

On the XOY plane below, region R is unshaded.



Write the three inequalities that define the region R.

Solution

(i) $x \geq -3$

(ii) $(-3, 10)$ *and* $(3, 4)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4-10}{3+3}$$

$$= \frac{-6}{6}$$

$$= -1$$

$$y - y_1 = m(x - x_1)$$

$$y - 10 = -1(x + 3)$$

$$y - 10 = -x - 3$$

$$y + x = -3 + 10$$

$$y + x = 7$$

$$x + y \leq 7$$

(iii) $(-3, 0)$ *and* $(3, 4)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4-0}{3+3}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{2}{3}(x + 3)$$

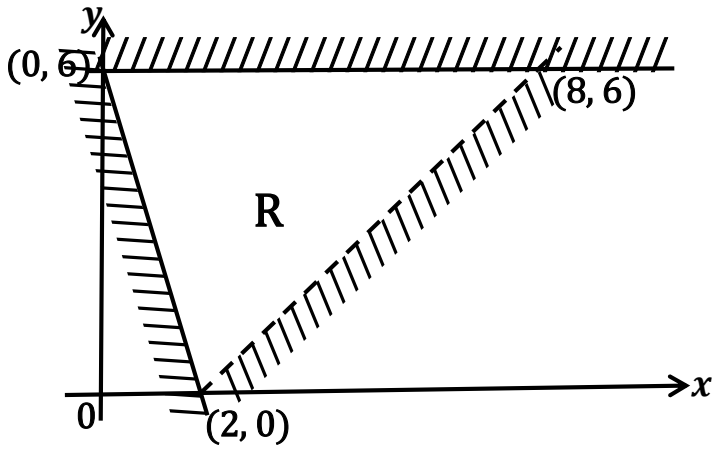
$$3y = 2(x + 3)$$

$$3y = 2x + 6$$

$$3y \geq 2x + 6$$

2019.P1.Nov

Write down three inequalities that define the unshaded region R, on the diagram below.



Solution

(ii) $y \leq 6$

(iii) $(0,6)\ and\ (2,0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{0-6}{2-0}$$

$$= \frac{-6}{2}$$

$$= -3$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -3(x - 0)$$

$$y - 6 = -3x + 0$$

$$y + 3x = 6 + 0$$

$$y + 3x = 6$$

$$3x + y \geq 6$$

(i) $(2,0)\ and\ (8,6)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{6-0}{8-2}$$

$$= \frac{6}{6}$$

$$= 1$$

$$y - y_1 = m(x - x_1)$$

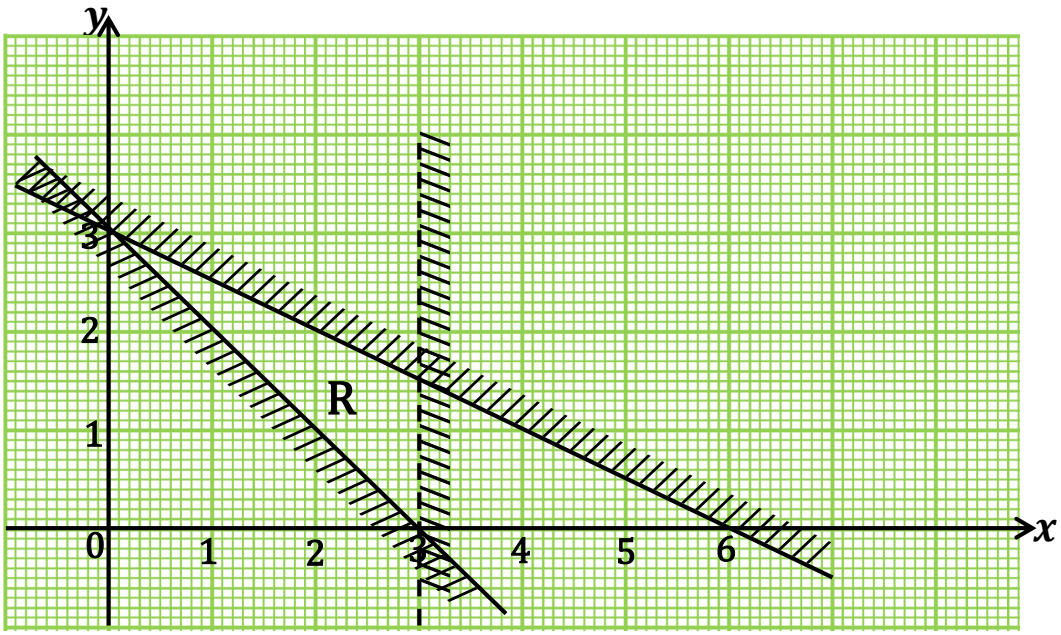
$$y - 0 = 1(x - 2)$$

$$y = x - 2$$

$$y > x - 2$$

2020.P1.Nov

Write three inequalities that define the unshaded region R.



Solution

(i) $x < 3$

(ii) $(0,3)\ and\ (3,0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{0-3}{3-0}$$

$$= \frac{-3}{3}$$

$$= -1$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -1(x - 0)$$

$$y - 3 = -x + 0$$

$$y + x = 3 + 0$$

$$y + x = 3$$

$$x + y \geq 3$$

(iii) $(0,3)\ and\ (6,\ 0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{0-3}{6-0}$$

$$= \frac{-3}{6}$$

$$= -\frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -\frac{1}{2}(x - 0)$$

$$2y - 6 = -1(x - 0)$$

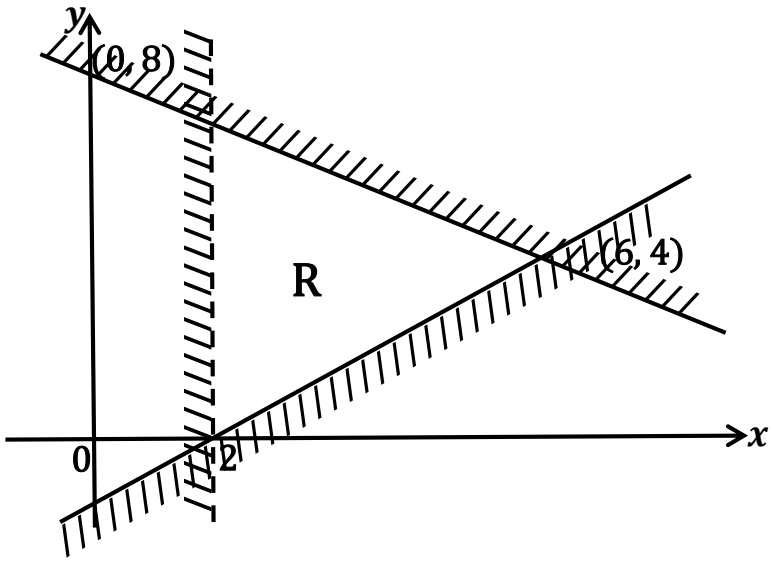
$$2y - 6 = -x$$

$$2y + x = 6$$

$$2y + x \leq 6$$

2021.P1.Nov

Write three inequalities that define the unshaded region R, on the diagram below.



Solution

(i) $x > 2$

(ii) $(0, 8)$ *and* $(6, 4)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4-8}{6-0}$$

$$= \frac{-4}{6}$$

$$= -\frac{2}{3}$$

$$y - y_1 = m(x - x_1)$$

$$y - 8 = -\frac{2}{3}(x - 0)$$

$$3y - 24 = -2(x - 0)$$

$$3y - 24 = -2x + 0$$

$$3y + 2x = 24$$

$$3y + 2x \leq 24$$

(iii) $(2, 0)$ *and* $(6, 4)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4-0}{6-2}$$

$$= \frac{4}{4}$$

$$= 1$$

$$y - y_1 = m(x - x_1)$$

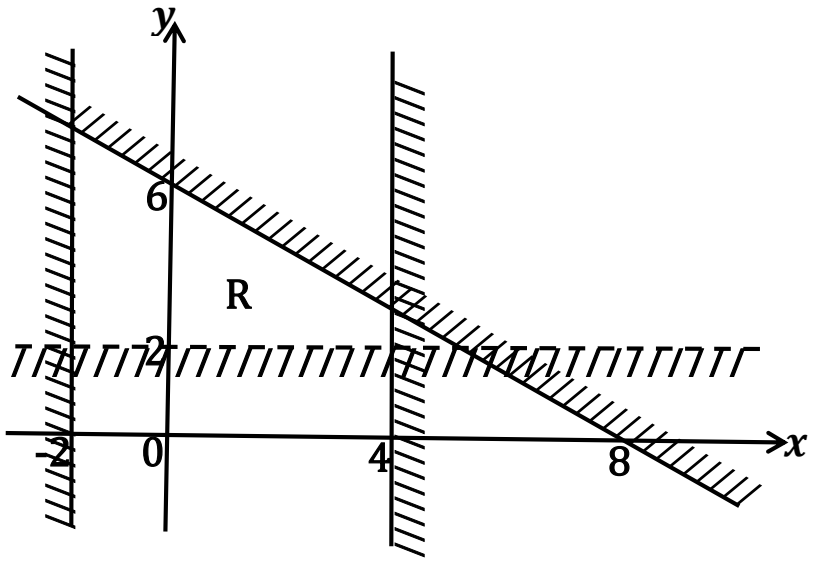
$$y - 0 = 1(x + 2)$$

$$y = x + 2$$

$$y \geq x + 2$$

2022.P1.Nov

Write the four inequalities that define the unshaded region R in the diagram below.



Solution

(i) $y > 2$

(ii) $x \geq -2$

(iii) $x \leq 4$

(iv) $(0, 6)\ and\ (8, 0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{0-6}{8-0}$$

$$= \frac{-6}{8}$$

$$= -\frac{3}{4}$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -\frac{3}{4}(x - 0)$$

$$4y - 24 = -3(x - 0)$$

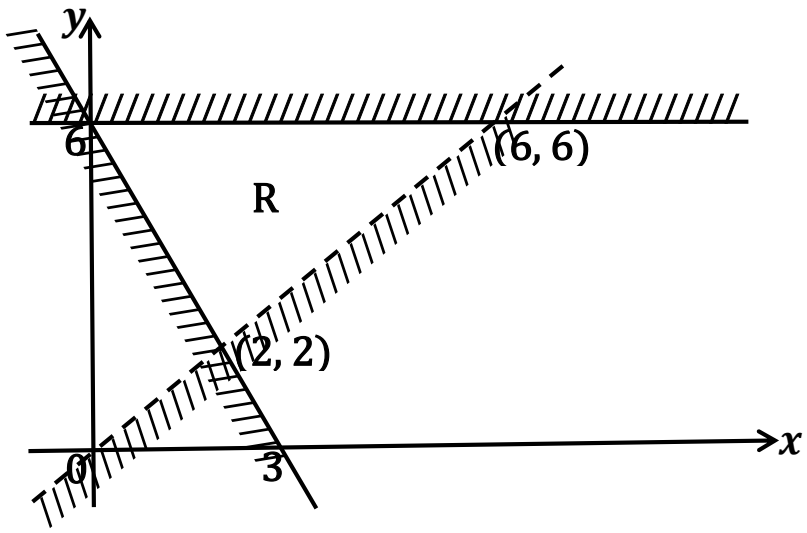
$$4y - 24 = -3x + 0$$

$$4y + 3x = 24$$

$$4y + 3x \leq 24$$

2023.P1.Nov

Write three inequalities that defines the unshaded region R.



Solution

(ii) $y \leq 6$

(iii) $(0, 6)\ and\ (3, 0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{0-6}{3-0}$$

$$= \frac{-6}{3}$$

$$= -2$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -2(x - 0)$$

$$y - 6 = -2x + 0$$

$$y + 2x = 6 + 0$$

$$y + 2x = 6$$

$$y + 2x \geq 6$$

(i) $(0, 0)\ and\ (2, 2)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{2-0}{2-0}$$

$$= \frac{2}{2}$$

$$= 1$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 0)$$

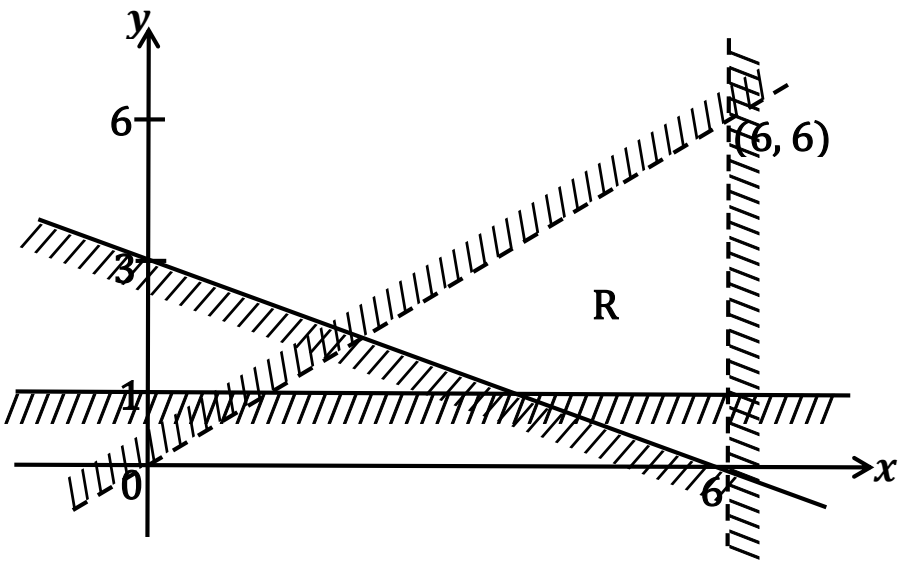
$$y = x - 0$$

$$y = x$$

$$y > x$$

2024.P1.GCE

Write the four inequalities that define the unshaded region R, on the diagram below.



Solution

(i) $y \geq 1$

(ii) $x < 6$

(iii) (0, 3) and (6, 0)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{0-3}{6-0}$$

$$= -\frac{3}{6}$$

$$= -\frac{1}{2}$$

$$y = mx + c$$

$$y = -\frac{1}{2}x + 3$$

$$2y = -x + 6$$

$$2y + x = 6$$

$$2y + x \geq 6$$

(iv) (0, 0) and (6, 6)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{6-0}{6-0}$$

$$= \frac{6}{6}$$

$$= 1$$

$$y = mx + c$$

$$y = x + 0$$

$$y = x$$

$$y < x$$

2018.P2.GCE

Answer the whole of this question on a sheet of graph paper.

A tailor at a certain market intends to make dresses and suits for sale.

(a) Let x represent the number of dresses and y the number of suits. Write the inequalities which represent each of the conditions below.

(i) The number of dresses should not exceed 50. [1]

(ii) The number of dresses should not be less than the number of suits. [1]

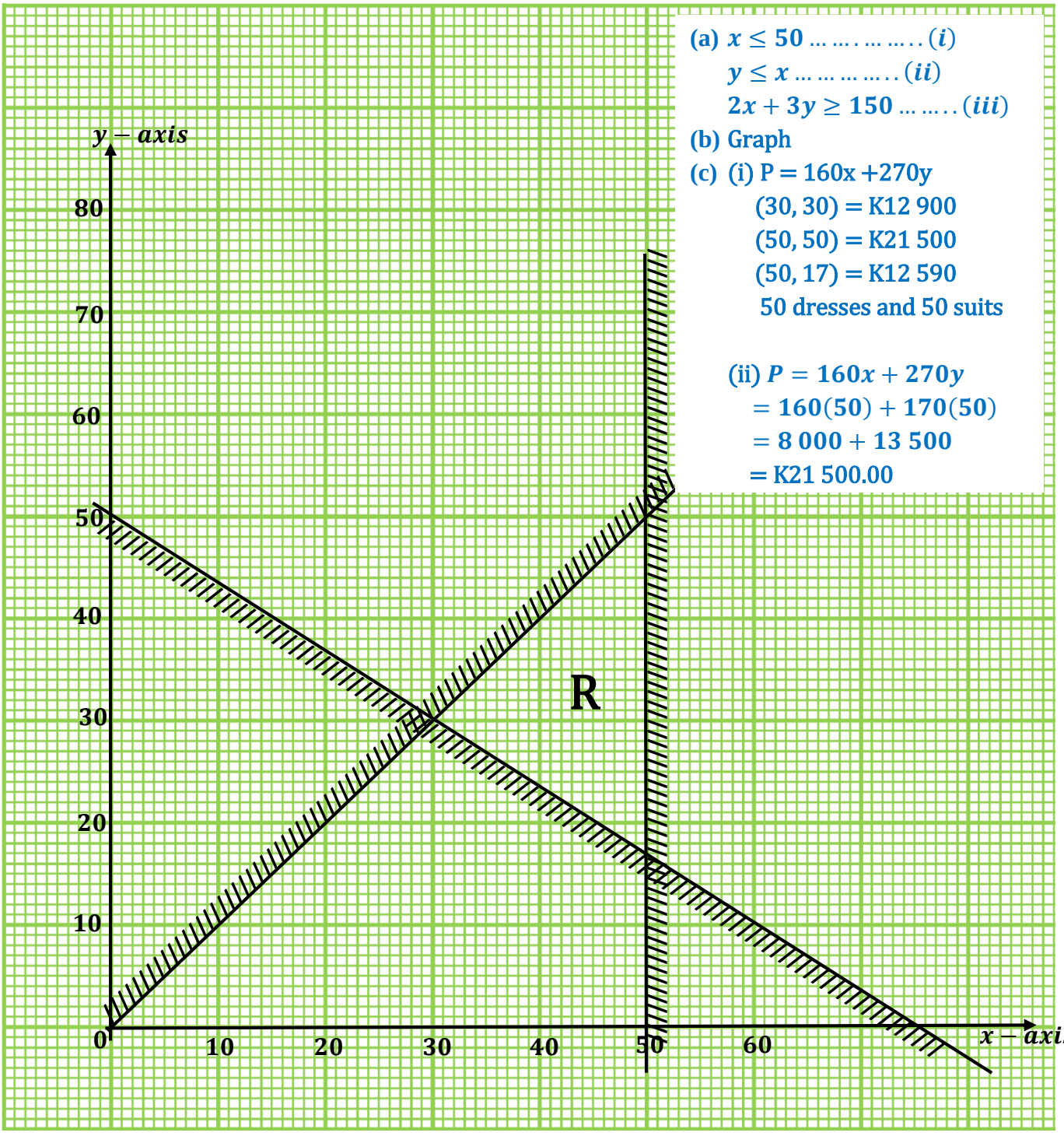
(iii) The cost of making a dress is K140.00 and that of a suit is K210.00. The total cost should be at least K10 500.00. [2]

(b) Using a scale of 2cm to represent 10 units on both axes, draw x and y axes for $0 \leq x \leq 60$ and $0 \leq y \leq 80$. Shade the unwanted region to indicate clearly the region where (x, y) must lie. [4]

(c) (i) The profit on a dress is K160.00 and on a suit it is K270.00. Find the number of dresses and suits the tailor must make for maximum profit. [2]

(ii) Calculate this maximum profit. [2]

Solution



(a) $x \leq 50$ … … … … … (i)

$y \leq x$ … … … … … (ii)

$2x + 3y \geq 150$ … … … (iii)

(b) Graph

(c) (i) P = 160x +270y

(30, 30) = K12 900

(50, 50) = K21 500

(50, 17) = K12 590

50 dresses and 50 suits

(ii) $P = 160x + 270y$

$= 160(50) + 170(50)$

$= 8\,000 + 13\,500$

= K21 500.00

2019.P2.Nov

Kuunika wishes to build a lodge with single and double rooms. He need to decide the number of each room type he should build to maximize profit.

Let x represent the number of single rooms and y the number of double rooms.

(a) Write the inequalities which represents each of the following conditions:
  (i) There must be at least one single room. [1]
  (ii) There must be at least 10 rooms altogether. [1]
  (iii) The total number of rooms should not exceed 15. [1]
  (iv) The number of double rooms must be at least twice the number of single rooms. [1]
  (v) The number of double rooms should not be more than 12. [1]

(b) Using a scale of 2cm to represent 5 units on both axes, draw x and y axes for $0 \leq x \leq 15$ and $0 \leq y \leq 15$ respectively and shade the unwanted region to indicate clearly the region where the solution of the inequalities lie. [5]

(c) The rate for a single room is K600.00 and K900.00 for a double room. How many rooms of each type should Kuunika build to maximize the income? [2]

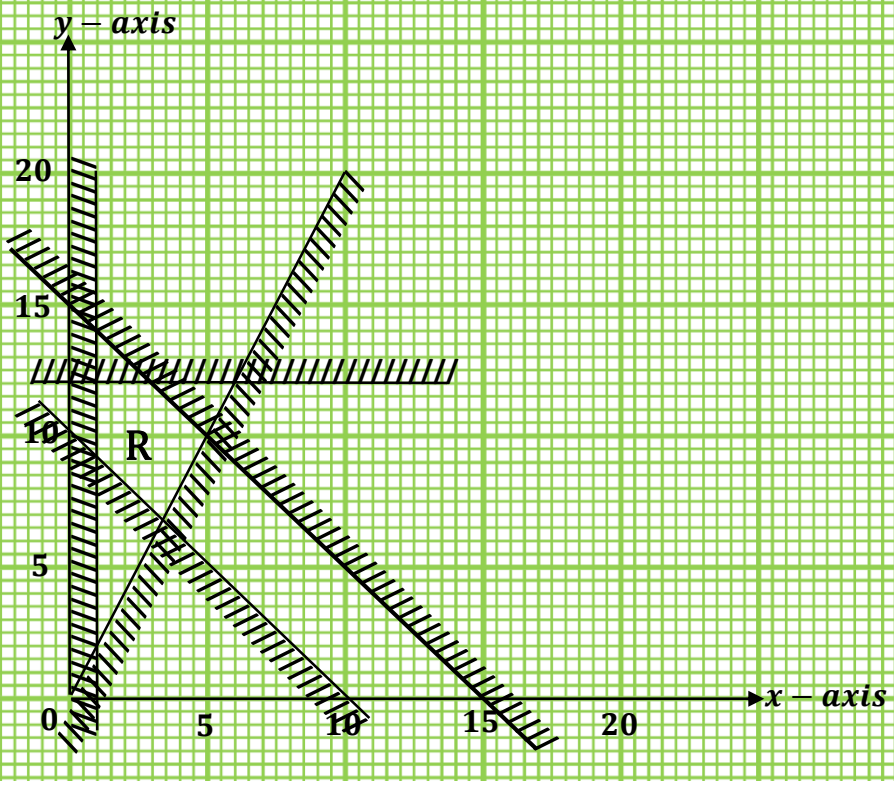
Solution

(a) $x \geq 1$ ……. (i)
$x + y \geq 10$ ……. (ii)
$x + y \leq 15$ ……. (iii)
$y \geq 2x$ ……. (iv)
$y \leq 12$ ……. (v)

(b) Graph

(c) R= 600x +900y
(3, 7) = K8 100
(1, 9) = K8 700
(1, 12) = K11 400
(3, 12) = K12 600
(5, 10) = K12 000

3 single rooms and 12 double rooms.



2020.P2.Nov

A carpenter intends to manufacture at least 10 tables and at least 20 chairs. Each table requires 4 hours of assembling and 2 hours of varnishing. Each chair requires 3 hours of assembling and 1 hour of varnishing. There are 240 hours available for assembling and 100 hours for varnishing.

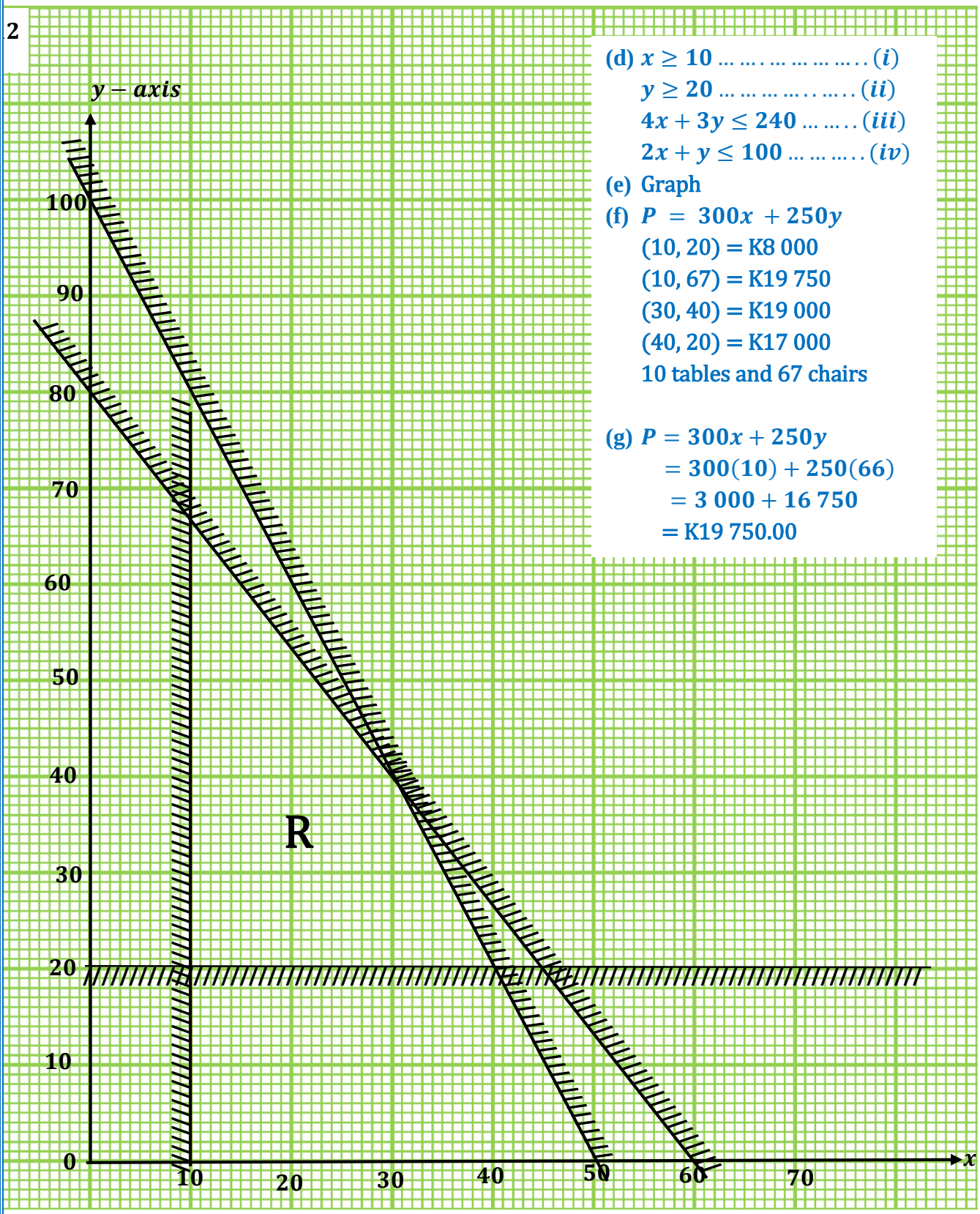
(a) Given that x represents the number of tables and y the number of chairs. Write four inequalities which represents these conditions. [4]

(b) Using a scale of 2cm to represent 10 pieces of furniture on each axis, draw the x and y axes for $\mathbf{0} \leq x \leq \mathbf{70}$ and $\mathbf{0} \leq y \leq \mathbf{100}$ respectively and shade the unwanted region to show clearly the region where the solution of the inequalities lie. [4]

(c) Each table sold yields a profit of K300.00 and each chair sold yields a profit of K250.00. find the best combination of the number of tables and chairs to gain the maximum profit. [2]

(d) Calculate this estimate of the maximum profit. [2]

2



(d) $x \geq 10$ … … . … … … … … . . (i)

$y \geq 20$ … … … … … . … . . (ii)

$4x + 3y \leq 240$ … … . . (iii)

$2x + y \leq 100$ … … … . . (iv)

(e) Graph

(f) $P = 300x + 250y$

(10, 20) = K8 000

(10, 67) = K19 750

(30, 40) = K19 000

(40, 20) = K17 000

10 tables and 67 chairs

(g) $P = 300x + 250y$

$= 300(10) + 250(66)$

$= 3\,000 + 16\,750$

= K19 750.00

2021.P1.Nov

A traditional drinks dealer stocks two brands of drinks, brand A and brand B, both of which are produced in bottles of the same size. He wishes to order fresh supplies and finds that he has room for up to 900 bottles. He knows that brand A is more popular and so decides to order at least twice as many bottles of brand A as brand B. he wishes, however, to have at least 100 bottles of brand B and not more than 700 bottles of brand A.

(a) Let x be the number of bottles of brand A and y the number of bottles of brand B. Write four inequalities to represent the information above. [4]

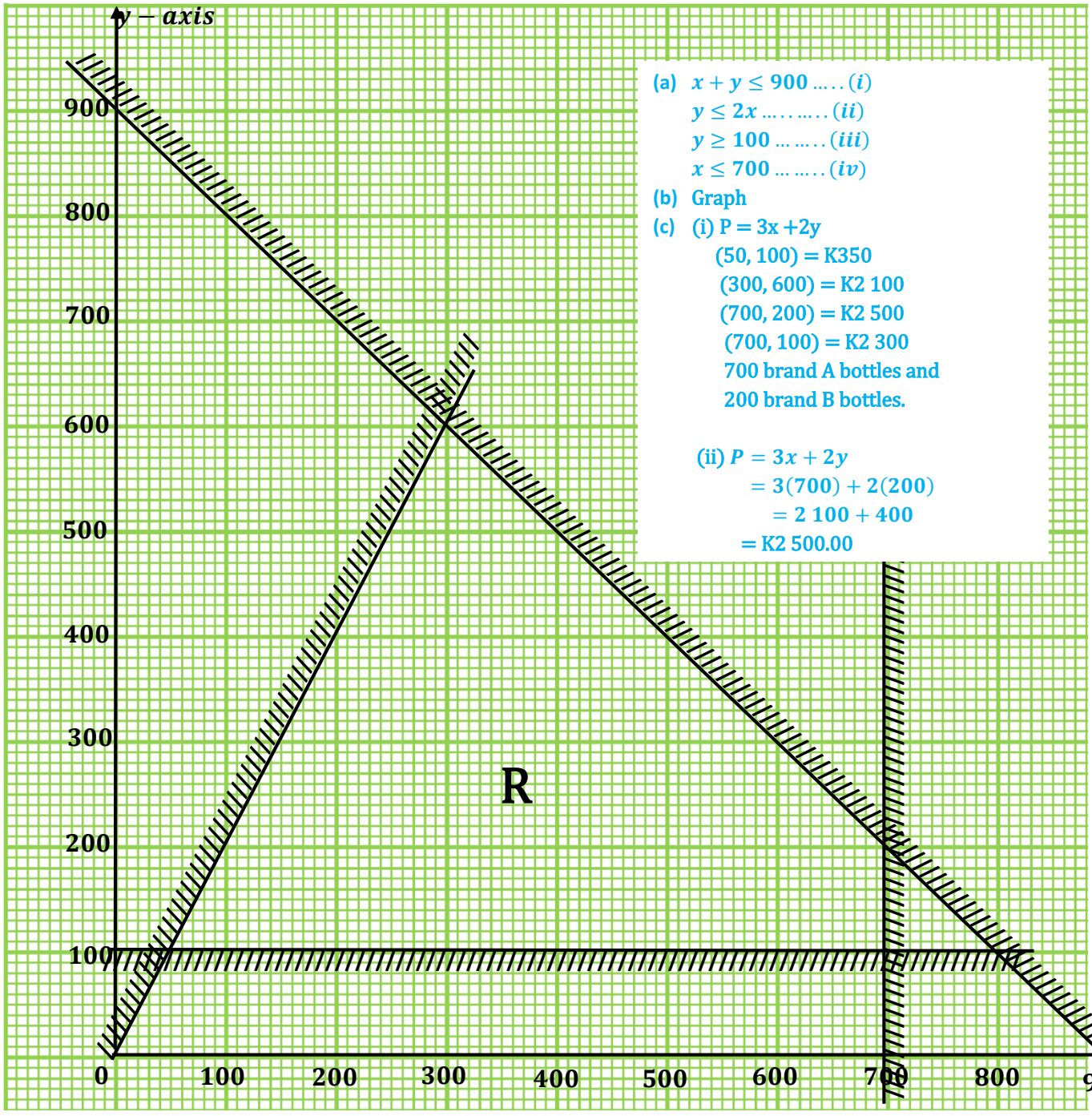
(b) Using a scale of 2cm to represent 100 bottles on each axis, draw x and y axes for $\mathbf{0} \leq x \leq \mathbf{900}$ and $\mathbf{0} \leq y \leq \mathbf{900}$ respectively and shade the unwanted region to show clearly the region where the solution of the inequalities lie. [4]

(c) Given that the profit on a bottle of brand A is K3.00 and on a bottle of brand B is K2.00, find

(i) the number of bottles of each brand that gives maximum profit, [2]

(ii) the maximum profit. [2]

Solution



(a) $x + y \leq 900$ ….. (i)
$y \leq 2x$ ……..…. (ii)
$y \geq 100$ ……..… (iii)
$x \leq 700$ ……..… (iv)

(b) Graph

(c) (i) P = 3x +2y
(50, 100) = K350
(300, 600) = K2 100
(700, 200) = K2 500
(700, 100) = K2 300
700 brand A bottles and
200 brand B bottles.

(ii) $P = 3x + 2y$
$= 3(700) + 2(200)$
$= 2\ 100 + 400$
= K2 500.00

2022.P2.Nov

A businessman intends to sell two types of markers, permanent and dry erase. He decides that there should be at least 20 permanent markers and at least 10 dry erase makers. In order to make his business stable, he decides to order not more than 70 markers altogether. The order price of a permanent marker is K15.00 and that of a dry erase marker is K20.00. He is prepared to spend not more than K1 200.00 altogether.

(a) Let x represent the number of permanent markers and y the number of dry erase markers. Write four inequalities which satisfy the above conditions. [4]

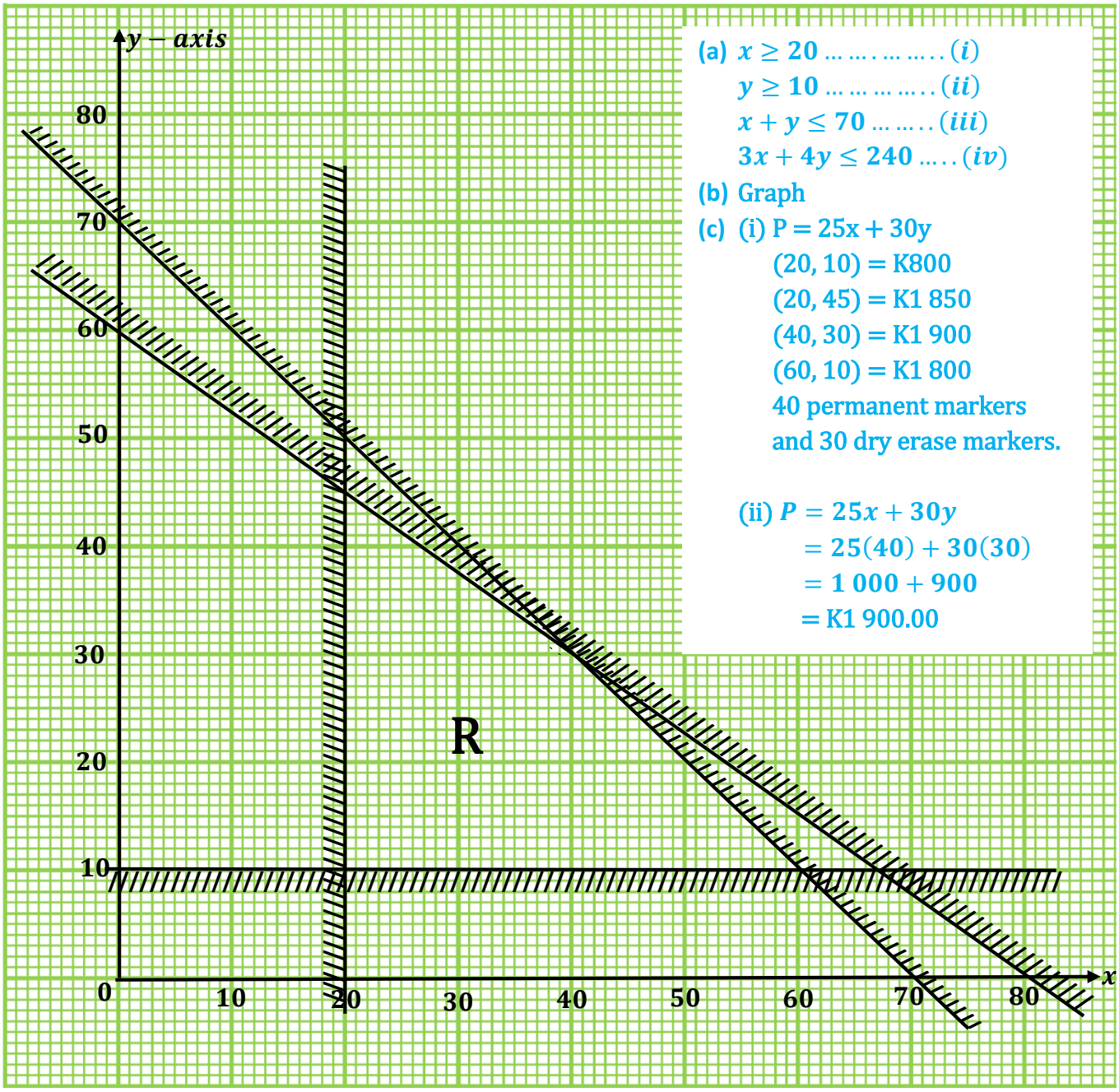
(b) Using a scale of 2cm to represent 10 markers on each axis, draw x and y axes for $0 \leq x \leq 80$ and $0 \leq y \leq 80$ respectively and shade the unwanted region to show clearly the region where the solution of the inequalities lie. [4]

(c) The profit on the sale of a permanent marker is K25.00 and on each dry erase marker profit is K30.00.

(i) How many markers of each type should be ordered to make maximum profit? [2]

(ii) Find the maximum profit. [2]

Solution



(a) $x \geq 20$ ... ... (i)
$y \geq 10$ ... ... (ii)
$x + y \leq 70$ ... ... (iii)
$3x + 4y \leq 240$ ... ... (iv)

(b) Graph

(c) (i) P = 25x + 30y
(20, 10) = K800
(20, 45) = K1 850
(40, 30) = K1 900
(60, 10) = K1 800
40 permanent markers
and 30 dry erase markers.

(ii) $P = 25x + 30y$
$= 25(40) + 30(30)$
$= 1\,000 + 900$
= K1 900.00

2023.P2.Nov

A man intends to buy goats and cows for sale. He decides to buy not more than 50 goats and not more than 40 cows. He plans to buy at least 60 animals altogether.

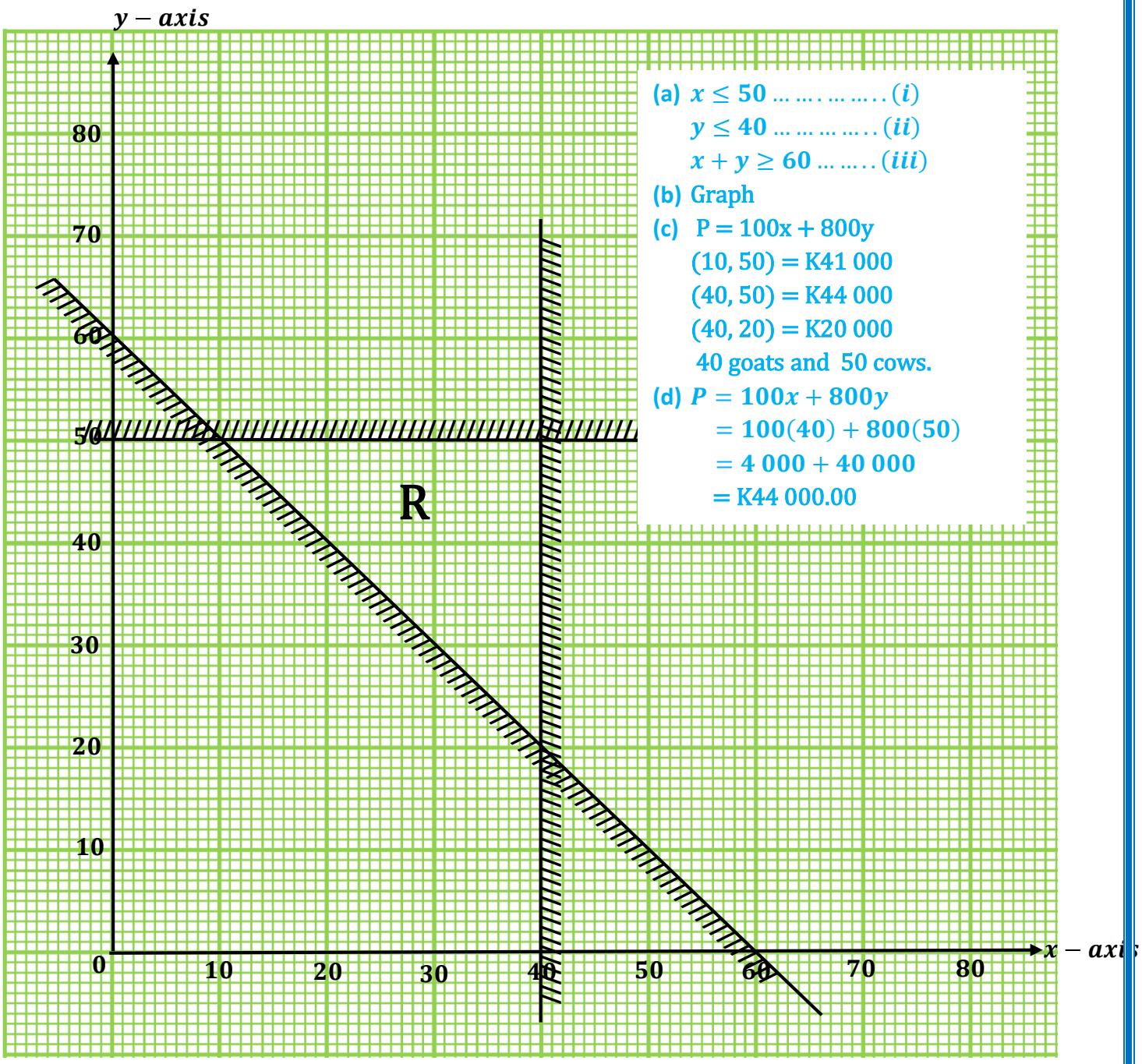
(a) Given that x represents the number of goats and y the number of cows, write four inequalities which represents these conditions. [4]

(b) Using a scale of 2cm to represent 10 animals on each axis, draw x and y axes for $\mathbf{0} \leq x \leq \mathbf{80}$ and $\mathbf{0} \leq y \leq \mathbf{80}$ respectively and shade the unwanted region to show clearly the region where the solution of the inequalities lie. [4]

(c) The profit on the sale of a goat is K100.00 and on a cow profit is K800.00. How many animals of each should he buy to make maximum profit? [2]

(d) What is this maximum profit? [2]

Solution


y − axis
80
70
60
50
40
30
20
10
0
10
20
30
40
50
60
70
80
x − axis
R
(a) x ≤ 50 … … … … … (i)
y ≤ 40 … … … … … (ii)
x + y ≥ 60 … … … (iii)
(b) Graph
(c) P = 100x + 800y
(10, 50) = K41 000
(40, 50) = K44 000
(40, 20) = K20 000
40 goats and 50 cows.
(d) P = 100x + 800y
= 100(40) + 800(50)
= 4 000 + 40 000
= K44 000.00

2024.P2.GCE

Answer the whole of this question on a sheet of graph paper.

Mapulanga's carpentry shop makes two types of chairs, type A and type B for sale. The cost of making one chair of each type is K2 000.00. The shop has K140 00.00 available for making chairs. The number of chairs of type A must be at least 20 while that of type B must be at least 10.

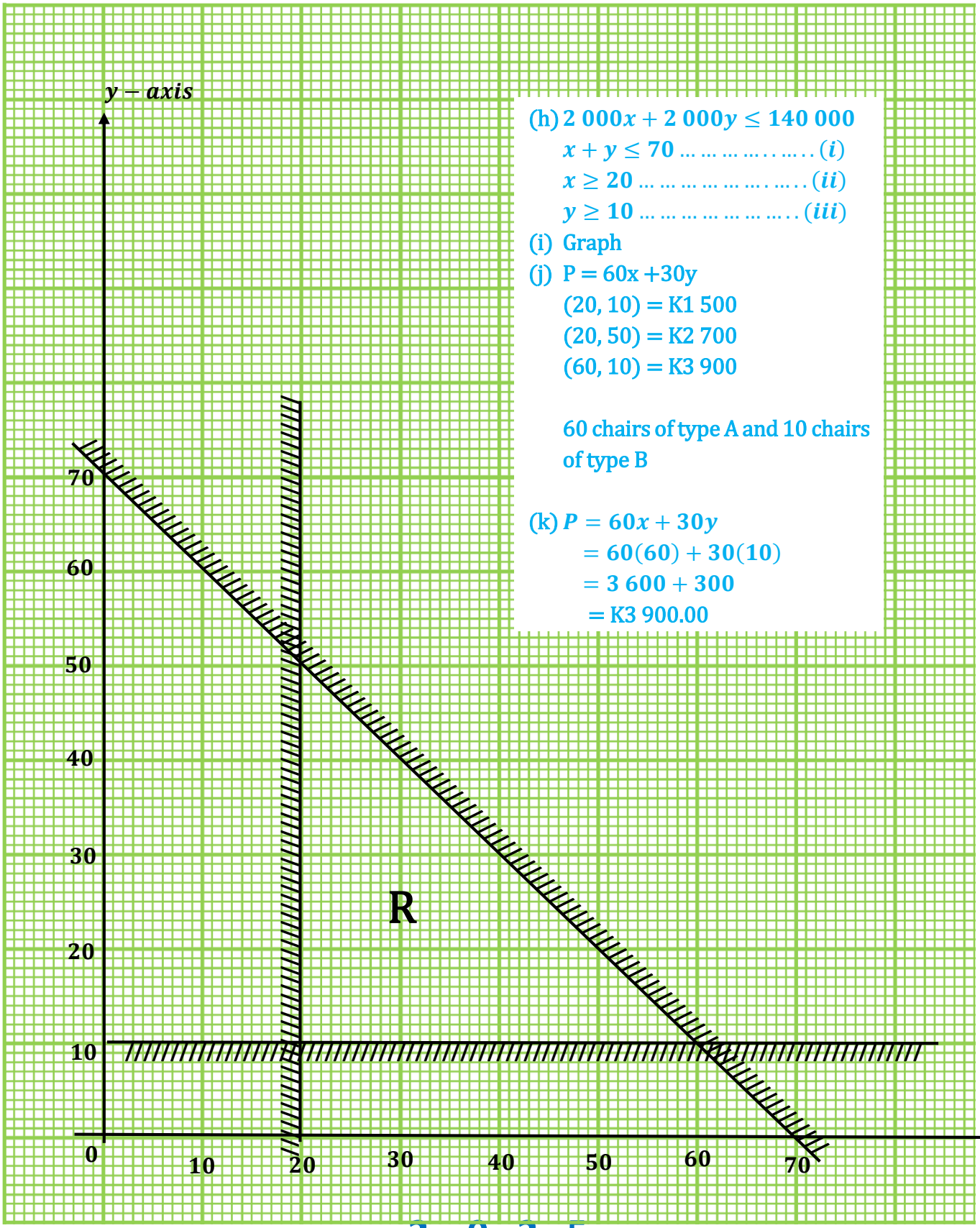
(a) If x is the number of chairs of type A and y is the number of chairs of type B, write three inequalities which represent these conditions. [4]

(b) Using a scale of 2cm to represent 10 chairs on each axis, draw x and y axes for $\mathbf{0} \leq x \leq \mathbf{70}$ and $\mathbf{0} \leq y \leq \mathbf{70}$ respectively and shade the unwanted region to show clearly the region where the solution of the inequalities lie. [4]

(c) Given that the profit on type A chair is K60.00 and on type B is K30.00, how many chairs of each type should be made for maximum profit? [2]

(d) Calculate the maximum profit. [2]

Solution



(h) $2\,000x + 2\,000y \leq 140\,000$

$x + y \leq 70$ ……. (i)

$x \geq 20$ ……. (ii)

$y \geq 10$ ……. (iii)

(i) Graph

(j) P = 60x +30y

(20, 10) = K1 500

(20, 50) = K2 700

(60, 10) = K3 900

60 chairs of type A and 10 chairs of type B

(k) $P = 60x + 30y$

$= 60(60) + 30(10)$

$= 3\,600 + 300$

= K3 900.00

GRAPHS OF CUBIC FUNCTIONS

Objectives

By the end of this topic, learners should

1 Define cubic functions and identify their general form.
2 Sketch and draw the graphs of cubic functions, including key features such as intercepts, turning points, and asymptotes.
3 Analyze the behavior of cubic functions, including increasing/decreasing intervals and concavity.
4 Solve problems involving cubic functions, including finding roots and determining the maximum/minimum value.
5 Solve cubic equations by using the curve.
6 Calculate area under the curve using trapezium rule.
7 Calculate the gradient of the curve by drawing a tangent at given point.

Definition

- A cubic function is a polynomial function of degree three, which means the highest power of the variable (usually x) is three.
- The general form of a cubic function is $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c, and d are constants (real numbers), a cannot be zero (if $a = 0$, the function becomes a quadratic function) and x is the variable (input).
- Examples of cubic functions are:
  (i) $f(x) = x^3 + 2x^2 - 3x + 1$
  (ii) $f(x) = 2x^3 - 5x^2 + x - 3$
  (iii) $f(x) = x^3 - 2x^2 - 5x + 2$
- Cubic functions can be used to model various real-world phenomena, such as population growth, electrical circuits, and mechanical systems.
- They are also used in computer graphics, game development, and other fields.

Introduction

- Review quadratic functions and their graphs.
- Introduce cubic functions and their general form: $f(x) = ax^3 + bx^2 + cx + d$.
- Discuss the differences between quadratic and cubic functions.

Key Features of Cubic Functions

Intercepts:

- x-intercepts (roots): values of x where the graph crosses the x-axis.
- y-intercept: value of y where the graph crosses the y-axis.

Turning Points:

- Local maxima: highest point on the graph.
- Local minima: lowest point on the graph.

Asymptotes:

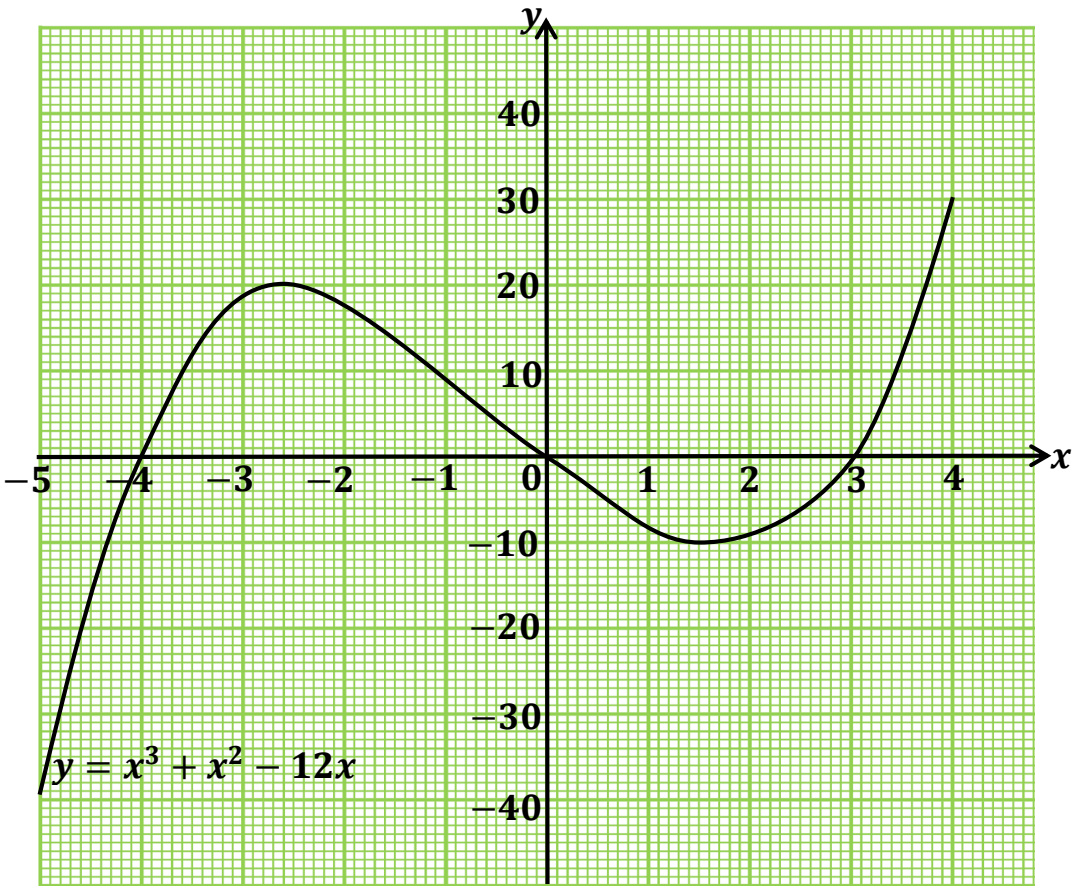
- None: cubic functions do not have asymptotes.
- Horizontal asymptote: may have a horizontal asymptote, depending on the leading coefficient.

End Behavior:

- Leading coefficient (a): determines the end behavior of the graph.

2018.P2.GCE

The diagram below shows the graph of $y = x^3 + x^2 - 12x$.



(i) Use the graph to solve the equations

(a) $x^3 + x^2 - 12x = 0$, [2]

(b) $x^3 + x^2 - 12x = x + 10$. [3]

(ii) Calculate an estimate of the

(a) gradient of the curve at the point where $x = -3$, [2]

(b) area bounded by the curve, $x = -3$, $x = -1$ and $y = -10$. [2]

Solution

(i) (a) $x^3 + x^2 - 12x = 0$

$y = 0$

$\therefore x = -4, x = 0 \text{ or } x = 3$

(b) $x^3 + x^2 - 12x = x + 10$

$y = x + 10$

$\therefore x = 3.51, x = -0.76 \text{ or } x = -3.75$

(ii) (a) (-3, 18) and (-4, 10)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 18}{-4 + 3}$$

$$= \frac{-8}{-1}$$

$$= 8$$

(b) Area $= (l \times b) + \frac{1}{2}(a + b)h$

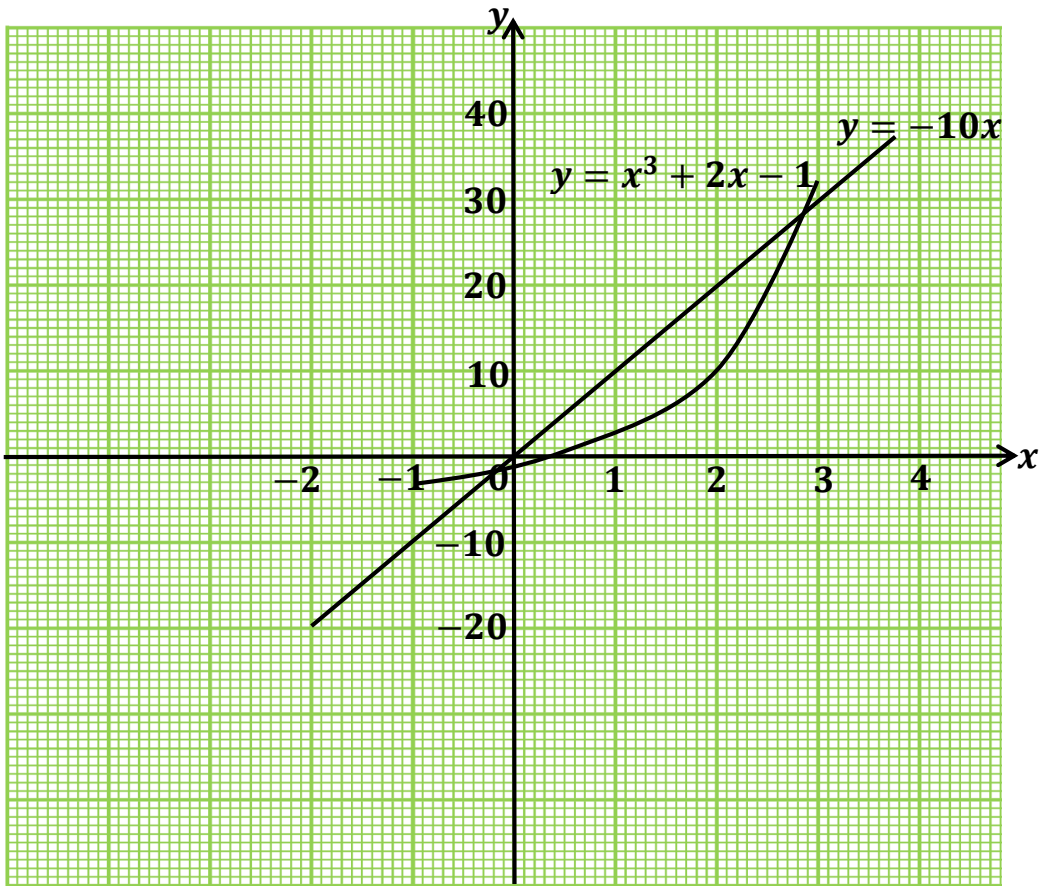
$$= (30 \times 1) + \frac{1}{2}(23 + 30)1$$

$$= 30 + 26.5$$

$$= 56.5 \ units^2$$

2019.P2.Nov

The diagram below shows part of the graph of $y = x^3 + 2x - 1$ and $y = -10x$.



(i) Use the graph to solve the equations

(a) $x^3 + 2x = 6$, [2]

(b) $x^3 + 2x - 1 = 10x$. [3]

(ii) Calculate an estimate of

(a) the gradient of the curve at the point (2, 11), [2]

(b) the area bounded by the curve, $x = -3$, $x = -1$ and $y = -10$. [2]

Solution

(i) (a) $x^3 + 2x = 6$

$x^3 + 2x - 1 = 6 - 1$

$x^3 + 2x - 1 = 5$

$y = 5$

$\therefore x = 1.5$

(b) $x^3 + 2x - 1 = 10x$

$y = 10x$

$\therefore x = -0.1 \; or \; x = 2.8$

(ii) (a) (2, 11) and (3, 24)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{24 - 11}{3 - 2}$

$= \frac{13}{1}$

$= 13$

(b) Area $= \frac{1}{2}bh + \frac{1}{2}bh + \frac{1}{2}bh$

$= \frac{1}{2} \times 1 \times 10 + \frac{1}{2} \times 10 \times 1 + \frac{1}{2} \times 1 \times 10$

$= 5 + 5 + 5$

$= 15 \; units^2$

2020.P2.Nov

The values of x and y are connected by the equation $y = x^3 + 4x^2 + x - 5$ as shown in the table below.

x	-4	-3	-2	-1	0	1	2
y	-9	1	1	-3	-5	1	p

(i) Calculate the value of p. [1]

(ii) Using a scale of 2cm to represent 1 unit on the x-axis and 2cm to represent 5 units on the y-axis for $-4 \leq x \leq 2$ and $-10 \leq y \leq 25$, draw the graph of $y = x^3 + 4x^2 + x - 5$. [3]

(iii) Use your graph to find the solutions of the equations

(a) $x^3 + 4x^2 + x - 5 = 0$, [2]

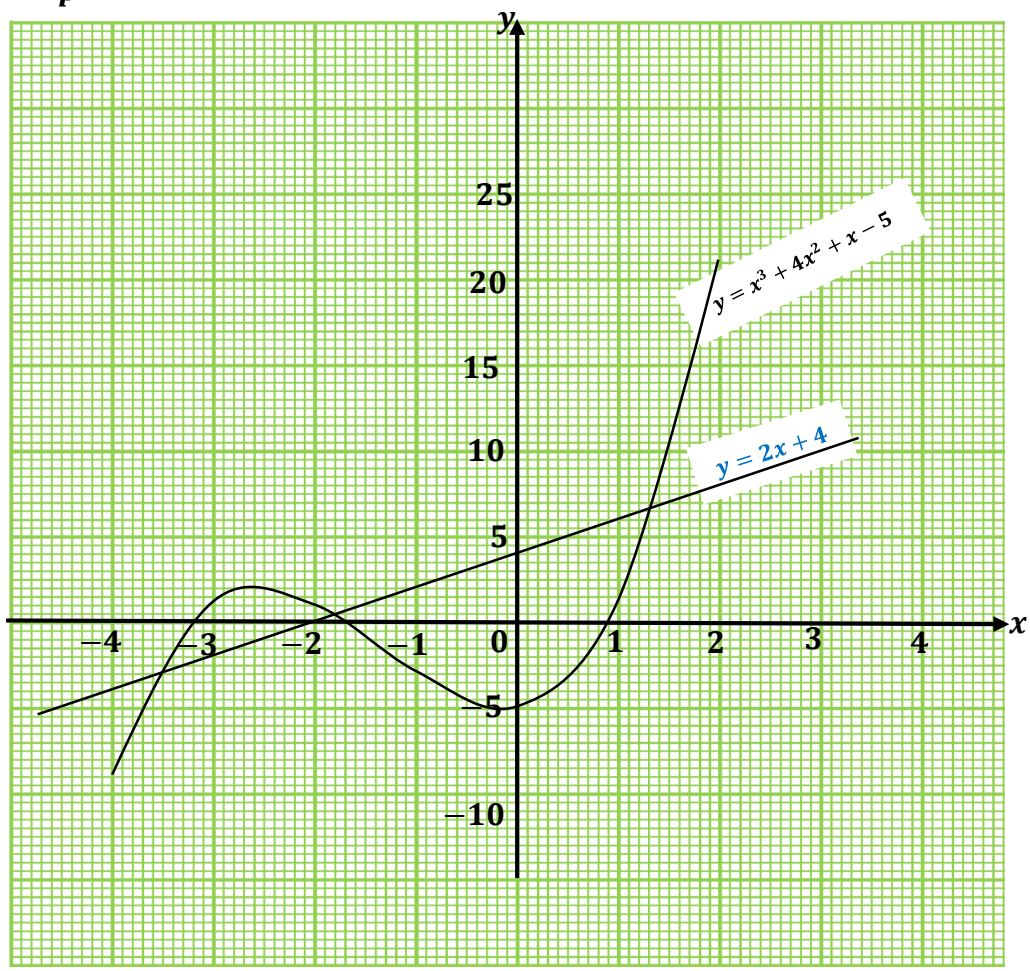
(b) $x^3 + 4x^2 + x - 5 = 2x + 4$. [3]

Solution

(a) $y = x^3 + 4x^2 + x - 5$

(i) $p = (2)^3 + 4(2)^2 + 2 - 5$

$p = 21$



(iii) (a) $x^3 + 4x^2 + x - 5 = 0$

$y = 0$

$\therefore x = -3.2$

$x = -1.7$

$x = 0.9$

(b) $x^3 + 4x^2 + x - 5 = 2x + 4$

$y = 2x + 4$

$\therefore x = -3.5$

$x = -1.8$

$x = 1.3$

2021.P2.Nov

The values of x and y are connected by the equation $y = x(x^2 - 4)$. Some corresponding values of x and y are given in the table below.

x	-3	-2	-1	0	1	2	3
y	-15	r	3	0	-3	0	15

(i) Calculate the value of r. [1]

(ii) Using a scale of 2cm to represent 1 unit on the x-axis for $-3 \leq x \leq 3$ and 2cm to represent 10 units on the y-axis for $-20 \leq y \leq 20$, draw the graph of $y = x(x^2 - 4)$. [3]

(iii) Use your graph to find the solutions of the equations

(a) $x(x^2 - 4) = 0$, [2]

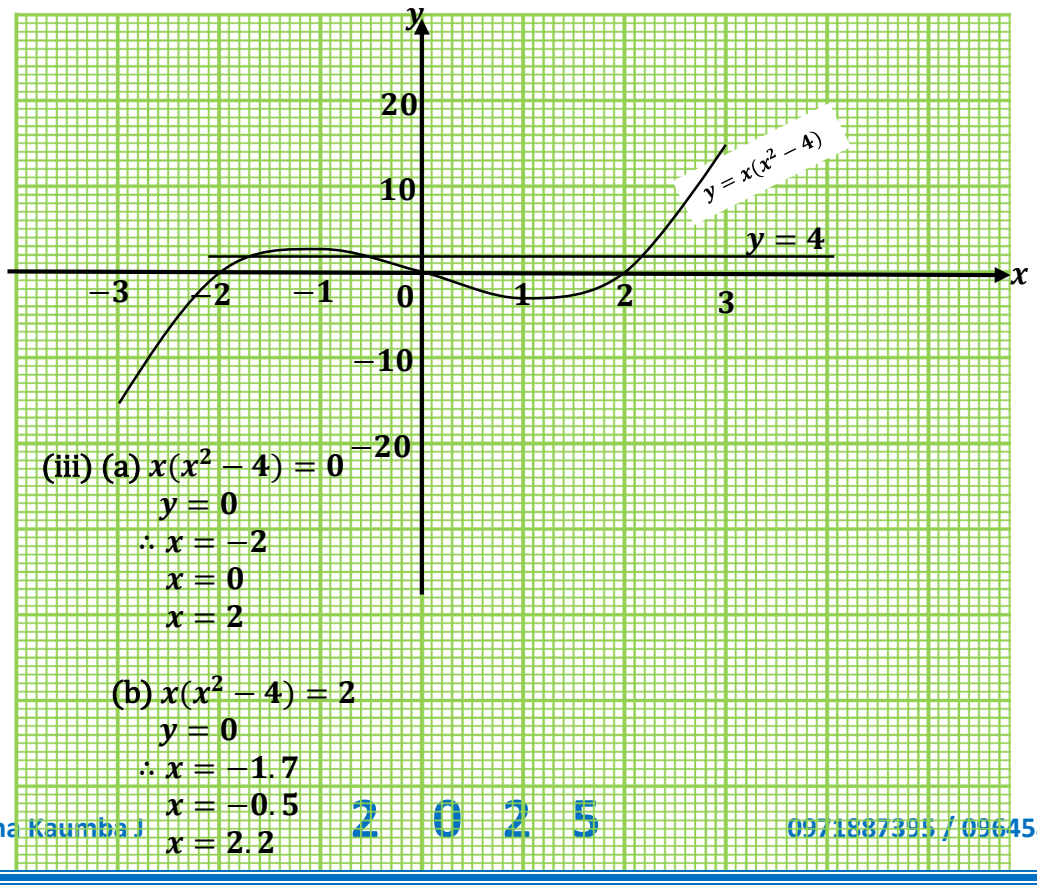
(b) $x(x^2 - 4) = 2$. [3]

Solution

(b) $y = x(x^2 - 4)$

(i) $r = -2[(-2)^2 - 4]$

$r = 0$



(iii) (a) $x(x^2 - 4) = 0$

$y = 0$

$\therefore x = -2$

$x = 0$

$x = 2$

(b) $x(x^2 - 4) = 2$

$y = 0$

$\therefore x = -1.7$

$x = -0.5$

$x = 2.2$

2022.P2.Nov

The variables x and y are connected by the equation $y = (x-3)(x-5)(x+2)$. Some corresponding values of x and y are shown in the table below.

x	-2	-1	0	1	2	3	4	5	6
y	0	24	30	24	12	0	-6	0	k

(i) Calculate the value of k. [1]

(ii) Using a scale of 2cm to represent 1 unit on the x-axis for $-2 \leq x \leq 6$ and 2cm to represent 10 units on the y-axis for $-10 \leq y \leq 40$, draw the graph of $y = (x-3)(x-5)(x+2)$. [3]

(iii) Use your graph to estimate the

(a) gradient of the curve at the point (-1, 24), [2]

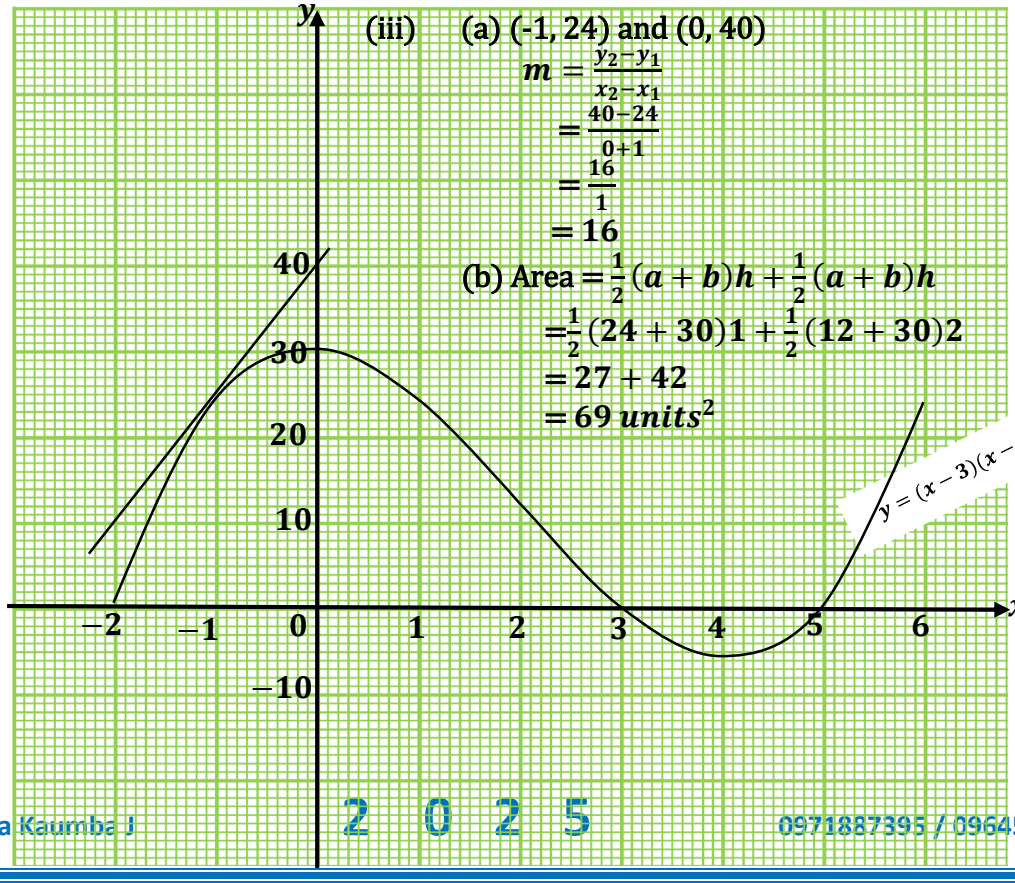
(b) area bounded by the curve, $x = -1, x = 2$ and $y = 0$. [3]

Solution

(c) $y = (x-3)(x-5)(x+2)$

(i) $k = (6-3)(6-5)(6+2)$

$k = 24$



(iii) (a) (-1, 24) and (0, 40)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{40-24}{0+1}$

$= \frac{16}{1}$

$= 16$

(b) Area $= \frac{1}{2}(a+b)h + \frac{1}{2}(a+b)h$

$= \frac{1}{2}(24+30)1 + \frac{1}{2}(12+30)2$

$= 27 + 42$

$= 69\ units^2$

2023.P2.Nov

The values of x and y are connected by the equation $y = x^3 - 9x + 5$. Some of the corresponding values of x and y are given in the table below.

x	-3.5	-3	-2	-1	0	1	2	2.5	3	3.5
y	-6.4	5	q	13	5	-3	-5	-1.9	5	16.4

(i) Calculate the value of q. [1]

(ii) Using a scale of 2cm to represent 1 unit on the x-axis and 2cm to represent 5 units on the y-axis for $-4 \leq x \leq 4$ and $-10 \leq y \leq 20$, draw the graph of $y = x^3 - 9x + 5$. [3]

(iii) Use your graph to find the solutions of the equations

(a) $x^3 - 9x + 5 = 0$, [2]

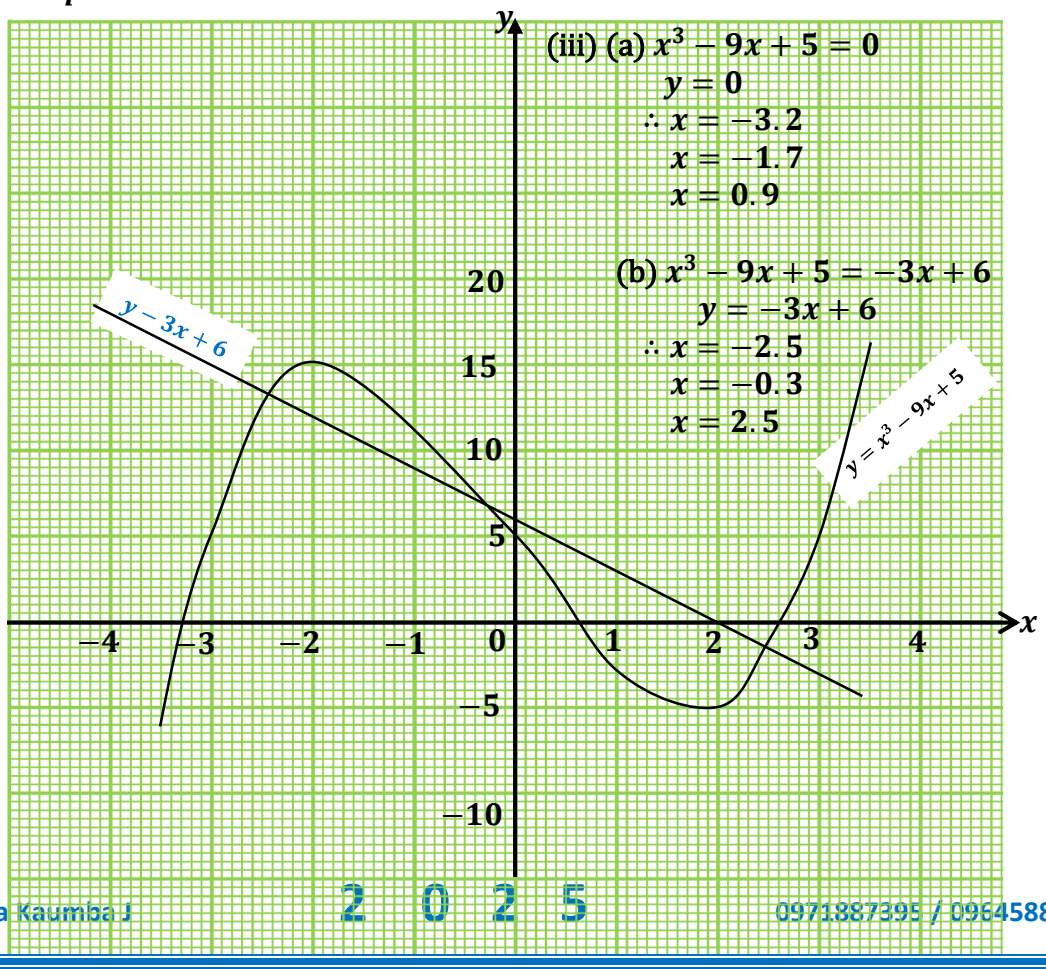
(b) $x^3 - 9x + 5 = -3x + 6$. [3]

Solution

(d) $y = x^3 - 9x + 5$

(i) $q = (-2)^3 - 9(-2) + 5$

$q = 15$



(iii) (a) $x^3 - 9x + 5 = 0$

$y = 0$

$\therefore x = -3.2$

$x = -1.7$

$x = 0.9$

(b) $x^3 - 9x + 5 = -3x + 6$

$y = -3x + 6$

$\therefore x = -2.5$

$x = -0.3$

$x = 2.5$

2024.P2.GCE

The values of x and y are connected by the equation $y = -x^3 + x^2 + 6x$. Some of the corresponding values of x and y are given in the table below.

x	-3	-2	-1	0	1	2	3	4
y	18	0	-4	0	6	8	0	q

(i) Calculate the value of q. [1]

(ii) Using a scale of 2cm to represent 1 unit on the x-axis for $-3 \leq x \leq 4$ and 2cm to represent 10 units on the y-axis for $-30 \leq y \leq 20$, draw the graph of $y = -x^3 + x^2 + 6x$. [3]

(iii) Use your graph to

(a) solve the equation $-x^3 + x^2 + 6x = 4 - x$, [2]

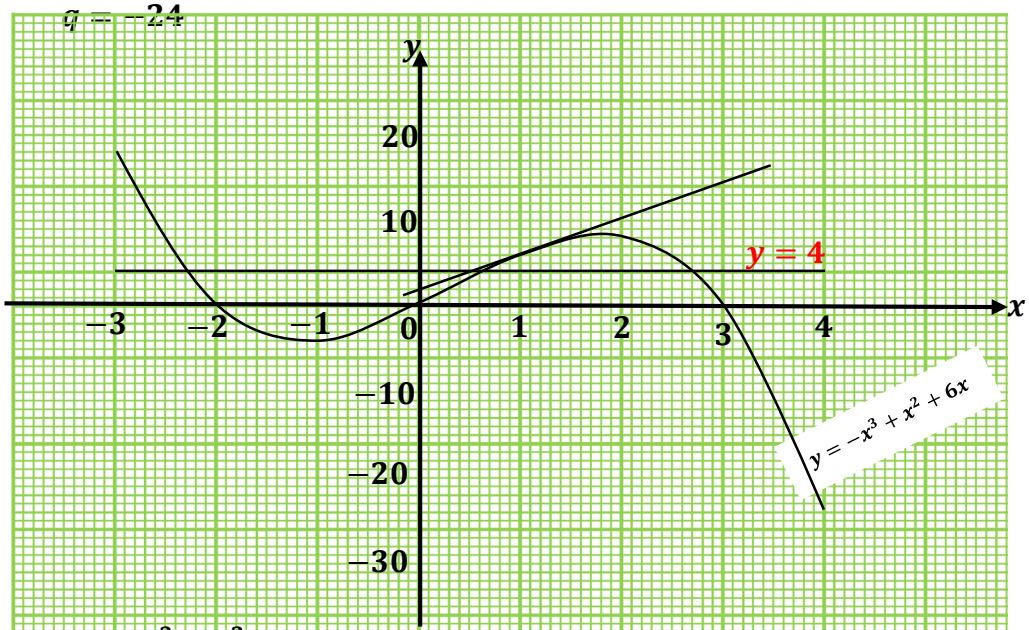
(b) estimate the gradient of the curve at the point where $x = 1$. [3]

Solution

(b) $y = -x^3 + x^2 + 6x$

(i) $q = -(4)^3 + (4)^2 + 6(4)$

$q = -24$



(iii) (a) $-x^3 + x^2 + 5x = 4 - x$

$-x^3 + x^2 + 5x + x = 4 - x + x$

$-x^3 + x^2 + 6x = 4$

$y = 4$

$\therefore x = -2.3$

$x = 0.6$

$x = 2.7$

(b) (1, 6) and (2, 10)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{10-6}{2-1}$

$= \frac{4}{1}$

4

EARTH GEOMETRY

Objectives

By the end of this topic, learners should

1. Define and explain the concept of Earth geometry, including the Earth's shape, size, and proportions.
2. Identify and describe the different types of circles on the Earth, including great circles, small circles, and parallel circles.
3. Explain the concept of latitude and longitude, including how to measure and calculate distances using these coordinates.
4. Apply Earth geometry concepts to solve problems involving distance, direction, and navigation.
5. Analyze and interpret maps and globes to understand the relationships between different geographical features.

Definition

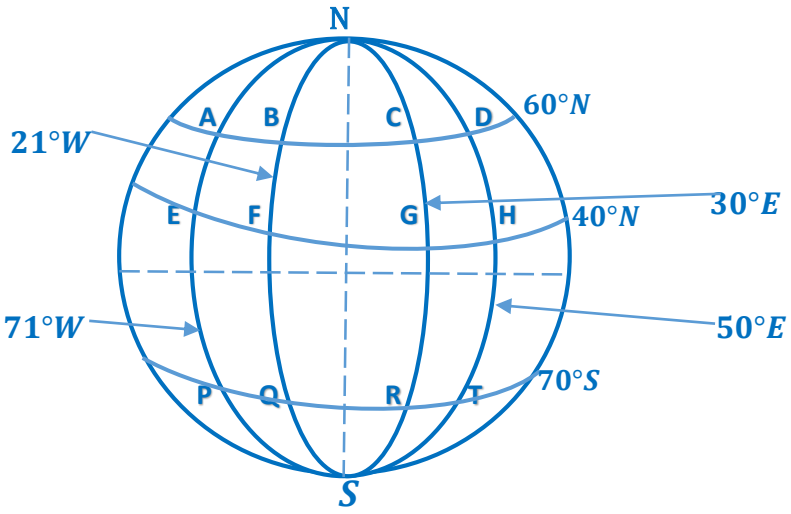
- Earth Geometry is a branch of Mathematics which deals with the study of the earth's surface using small circles (latitudes) and great circles (longitudes).
- Lines of latitudes run in an East-West direction on the earth's surface, while lines of longitudes run in a North-South direction on the earth's surface.

The position of a place on the earth's surface

- The position of a place on the surface of the earth is stated (determined) by identifying its latitude and longitude in coordinate form, e.g. (latitude, longitude)

Examples

The diagram below shows the sketch of the earth's surface.



Using the diagram above, state the positions of the points A, B, C, D, E, F, G, H, P, Q, R and T.

Solution

A(60°N, 71°W)

B(60°N, 21°W)

C(60°N, 30°E)

D(60°N, 50°E)

E(40°N, 71°W)

F(40°N, 21°W)

G(40°N, 30°E)

H(40°N, 50°E)

P(70°S, 71°W)

Q(70°S, 21°W)

R(70°S, 30°E)

T(70°S, 50°E)

The difference between two places on the earth's surface

- ✓ When two points are on opposite sides of the equator or GMT, we add the angles of either the latitudes or the longitudes to get their differences.
- ✓ When two points are on the same side of the equator or GMT, we subtract angles of either latitudes or longitudes to get their differences.

The distance between two places on the surface of the earth.

(a) Distance along a great circle (longitude)

- ➢ The distance between two points on the earth's surface along the parallel of great circles (longitudes) is given by $\mathbf{D} = \frac{\theta}{360^\circ} \times 2\pi R$.
- ➢ If the radius (R) of the earth is given in nautical miles, then the distance between two points on the earth's surface along the parallel of great circles (longitudes) can also be given by $\mathbf{D} = \theta \times 60$.

(b) Distance along a small circle (latitude)

- ➢ The distance between two points on the earth's surface along the parallel of small circles (latitudes) is given by $\mathbf{D} = \frac{\alpha}{360^\circ} \times 2\pi \mathbf{R} \cos \theta$.
- ➢ If the radius (R) of the earth is given in nautical miles, then the distance between two points on the earth's surface along the parallel of small circles (latitudes) can also be given by $\mathbf{D} = \alpha \times 60 \cos \theta$.

(c) The radius of a circle of latitude

$r = \mathbf{R} \cos \theta$

(d) The circumference of the earth

- It is also called the length of a circle of latitude at a given angle.
- It is given by the formula:

$$c = 2\pi r$$

$$c = 2\pi R \cos \theta$$

Time

1 Earth's Rotation

- The Earth rotates once on its axis every 24 hours, which is the basis for our modern concept of time.

2 Time Zones

- The Earth is divided into 24 time zones, each representing a one-hour difference from Coordinated Universal Time (UTC).

3 Longitude and Time

- As the Earth rotates from west to east, different longitudes experience different local times.
- The Earth is divided into 24 time zones, each representing a one-hour difference from Coordinated Universal Time (UTC).
- These time zones are determined by the longitude, with each time zone representing a 15-degree range of longitude.

Speed

1 Earth's Circumference
   - The Earth's circumference is approximately 40 075 kilometers.
2 Speed of the Earth
   - The Earth rotates at a speed of approximately 1 674 kilometers per hour (km/h) at the equator.
3 Speed and Latitude
   - As you move north or south of the equator, the speed of the Earth decreases due to the Earth's slightly ellipsoidal shape.

Distance

1 Earth's Radius
   - The Earth's radius is approximately 6 371 kilometers.
2 Distance and Latitude
   - As you move north or south of the equator, the distance between two points on the Earth's surface changes due to the Earth's slightly ellipsoidal shape.
3 Distance and Longitude
   - As you move east or west of the prime meridian, the distance between two points on the Earth's surface changes due to the Earth's rotation.

Relationships between Time, Speed, and Distance

1 $\textbf{Time} = \frac{\textbf{Distance}}{\textbf{Speed}}$

- This formula can be used to calculate the time taken to travel a certain distance at a constant speed.

2 $\textbf{Speed} = \frac{\textbf{Distance}}{\textbf{Time}}$

- This formula can be used to calculate the speed at which an object travels a certain distance in a given time.

3 $\textbf{Distance} = \textbf{Speed} \times \textbf{Time}$

- This formula can be used to calculate the distance traveled by an object at a constant speed over a given time period.

2018.P1.Oct

A plane leaves a town P(0°, 30°W) and flies due East to a town Q(0°, 50°E) in 3 hours. The distance between P and Q is 4 800nm.

(a) Calculate its average speed.

(b) If the plane leaves P at 06 00 hours, what time will it arrive at Q?

Solution

(b) $S = \frac{D}{T}$

$= \frac{4\,800}{3}$

$= 1\,600\ knots$

(a) Longitude difference $= 30°W + 50°E$

$= 80°$

Time difference $= \frac{80}{15}$

$= 5$ hours 20 minutes

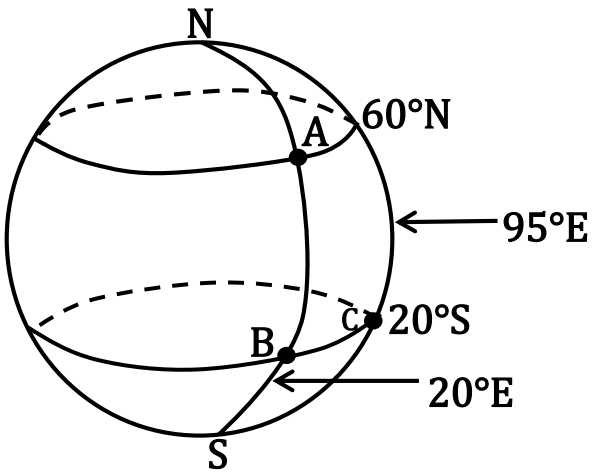
$\therefore$ 06 00

$+$ 05 20

11 20 hours

2019.P1.Nov

The diagram below shows the positions of towns A, B and C on the earth's surface.



(a) If it is 08 20 at A, what time is it at C?

(b) A plane flies from A to B at a speed of 400 knots. How long does the journey take if AB = 4 800nm?

Solution

(a) Longitude difference $= 95°E - 20°E$

$= 75°$

Time difference $= \frac{75}{15}$

$= 5$ hours

$\therefore$ 08 20

$+$ 05 00

13 20 hours

(b) $S = \frac{D}{T}$

$400 = \frac{4800}{T}$

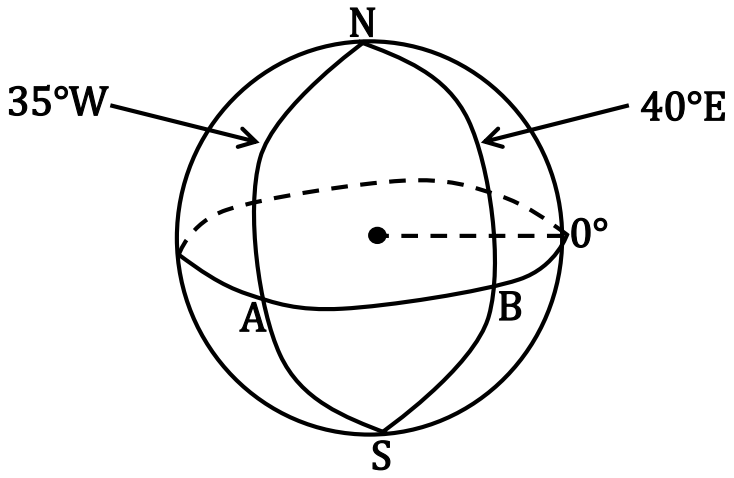
$400T = 4\,800$

$T = \frac{4800}{400}$

$T = 12\ hours$

2020.P1.Nov

The diagram below shows two towns A and B on the equator. A is on longitude 35°W and B is on longitude 40°E.



(a) Find the time at A when it is 18 00 hours at B.

(b) A plane flying from A to B takes 5 hours. Find its speed.

Solution

(a) Longitude difference $= 35°W + 40°E$

$= 75°$

Time difference $= \frac{75}{15}$

$= 5$ hours

$\therefore$ 18 00

$-$05 00

1300 hours

(b) $AB = \theta \times 60$

$AB = 75 \times 60$

$AB = 4\,500nm$

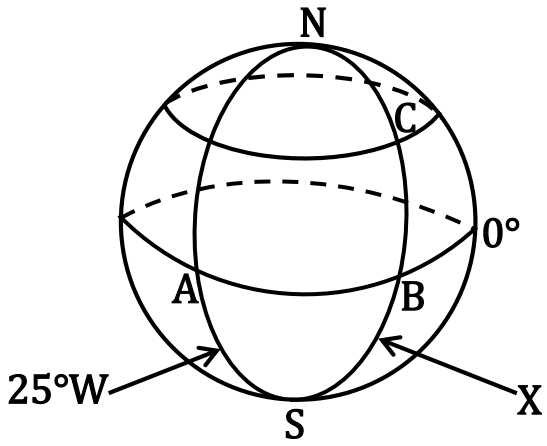
$S = \frac{D}{T}$

$= \frac{4\,500}{5}$

$= 900\ knots$

2021.P1.Nov

The positions of three towns A, B and C on the earth's surface are shown in the diagram below.



(a) A plane flying a distance of 4 500 nautical miles directly from town A to B takes 5 hours. Find the speed of the plane in knots.

(b) If town C is 5 hours ahead of town A, find the longitude marked X.

Solution

(b) $S = \frac{D}{T}$

$= \frac{4\,500}{5}$

$= 900$ knots

(a) Longitude difference $= 5 \times 15$

$= 75°$

$\therefore 25 + X = 75$

$X = 75 - 25$

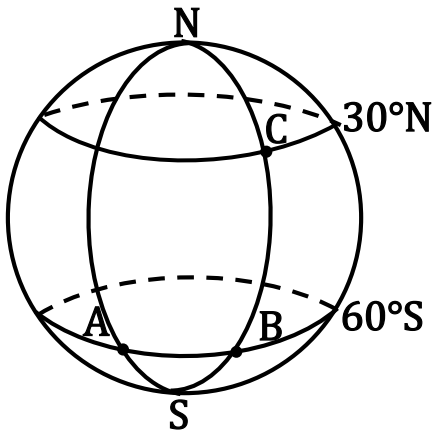
$X = 50°E$

2022.P1.Nov

In the following diagram, points A and B lie on latitude 60°S while point C is on latitude 30°N.

(a) The local time at A is 10 00 hours when it is 13 00 hours at B. find the difference in longitudes between A and B.

(b) A plane flew from B to C at a speed of 600 knots. How long did the plane take?



Solution

(a) Longitude difference $= 3 \times 15$

$= 45°$

Time difference $= 13 - 10$

$= 3$ hours

$$\begin{array}{r} 13\ 00 \\ -10\ 00 \\ \hline 0300 \text{ hours} \end{array}$$

(b) $BC = \theta \times 60$

$BC = 90 \times 60$

$AB = 5\ 400\text{nm}$

$S = \frac{D}{T}$

$600 = \frac{5\ 400}{T}$

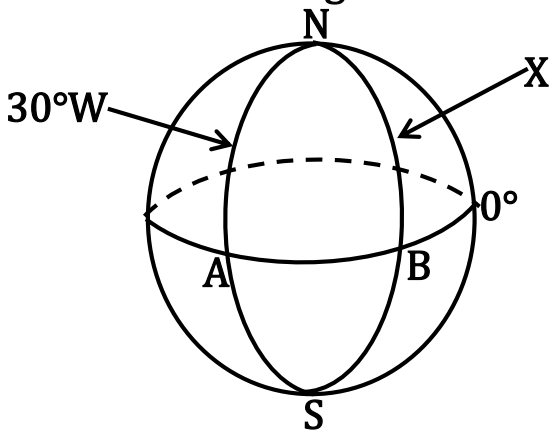
$600T = 5\ 400$

$T = \frac{5\ 400}{600}$

$T = 9$ hours

2023.P1.Nov

The diagram below shows two towns A and B on the equator. A is on longitude 30°W and B is on longitude X.



(a) The time at B is 17 05 hours when it is 14 05 hours at A. Find the longitude X.

(b) A ship sailed due east from A to B at an average speed of 450 knots. Find the time it took the ship to travel from A to B.

Solution

(a) Longitude difference $= 3 \times 15$

$= 45°$

Time difference $= 13 - 10$

$= 3$ hours

$$\begin{array}{r} 17\ 05 \\ -14\ 05 \\ \hline 03\ \ 00 \text{ hours} \end{array}$$

$X + 30 = 45$

$X = 45 - 30$

$X = 10°E$

(b) $AB = \theta \times 60$

$AB = 45 \times 60$

$AB = 2\,700 \text{nm}$

$S = \frac{D}{T}$

$450 = \frac{2\,700}{T}$

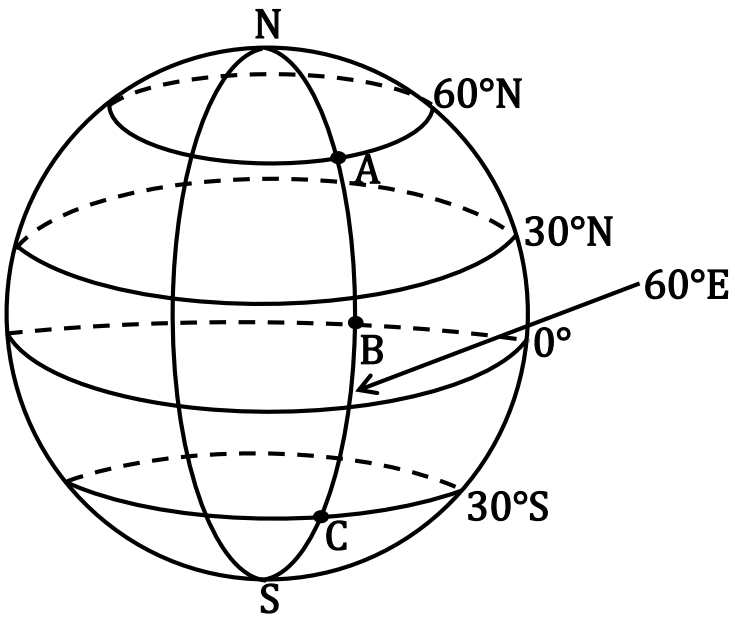
$450T = 2\,700$

$T = \frac{2\,700}{4500}$

$T = 6$ hours

2024.P1.GCE

The diagram below shows positions of towns A, B and C.



(a) What is the time at A if it is 05 00 hours at B?

(b) A plane flew from town A to town C in 5 hours. What was the speed of the plane in knots?

Solution

(a) 05 00 hours

(b) $D = \theta \times 60$

$= 90 \times 60$

$= 5\,400\text{nm}$

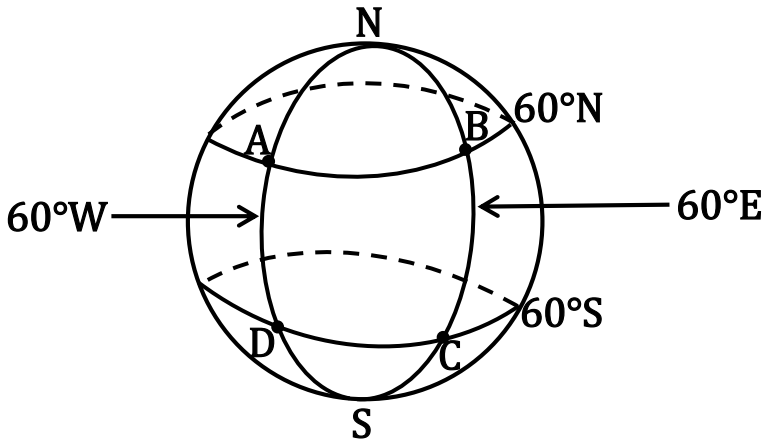
$S = \frac{D}{T}$

$= \frac{5\,400\text{nm}}{5\text{ hrs}}$

$= 1\,080$ knots

2018.P2.GCE

In the diagram below, A and B are points on latitude 60°N while C and D are points on latitude 60°S. [$\pi = 3.142$ and R = 3437nm].



(i) Calculate the distance BC along the longitude 60°E in nautical miles.[2]

(ii) A ship sails from C to D in 12 hours. Find its speed in knots. [4]

Solution

(i) $BC = \theta \times 60$

$= 120 \times 60$

$= 7\ 200nm$

(ii) $CD = \alpha \times 60\ \cos\theta$

$= 120 \times 60\ \cos 60$

$= 3\ 600nm$

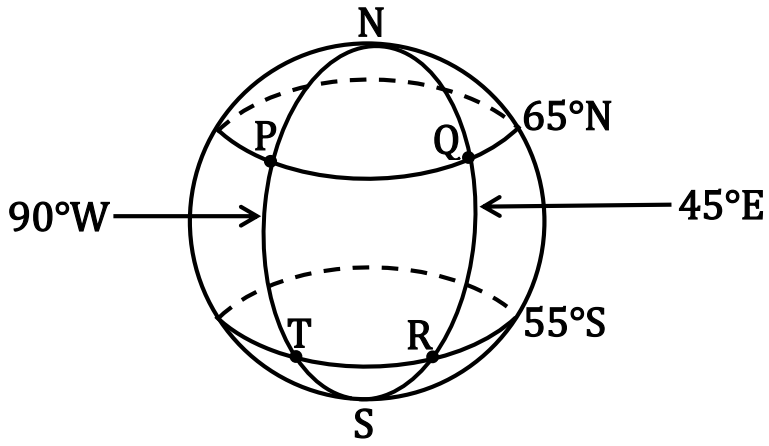
$S = \frac{D}{T}$

$= \frac{3\ 600}{12}$

$= 300$ knots

2019.P2.Nov

The points P, Q , R and T are on the surface of the earth as shown in the diagram below. [Take π as 3.142 and R = 3437nm]



(i) Find the difference in longitude between the points T and R. [2]

(ii) Find, in nautical miles, the distance between

(a) P and Q along the latitude 65°N, [2]

(b) P and T along the longitude 90°W. [2]

Solution

(i) $90°W + 45°E$

$135°$

(ii) (a) PQ $= \alpha \times 60 \cos\theta$

$= 135 \times 60 \cos 65$

$= 3\,423.207\,920\,099\,67$

$= 3\,420nm$

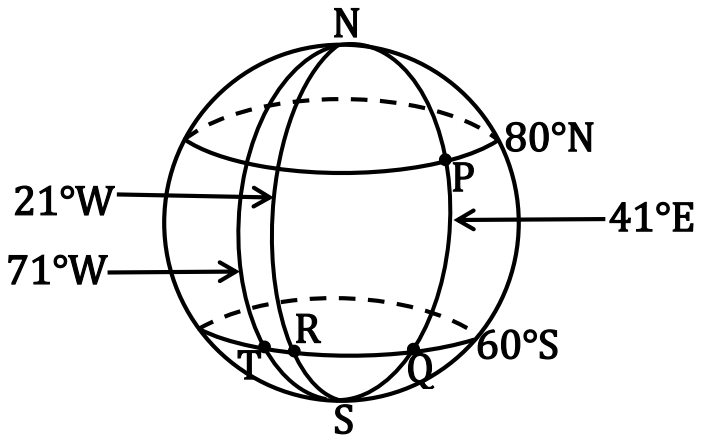
(b) PT $= \theta \times 60$

$= 120 \times 60$

$= 7\,200$nm

2020.P2.Nov

P, Q, R and T are points on the surface of the earth as shown in the diagram below. (Take π as 3.142 and R = 3437nm).



(i) Find the difference in longitude between points Q and R. [2]

(ii) Calculate the distance, in nautical miles, of

(a) PQ, [2]

(b) RT. [2]

Solution

(i) $21°W + 41°E$

$= 62°$

(ii) (a) PQ $= \theta \times 60$

$= 140 \times 60$

$= 8\,400nm$

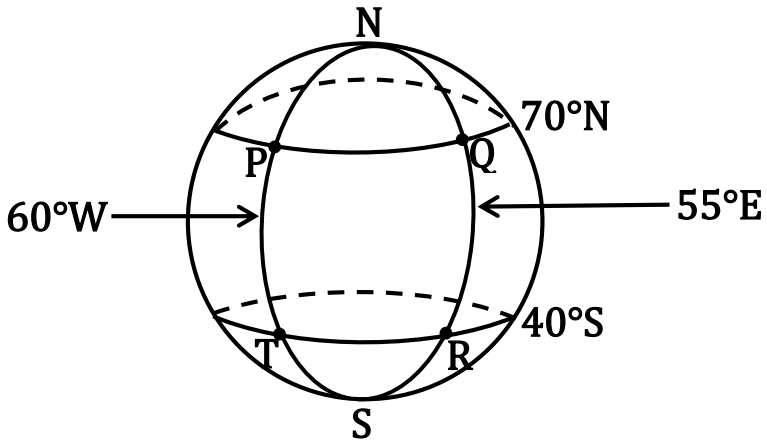
(b) RT $= \alpha \times 60 \cos\theta$

$= 50 \times 60 \cos 60$

$= 1\,500nm$

2021.P2.Nov

The points P, Q , R and T are on the surface of the earth as shown in the diagram below. [Take π as 3.142 and R = 3437nm]



(i) Determine the difference in longitudes between points Q and R. [2]
(ii) Find the length of the circle latitude 40°S in nautical miles. [2]
(iii) Calculate the distance TR in nautical miles. [2]

Solution

(i) $55°E - 55°E$

$0°$

(ii) $l = 2\pi R \cos\theta$

$= 2 \times 3.142 \times 3437 \cos 40$

$= 16\,545.110\,62$

$= 16\,500nm$

(iii) $TR = \alpha \times 60 \cos\theta$

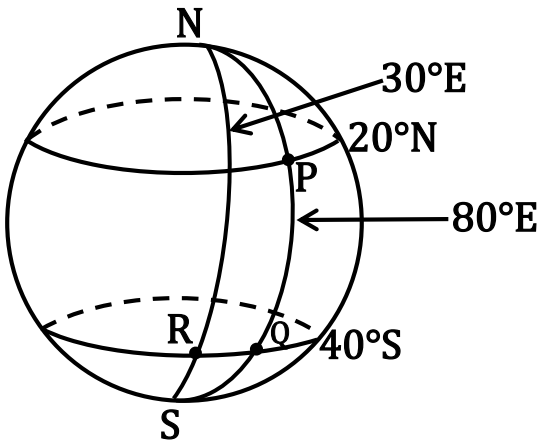
$= 115 \times 60 \cos 40$

$= 5\,285.706\,658$

$= 5\,290nm$

2022.P2.Nov

The diagram below shows the points P(20°N, 80°E), Q(40°S, 80°E) and R(40°S, 30°E) are on the surface of the earth. (Take π as 3.142 and R = 6370km).



(i) Find the difference in longitudes between points R and Q. [2]
(ii) Calculate the distance PQ in kilometres. [2]
(iii) Find the length of the circle latitude 40°S in kilometres. [2]

Solution

(i) $80°E - 30°E$

$50°$

(ii) $PQ = \frac{\theta}{360} \times 2\pi R$

$= \frac{60}{360} \times 2 \times 3.242 \times 6370$

$= 6\,671.513333$

$= 6\,670km$

(iii) $l = 2\pi R \cos\theta$

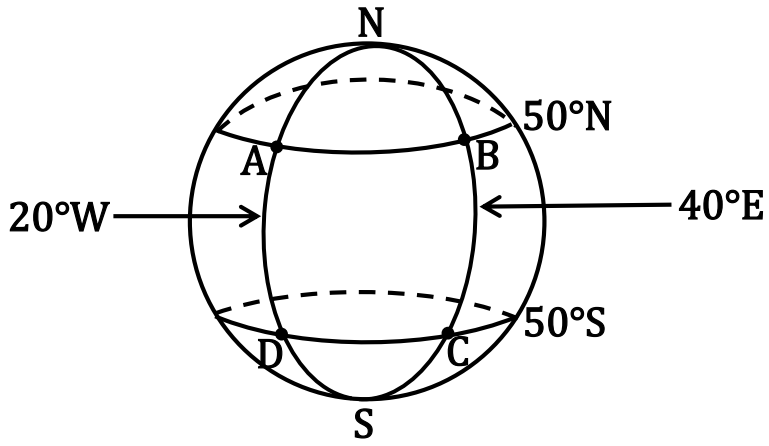
$= 2 \times 3.142 \times 6370 \cos 40$

$= 30\,664.0543$

$= 30\,664km$

2023.P2.Nov

The points A, B , C and D are on the surface of the earth as shown in the diagram below. (Take π as 3.142 and R = 6 370km)



(i) Find the difference in latitude between points A and D. [2]

(ii) Calculate in kilometres, the distance between

(a) A and B along latitude 50°N, [2]

(b) B and C along longitude 40°E. [2]

Solution

(i) $50°N + 50°S$

$100°$

(ii) (a) $AB = \frac{\alpha}{360} \times 2\pi R \cos\theta$

$= \frac{60}{360} \times 2 \times 3.142 \times 6370 \cos 50$

$= 4\,288.366\,109$

$= 4\,290km$

(b) $BC = \frac{\theta}{360} \times 2\pi R$

$= \frac{100}{360} \times 2 \times 3.142 \times 6370$

$= 11\,119.188\,89$

$= 11\,100km$

2024.P2.GCE

The points P(80°N, 50°W), Q(80°N, 80°E), R(50°N, 80°E) and T(50°N, 50°W) are on the surface of the earth. [Take π as 3.142 and R = 6 370km].

(i) Draw a sketch to represent the above information. [2]

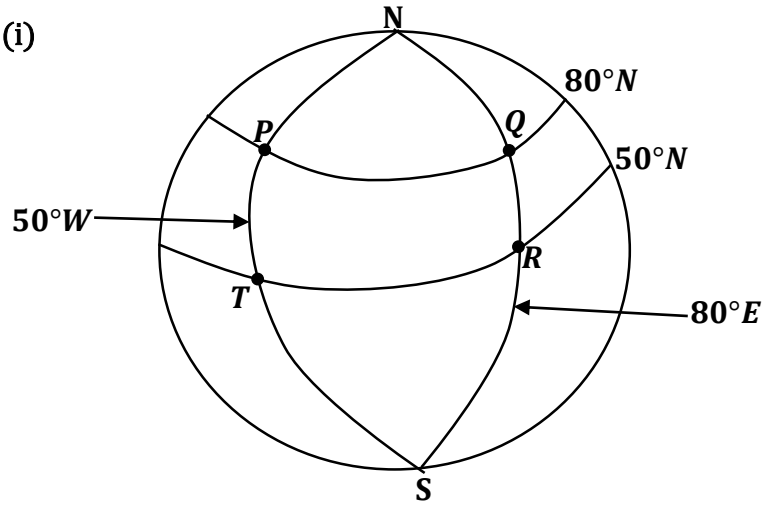
(ii) Calculate the distance

(a) QR along longitude 80°E in kilometres, [2]

(b) RT along latitude 50°N in kilometres. [2]

Solution

(i)



(ii) (a) $QR = \frac{\theta}{360} \times 2\pi R$

$= \frac{30}{360} \times 2 \times 3.142 \times 6370$

$= 3\,335.7566666$

$= 3\,340\text{km}$

(b) $RT = \frac{\alpha}{360} \times 2\pi R \cos\theta$

$= \frac{130}{360} \times 2 \times 3.142 \times 6370 \cos 50$

$= 9\,291.459\,901\,804\,62$

$= 9\,290\text{km}$

TRANSFORMATION

Objectives

By the end of this topic, learners should

1 Define Geometric Transformation.
2 Identify and describe the different types of geometric transformations, including translation, rotation, reflection, enlargement, stretch and shear.
3 Apply geometric transformations to solve problems involving congruent and similar figures.
4 Use geometric transformations to prove theorems and properties of geometric shapes.
5 Analyze and interpret geometric transformations in real-world contexts.

Definition

- Transformation is the change in position, shape or size of an object.
- Basically, there are six (6) types of transformations namely:
  (i) Translation
  (ii) Reflection
  (iii) Rotation
  (iv) Enlargement
  (v) Stretch
  (vi) Shear

TRANSLATION

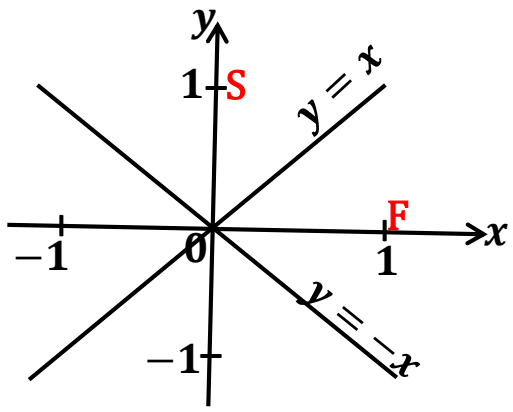
- Translation is a movement in a straight line without turning.
- It can be expressed as a column vector, e.g. $\mathbf{T} = \begin{pmatrix} x \\ y \end{pmatrix}$.
- Translation can be fully described by stating its column vector.
- If the object and the translation are given, we can find the image using the formula $I = T + O$.
- Under translation, the shape and size of an object does not change but the position may change.

REFLECTION

- Reflection can fully be described by stating the mirror line.
- The object distance from the mirror line is equal to the image distance behind the mirror line.
- If the object is in the mirror, the image is also in the mirror at the same point.
- The line joining the object and the image is perpendicular to the mirror line.
- Under reflection, the object and the image are of the same size and the same shape.
- Under reflection, the object and the image face each other.
- If the object and the matrix (M) are given, we can find the image using the formula $I = M(O)$.

Transformation Matrix under Reflection

- Consider the square unit below



(a) Reflection in the x-axis

$$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

(b) Reflection in the y-axis

$$\mathbf{M} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

(c) Reflection in the origin

$$\mathbf{M} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

(d) Reflection in the line $y = x$

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(e) Reflection in the line $y = -x$

$$\mathbf{M} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

ROTATION

- It is a circular transformation about a fixed point through a given angle in a given direction.
- Under rotation, the shape and size of an object does not change but the position and direction may change.
- If the object and the matrix (M) are given, we can find the image using the formula I = M(O).
- Rotation can fully be described by stating the following:
  (i) Direction of rotation
  (ii) Angle of rotation
  (iii) Centre of rotation

How to Find the Centre of Rotation?

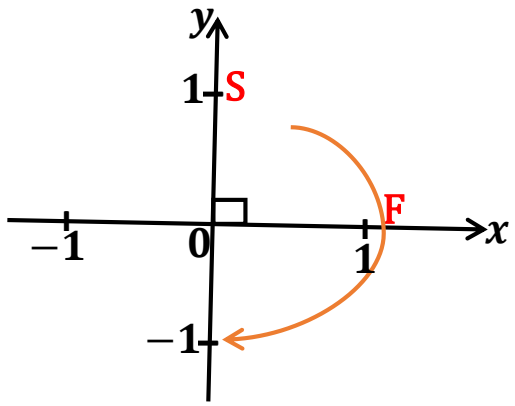
- Join any two pairs of corresponding points.
- Construct perpendicular bisectors of the two lines formed, and extend the bisectors until they meet.
- The point of intersection is the centre of rotation.

How to Find the angle of Rotation?

- Given the centre, the angle between the two bisectors from two corresponding points is the angle of rotation.

Transformation Matrix Associated with Rotation

- Transformation matrix for rotation can only be derived if the centre of rotation is the origin.
- Consider the unity square below.



(a) 90° clockwise about (0, 0)

$$M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

(b) 90° anti-clockwise about (0, 0)

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(c) 180° clockwise about (0, 0)

$$M = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

(d) 180° anti-clockwise about (0, 0)

$$M = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

(e) 270° clockwise about (0, 0)

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(f) 270° anti-clockwise about (0, 0)

$$M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

(g) 360° clockwise about (0, 0)

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(h) 360° anti-clockwise about (0, 0)

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Finding the image vertices when the centre is not the origin (0, 0) under Rotation

- If the centre is not the origin, we cannot use the matrix method to find the vertices of the image.
- Follow the following procedures to find the coordinates of the image:
  (i) Translate the center to the origin (0, 0) by subtracting the center coordinates from the vertices.
  (ii) Rotate the vertices using the rotation matrix.
  (iii) Translate the rotated vertices back to their original position by adding the center coordinates.

Example

Triangle ABC of vertices A(3, 2), B(6, 2) and C(6, 0) is mapped onto triangle $A_1B_1C_1$ under an anti-clockwise rotation of 90° about (1, 1). Find the coordinates of triangle $A_1B_1C_1$.

Solution

- Plot the coordinates of the centre of rotation as P(1, 1)
- Join point A to the centre P so that we have AP.
- Measure a $90°$ angle at the centre by using a protractor so that we have PA_1.
- Use a compass to draw an arc and let it cut PA_1 at A_1.
- Do the same to B and C.

∴ The coordinates of triangle $A_1B_1C_1$ are:

$A_1(0, 3)$

$B_1(0, 6)$

$C_1(2, 6)$

ENLARGEMENT

- Enlargement is a kind of transformation where an object changes in size in all the directions, e.g putting pressure in a balloon.
- An enlargement can fully be described by stating the following:
  (i) Centre of enlargement.
  (ii) Scale factor.

Transformation Matrix Associated with Enlargement

- A matrix can be derived if a centre of enlargement is the origin given that the scale factor is known.
- $\mathbf{M} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$

Finding the centre of Enlargement

- Join any two pairs of corresponding points.
- The point of intersection is the centre of enlargement.

Enlargement by Construction

- To construct the image of an object under an enlargement, the following procedure applies:
  1. Join the object points to the centre of enlargement with straight lines and prolong them.
  2. Measure the distance between the object point and the centre of enlargement (object distance).
  3. Calculate the corresponding image distance by multiplying the object distance by the given scale factor.
  4. Mark the images and join them to obtain the required image.

Stretch

- A stretch is the enlargement of a plane shape in one direction only.
- A stretch is fully described when we are given:
  (a) its direction,
  (b) the scale factor (stretch factor),
  (c) the invariant line.
- Stretch (scale) factor $= \frac{distance\ from\ the\ invariant\ line\ to\ an\ image\ point}{distance\ from\ the\ invariant\ line\ to\ the\ corresponding\ point\ on\ the\ object}$.
- The matrix of stretch parallel to the x-axis with scale factor k is given by $\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$ where the y-axis is the invariant line.
- The matrix of stretch parallel to the y-axis with scale factor k is given by $\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ where the x-axis is the invariant line.

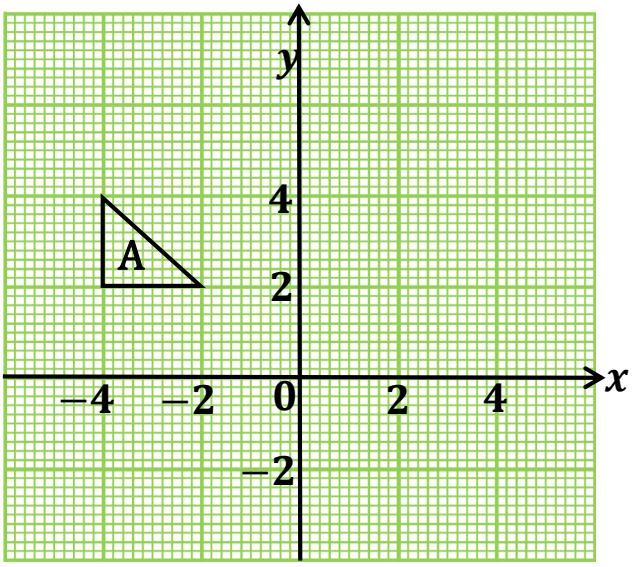
Shear

- A shear is a transformation which has the following properties:
  (a) There is a fixed line in the plane.
  (b) A point which is d units away from the fixed line is displaced parallel to the fixed line, through a distance which is proportional to d.
- A shear is fully described if we are given:
  (a) the invariant line,
  (b) the image of a point not on the invariant line.
- The matrix for a shear parallel to the x-axis is given by $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$, where k is the shear factor obtained as $\frac{d}{y}$, d being the distance that a point is displaced, y units away from the x-axis.

- If the shear is parallel to the y-axis, the transformation matrix is given by $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$, where $k = \frac{d}{x}$, d being the distance that a point is displaced, x units away from the y-axis.
- The sign of d must be included in the evaluation of k.
- The invariant line of each of these shears must pass through the origin.

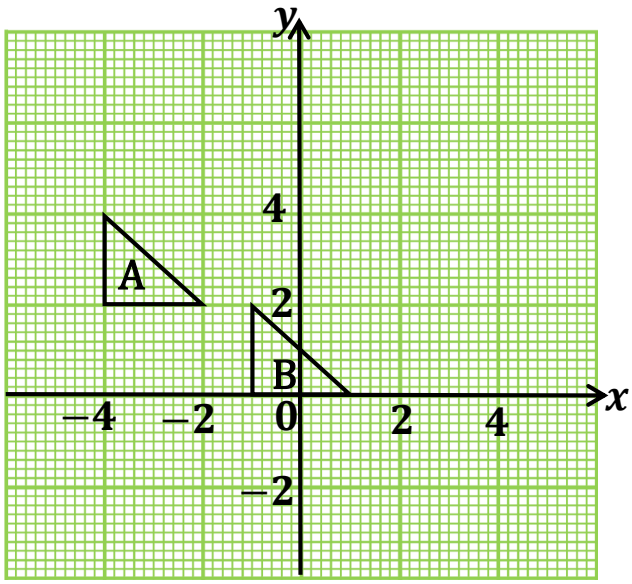
2018.P1.Oct

The diagram below shows ΔA. If ΔA is mapped onto ΔB by a translation $T = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$, draw ΔB on the diagram in the answer space below.



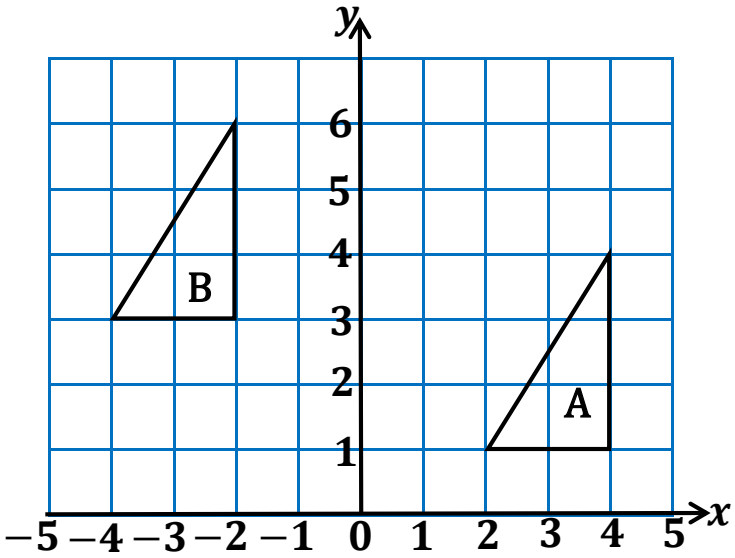
Solution

$$I = T + O$$
$$= \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} -2 & -4 & -4 \\ 2 & 2 & 4 \end{pmatrix}$$
$$= \begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & 2 \end{pmatrix}$$
$$\therefore (1, 0), (-1, 0) \; and \; (-1, 2)$$



2019.P1.Nov

The diagram below shows triangle A and B.



Describe fully the single transformation which maps triangle A onto triangle B.

Solution

$$I = T + O$$

$$T = I - O$$

$$T = B - A$$

$$T = \begin{pmatrix} -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$T = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$$

$\therefore$ It is a translation with a column vector $T = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$

2020.P1.Nov

A point A(2, 4) is mapped onto $A'(-3, 5)$ by a translation T. Find the translation T.

Solution

$$I = T + O$$

$$T = I - O$$

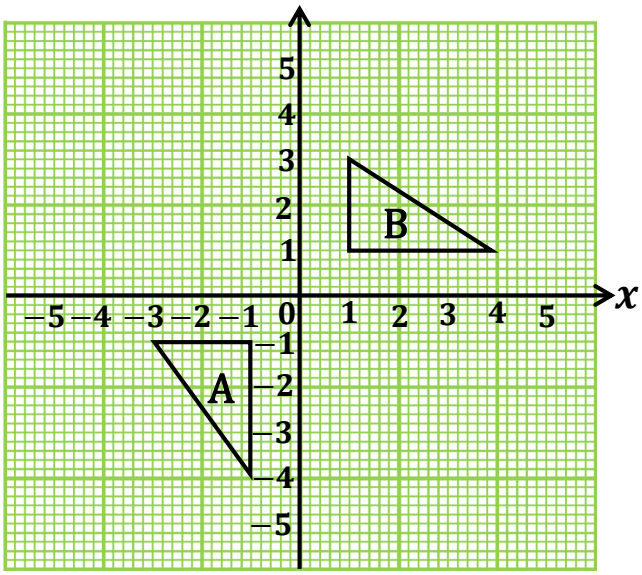
$$T = A' - A$$

$$T = \begin{pmatrix} -2 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$T = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

2021.P1.Nov

The diagram below shows triangle A which is mapped onto triangle B by a single transformation P.



Describe fully the transformation P.

Solution

$$\mathbf{I} = \mathbf{M(O)}$$

$$\mathbf{I} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -1 & -1 & -3 \\ -1 & -4 & -1 \end{pmatrix}$$

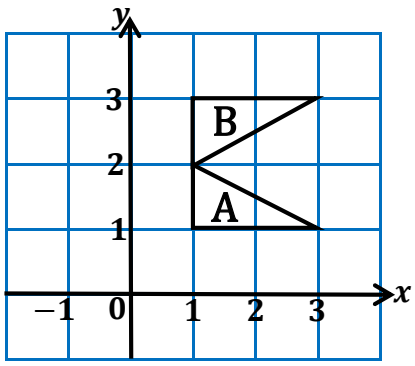
$$\mathbf{I} = \begin{pmatrix} 1 & 4 & 1 \\ 1 & 1 & 3 \end{pmatrix}$$

$\Delta B = (1, 1), (4, 1)$ and $(1, 3)$

$\therefore$ It is a reflection in the line $y = -x$.

2022.P1.Nov

The diagram below shows two triangles A and B.



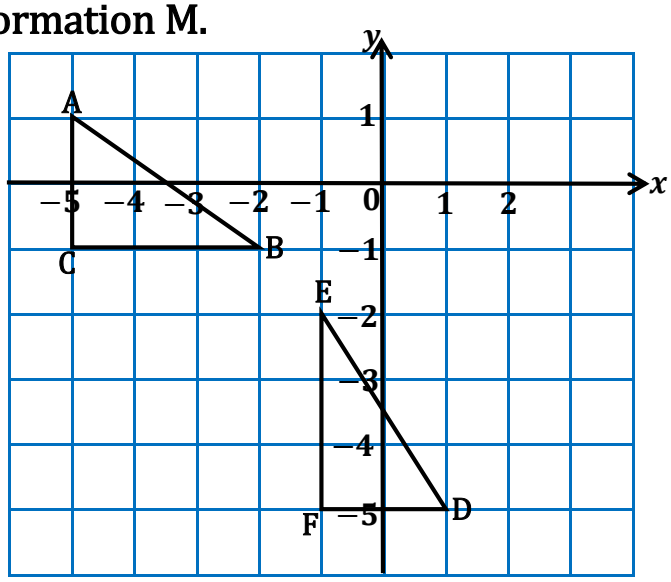
Describe fully the single transformation which maps triangle A onto triangle B.

Solution

It is a reflection in the line $y = 2$.

2023.P1.Nov

In the diagram below, triangle ABC is mapped onto triangle DEF by a single transformation M.



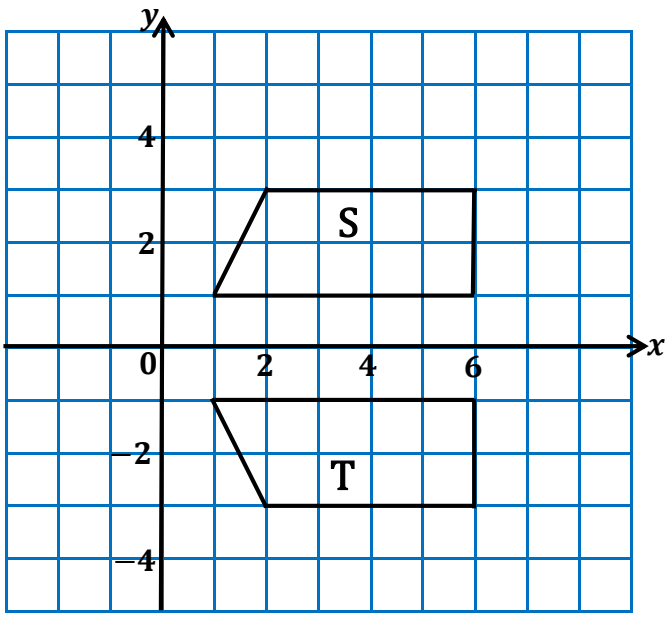
Describe fully the transformation M.

Solution

It is a reflection in the line $y = x$

2024.P1.GCE

The diagram below shows trapezium S and trapezium T.



Describe fully a single transformation which maps trapezium S onto trapezium T.

Solution

$$\mathbf{I} = \mathbf{M}(\mathbf{O})$$

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 6 & 6 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

$$\mathbf{I} = \begin{pmatrix} 1 & 2 & 6 & 6 \\ -1 & -3 & -3 & -1 \end{pmatrix}$$

(1, -1), (2, -1), (6, -3) and (6, -1)

$\therefore$It is a reflection in the x-axis or in the line $y = 0$.

2018.P2.GCE

Answer the whole of this question on a sheet of graph paper.

Using a scale of 1cm to represent 1 unit on both axes, draw x and y axes for $-8 \leq x \leq 12$ and $-6 \leq y \leq 14$.

(a) Draw and label triangle X with vertices (2, 4), (4, 4) and (4, 1). [1]

(b) Triangle X is mapped onto triangle U with vertices (6, 12), (12, 12) and (12, 3) by a single transformation.

(i) Draw and label triangle U. [1]

(ii) Describe fully this transformation. [3]

(c) A 90° clockwise rotation about the origin maps triangle X onto triangle W. Draw and label triangle W. [2]

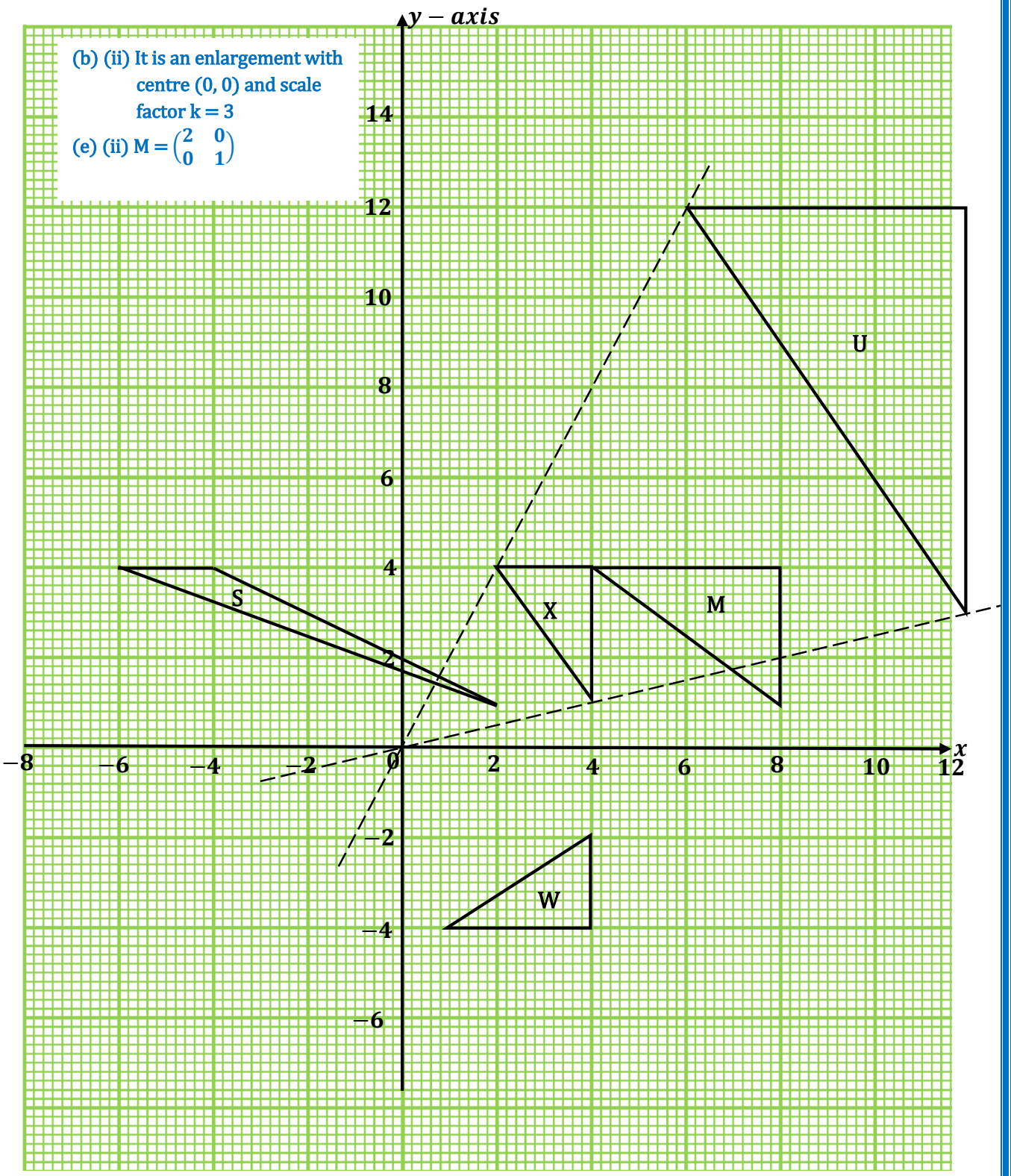
(d) A shear with x-axis as the invariant line and shear factor – 2 maps Triangle X onto triangle S. Draw and label triangle S. [2]

(e) Triangle X is mapped onto triangle M with vertices (4, 4), (8, 4) and (8, 1).

(i) Draw and label triangle M. [1]

(ii) Find the matrix which represents this transformation. [2]

Solution


y − axis
(b) (ii) It is an enlargement with centre (0, 0) and scale factor k = 3
(e) (ii) M = (2 0 / 0 1)
14
12
10
8
6
4
2
−2
−4
−6
−8
−6
−4
−2
0
2
4
6
8
10
12
x
S
X
M
U
W

2019.P2.Nov

Answer the whole of this question on a sheet of graph paper.

The vertices of triangle ABC are A(1, 1), B(1, 3) and C(3, 3). The vertices triangle $A_1B_1C_1$ are $A_1(-1, 1)$, $B_1(-3, 1)$ and $C_1(-3, 3)$.

(a) Using a scale of 1cm to represent 1 unit on both axes, draw x and y axes for $-6 \leq x \leq 6$ and $-6 \leq y \leq 6$. Draw and label triangles ABC and $A_1B_1C_1$. [2]

(b) Describe fully a single transformation that maps triangle ABC of triangle $A_1B_1C_1$. [2]

(c) An enlargement maps triangle ABC onto triangle $A_2B_2C_2$ with vertices $A_2(-2, -2)$, $B_2(-2, -6)$ and $C_2(-6, -6)$.

(i) Draw and label triangle $A_2B_2C_2$. [1]

(ii) Find the scale factor. [1]

(d) The transformation represented by the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ maps triangle ABC onto triangle $A_3B_3C_3$.

(i) Find the coordinates of the vertices A_3, B_3 and C_3. [2]

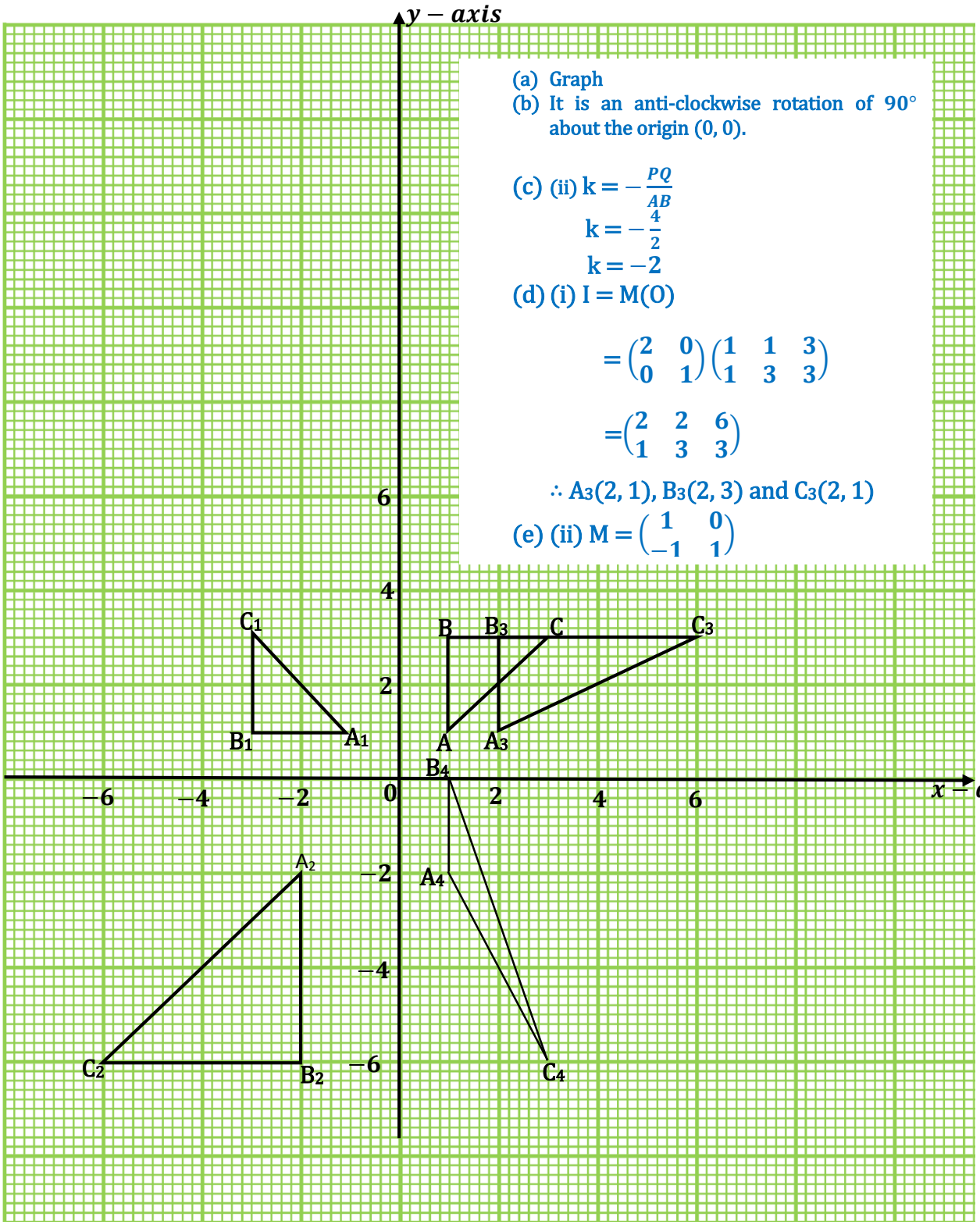
(ii) Draw and label triangle $A_3B_3C_3$. [1]

(e) Triangle ABC is mapped onto triangle $A_4B_4C_4$ with vertices $A_4(1, -2)$, $B_4(1, 0)$ and $C_4(3, -6)$.

(i) Draw and label triangle $A_4B_4C_4$. [1]

(ii) Find the matrix representing this transformation. [2]

Solution



(a) Graph

(b) It is an anti-clockwise rotation of 90° about the origin (0, 0).

(c) (ii) $k = -\frac{PQ}{AB}$

$k = -\frac{4}{2}$

$k = -2$

(d) (i) $I = M(O)$

$$= \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & 3 \end{pmatrix}$$

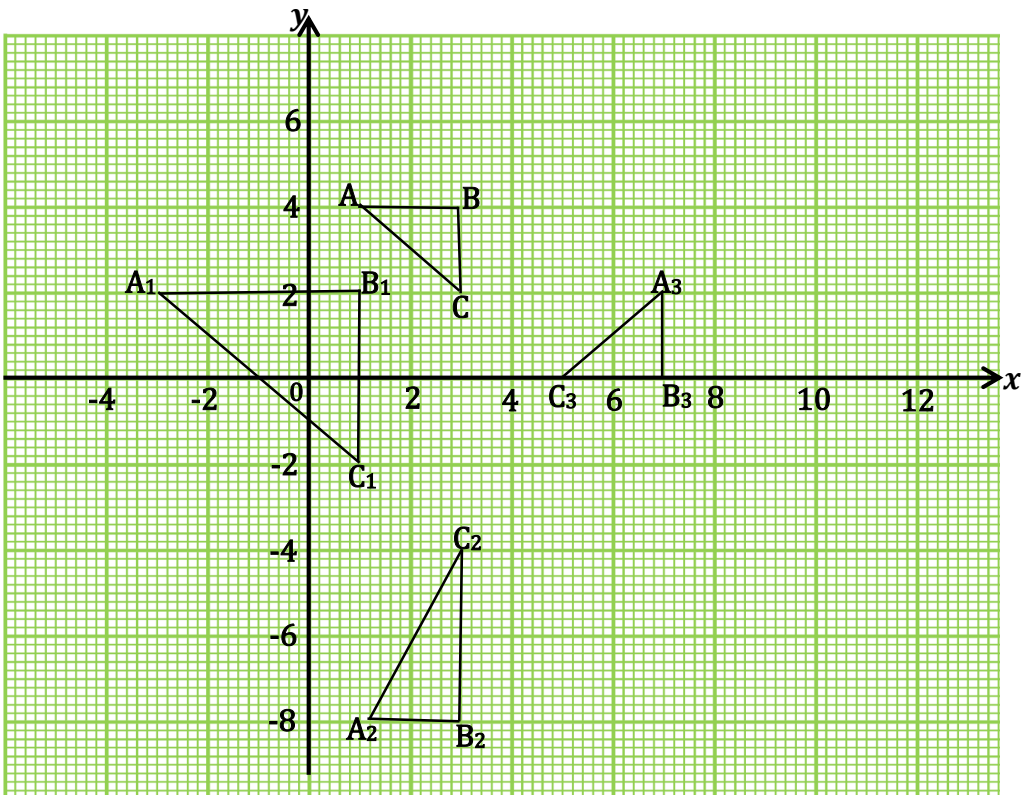
$$= \begin{pmatrix} 2 & 2 & 6 \\ 1 & 3 & 3 \end{pmatrix}$$

$\therefore A_3(2, 1), B_3(2, 3)$ and $C_3(2, 1)$

(e) (ii) $M = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$

2020.P2.Nov

Study the diagram below to answer the questions that follow.



(a) Triangle ABC is mapped onto triangle $A_1B_1C_1$ by an enlargement. Find its centre and scale factor. [2]

(b) Triangle ABC is mapped onto triangle $A_2B_2C_2$ by a single transformation. Find

(i) the matrix representing this transformation, [2]

(ii) the area scale factor of the transformation. [2]

(c) Triangle ABC is mapped onto triangle $A_3B_3C_3$ by a single transformation. Describe fully this transformation. [3]

(d) The matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$, maps triangle ABC is mapped onto triangle $A_4B_4C_4$ (not on the diagram). Find the coordinates of A_4, B_4 and C_4. [3]

Solution

(a) Centre of enlargement (5, 6)

Scale factor (k) $= \frac{Image\ length}{Object\ length}$

$= \frac{A_1B_1}{AB}$

$= \frac{4}{2}$

$= 2$

(b) (i) $I = M(O)$

$$\begin{pmatrix} 1 & 3 \\ -8 & -8 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 4 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 \\ -8 & -8 \end{pmatrix} = \begin{pmatrix} a+4b & 3a+4b \\ c+4d & 3c+4d \end{pmatrix}$$

$a + 4b = 1$

$3a + 4b = 3$

$-2a = -2$

$a = 1$

$1 + 4b = 1$

$4b = 1 - 1$

$4b = 0$

$b = 0$

$c + 4d = -8$

$3c + 4d = -8$

$-2c = 0$

$c = 0$

$0 + 4d = -8$

$4d = -8$

$d = -2$

∴The transformation matrix is $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$

(ii) Area scale factor $(K^2) = (-2)^2$

$= 4$

(c) Transformation: Rotation

Direction of rotation: Clockwise

Centre of rotation: (3, 0)

Angle of rotation: $90°$

∴ It is a clockwise rotation of $90°$ about (3, 0).

(d) $I = M(O)$

$$= \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 3 & 3 \\ 4 & 4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 13 & 15 & 9 \\ 4 & 4 & 2 \end{pmatrix}$$

∴$A_4(13, 4)$, $B_4(15, 4)$ and $C_4(9, 2)$

2021.P1.Nov

(a) A point C is mapped onto a point C'(-2, 3) by a single transformation whose matrix is $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$,

(i) Find the coordinates of the point C. [2]

(ii) Describe fully the single transformation represented by the matrix above. [3]

(b) Triangle PQR with vertices P(2, 2), Q(2, 0) and R(0, 1) is mapped onto triangle P'Q'R' whose vertices are P'(-4, 2), Q'(-4, 0) and R'(0, 1).

(i) Find the matrix which represents this transformation. [3]

(ii) Hence, describe fully the transformation. [3]

(c) F is a transformation represented by the matrix $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$. The images of A(2, 2) and B(-2, 4) under F are A'(5, 5) and B'(-5, 10) respectively. Find the value of k. [1]

Solution

(a) (i) I = M(O)

$$\begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} x - 2y \\ y \end{pmatrix}$$

$$y = 3$$

$$x - 2y = -2$$

$$x - 2(3) = -2$$

$$x - 6 = -2$$

$$x = -2 + 6$$

$$x = 4$$

$$\therefore C(4, 3)$$

(ii) It is a shear with y-axis invariant line and a shear factor K = -2.

(b) (i) I = M(O)

$$\begin{pmatrix} -4 & 4 \\ -2 & -4 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} -1 & -4 \\ -2 & -4 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 4 \\ -2 & -4 \end{pmatrix} = \begin{pmatrix} -a-2b & -4a-4b \\ -c-2d & -4c-4d \end{pmatrix}$$

$$-a-2b=3$$

$$-4a-4b=4$$

$$a+b=-1$$

$$-b=2$$

$$b=-2$$

$$a-2=-1$$

$$a=-1+2$$

$$a=1$$

$$-c-2d=-2$$

$$-4c-4d=-4$$

$$c+d=1$$

$$-d=-1$$

$$d=1$$

$$c+1=1$$

$$c=1-1$$

$$c=0$$

∴The transformation matrix is $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$

(ii) It is a stretch with the y-axis invariant line and a stretch factor K = -2.

(c) I = M(O)

$$\begin{pmatrix} 5 & -5 \\ 5 & 10 \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 2 & -2 \\ 2 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 5 & -5 \\ 5 & 10 \end{pmatrix} = \begin{pmatrix} 2k & -2k \\ 2k & 4k \end{pmatrix}$$

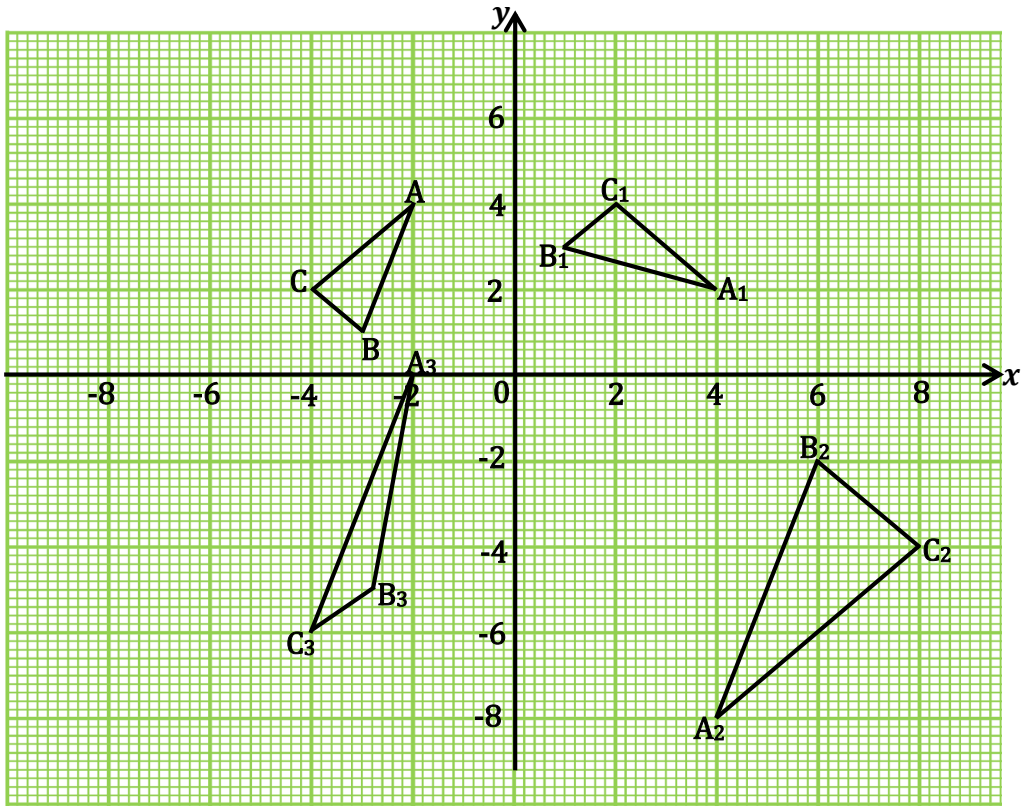
$$2k=5$$

$$k=\frac{5}{2}$$

$$k=2\frac{1}{2}$$

2022.P2.Nov

Study the following diagram and answer the questions that follow.



(a) Triangle ABC is mapped onto triangle $A_1B_1C_1$ by a single transformation. Describe fully this transformation. [3]

(b) An enlargement maps triangle ABC onto triangle $A_2B_2C_2$. Find the

(i) centre of enlargement, [1]

(ii) scale factor. [2]

(c) Triangle ABC is mapped onto triangle $A_3B_3C_3$ by a single transformation. Find the matrix representing this transformation. [3]

(d) The transformation with matrix $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$ maps triangle ABC onto triangle $A_4B_4C_4$ not shown on the diagram. Find the coordinates of A_4, B_4 and C_4. [3]

Solution

(a) It is a clockwise rotation of $90°$ about the origin $(0, 0)$

(b)

(i) Centre $(0, 0)$

(ii) Scale factor (k) $= \frac{Image\ length}{Object\ length}$

$$= -\frac{A_2C_2}{AC}$$
$$= -\frac{4}{2}$$
$$= -2$$

(c)

$I = M(O)$

$$\begin{pmatrix} -2 & -3 \\ 0 & -5 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} -2 & -3 \\ 4 & 1 \end{pmatrix}$$
$$\begin{pmatrix} -2 & -3 \\ 0 & -5 \end{pmatrix} = \begin{pmatrix} -2a + 4b & -3a + b \\ -2c + 4d & -3c + d \end{pmatrix}$$
$$-2a + 4b = -2$$
$$-3a + b = -3$$
$$b = 3a - 3$$
$$-2a + 4(3a - 3) = -2$$
$$-2a + 12a - 12 = -2$$
$$10a = 10$$
$$a = 1$$
$$b = 3(1) - 3$$
$$b = 0$$
$$-2c + 4d = 0$$
$$-3c + d = -5$$
$$d = 3c - 5$$
$$-2c + 4(3c - 5) = 0$$
$$-2c + 12c - 20 = 0$$
$$10c = 20$$
$$c = 2$$
$$d = 3(2) - 5$$
$$d = 1$$

$\therefore$The transformation matrix is $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$

(d)

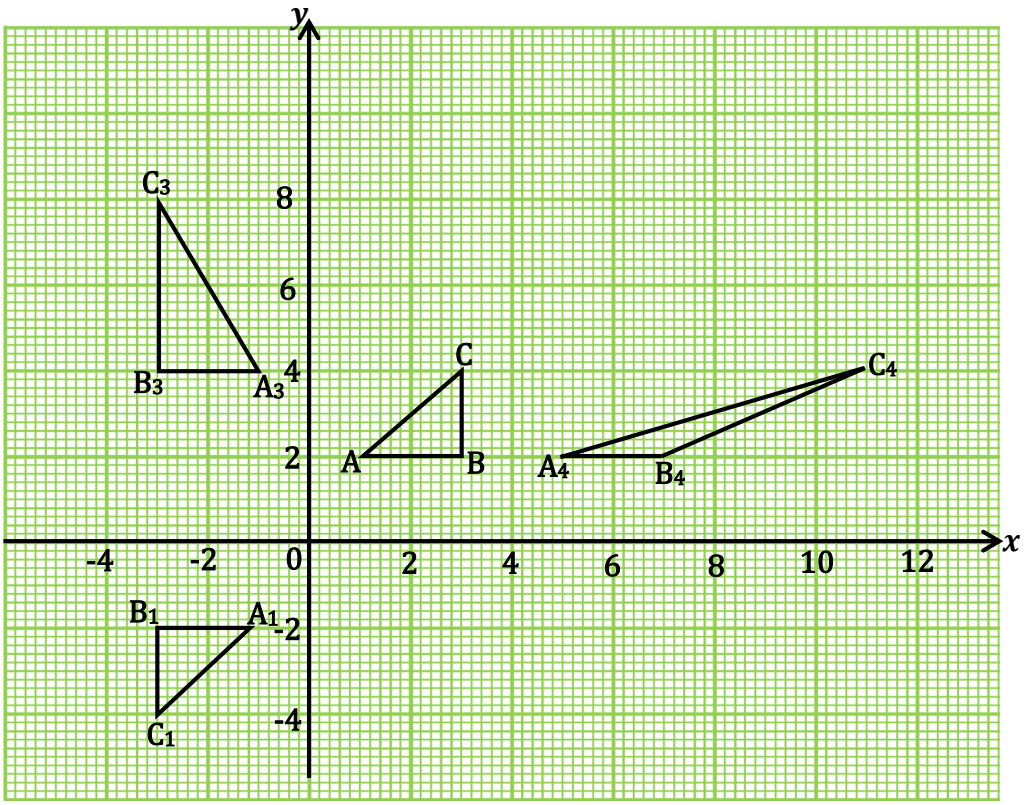
$I = M(O)$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} -2 & -3 & -4 \\ 4 & 1 & -2 \end{pmatrix}$$
$$= \begin{pmatrix} -2 & -3 & -4 \\ -8 & -2 & 4 \end{pmatrix}$$

$\therefore A_4(-2, -8), B_4(-3, -2)$ and $C_4(-4, 4)$

2023.P2.Nov

Study the diagram below and answer the questions that follow.



(a) Describe fully a single transformation which maps triangle ABC onto Triangle $A_1B_1C_1$. [3]

(b) Tringle ABC is mapped onto triangle $A_2B_2C_2$ (not on the diagram) by an enlargement with centre (1, 0) and scale factor 2. Find the coordinates of A_2, B_2 and C_2. [3]

(c) Given that triangle $A_1B_1C_1$ is mapped onto triangle $A_3B_3C_3$ by a stretch, find the

(i) invariant line, [1]

(ii) scale factor. [2]

(d) Triangle ABC is mapped onto triangle $A_4B_4C_4$ by a shear. Find the matrix of this transformation. [3]

Solution

(a) It is a rotation of $180°$ about the origin $(0, 0)$

(b)
- ✓ Let the centre be X(1, 0)
- ✓ Express each point from the centre to the object points as a column vector.
- ✓ Multiply each column vector by the scale factor k =2.
- ✓ Use corresponding points, e.g. A and A_2 to determine the coordinates of the image.

$$\overrightarrow{XA} = \overrightarrow{XO} + \overrightarrow{OA}$$
$$= \begin{pmatrix}-1\\0\end{pmatrix} + \begin{pmatrix}1\\2\end{pmatrix}$$
$$= \begin{pmatrix}0\\2\end{pmatrix}$$
$$\overrightarrow{AA_2} = 2\begin{pmatrix}0\\2\end{pmatrix}$$
$$= \begin{pmatrix}0\\4\end{pmatrix}$$
$$\overrightarrow{XB} = \overrightarrow{XO} + \overrightarrow{OB}$$
$$= \begin{pmatrix}-1\\0\end{pmatrix} + \begin{pmatrix}3\\2\end{pmatrix}$$
$$= \begin{pmatrix}2\\2\end{pmatrix}$$
$$\overrightarrow{BB_2} = 2\begin{pmatrix}2\\2\end{pmatrix}$$
$$= \begin{pmatrix}4\\4\end{pmatrix}$$

$$\overrightarrow{XC} = \overrightarrow{XO} + \overrightarrow{OC}$$
$$= \begin{pmatrix}-1\\0\end{pmatrix} + \begin{pmatrix}3\\4\end{pmatrix}$$
$$= \begin{pmatrix}2\\4\end{pmatrix}$$
$$\overrightarrow{CC_2} = 2\begin{pmatrix}2\\4\end{pmatrix}$$
$$= \begin{pmatrix}4\\8\end{pmatrix}$$

$$\overrightarrow{AA_2} = \overrightarrow{AO} + \overrightarrow{OA_2}$$
$$\begin{pmatrix}0\\4\end{pmatrix} = \begin{pmatrix}-1\\-2\end{pmatrix} + \overrightarrow{OA_2}$$
$$\begin{pmatrix}0\\4\end{pmatrix} + \begin{pmatrix}1\\2\end{pmatrix} = \overrightarrow{OA_2}$$
$$\overrightarrow{OA_2} = \begin{pmatrix}1\\6\end{pmatrix}$$
$$\therefore A_2(1, 6)$$

$$\overrightarrow{BB_2} = \overrightarrow{BO} + \overrightarrow{OB_2}$$
$$\begin{pmatrix}4\\4\end{pmatrix} = \begin{pmatrix}-3\\-2\end{pmatrix} + \overrightarrow{OB_2}$$
$$\begin{pmatrix}4\\4\end{pmatrix} + \begin{pmatrix}3\\2\end{pmatrix} = \overrightarrow{OA_2}$$
$$\overrightarrow{OB_2} = \begin{pmatrix}7\\6\end{pmatrix}$$
$$\therefore B_2(7, 6)$$

$$\overrightarrow{CC_2} = \overrightarrow{CO} + \overrightarrow{OC_2}$$
$$\begin{pmatrix}4\\8\end{pmatrix} = \begin{pmatrix}-3\\-4\end{pmatrix} + \overrightarrow{OC_2}$$
$$\begin{pmatrix}4\\8\end{pmatrix} + \begin{pmatrix}3\\4\end{pmatrix} = \overrightarrow{OC_2}$$
$$\overrightarrow{OC_2} = \begin{pmatrix}7\\12\end{pmatrix}$$
$$\therefore C_2(7, 12)$$

(c) (i) Since the stretch is parallel to the y-axis, then the invariant line is the x-axis or the line $y = 0$.

(ii) Stretch (scale) factor $= \frac{\textit{distance from the invariant line to an image point}}{\textit{distance from the invariant line to the corresponding point on the object}}$

$$= \frac{X - A_3}{A_1 - X}$$
$$= \frac{0-4}{2-0}$$
$$= -\frac{4}{2}$$
$$= -2$$

(d) $I = M(O)$

$$\begin{pmatrix}5 & 7\\2 & 2\end{pmatrix} = \begin{pmatrix}a & b\\c & d\end{pmatrix}\begin{pmatrix}1 & 3\\2 & 2\end{pmatrix}$$
$$\begin{pmatrix}5 & 7\\2 & 2\end{pmatrix} = \begin{pmatrix}a+2b & 3a+2b\\c+2d & 3c+2d\end{pmatrix}$$
$$a + 2b = 5$$
$$3a + 2b = 7$$
$$-2a = -2$$
$$a = 1$$
$$2b = 5 - 1$$
$$2b = 4$$
$$b = 2$$

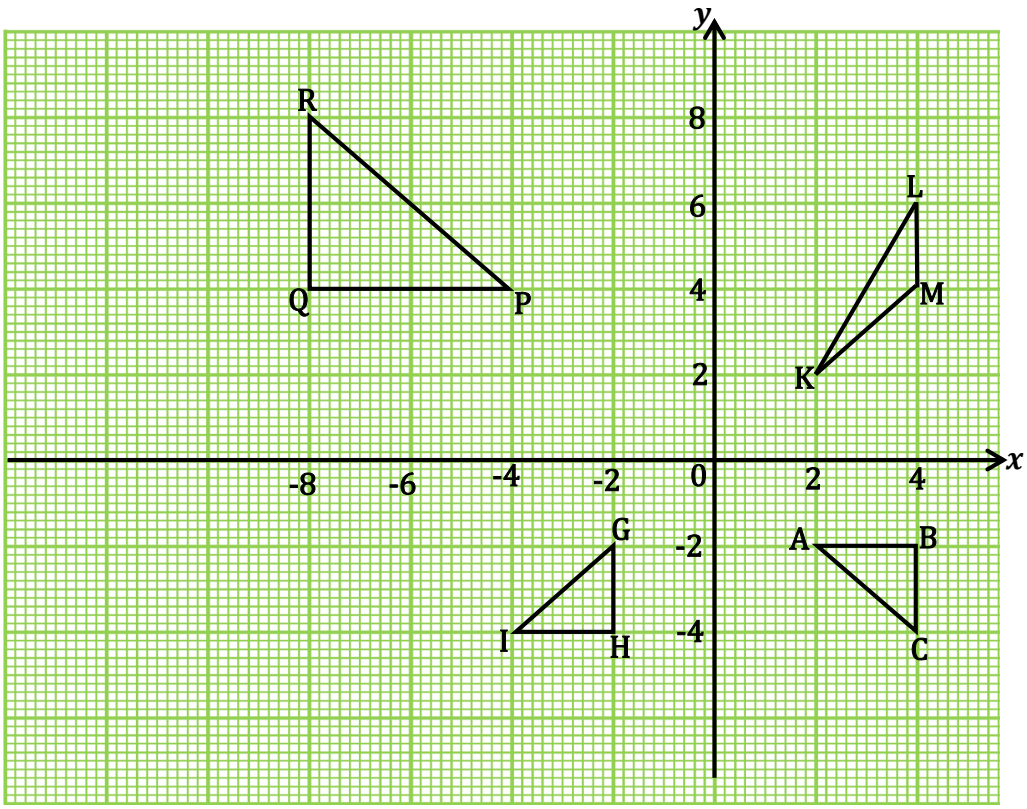
$$c + 2d = 2$$
$$3c + 2d = 2$$
$$-2c = 0$$
$$c = 0$$
$$0 + 2d = 2$$
$$2d = 2$$
$$d = 1$$

∴The transformation matrix is

$$M = \begin{pmatrix}1 & 2\\0 & 1\end{pmatrix}$$

2024.P2.GCE

Study the following diagram and answer the questions that follow.



(a) Describe fully a single transformation which maps triangle ABC onto triangle GHI. [3]

(b) Triangle ABC is mapped onto triangle PQR by an enlargement. Find the matrix representing this transformation. [3]

(c) A stretch represented by a matrix $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ maps triangle ABC onto triangle DEF (not in the diagram). Find the coordinates of D, E and F. [3]

(d) Triangle KLM is the image of triangle ABC, under a shear. Find the shear factor and the invariant line. [3]

Solution

(a) It is a clockwise rotation of $90°$ about the origin $(0, 0)$.

(b) $K = \frac{Image\ length}{Object\ length}$

$$K = -\frac{PQ}{AB}$$

$$K = -\frac{4}{2}$$

$$K = -2$$

$$M = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$$

$$M = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$$

(c) $I = M(O)$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 & 4 & 4 \\ -2 & -2 & -4 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 4 & 4 \\ -4 & -4 & -8 \end{pmatrix}$$

$\therefore D(2, -4), E(4, -4)\ and\ F(4, -8)$

(d) I = M(O)

$$\begin{pmatrix} 2 & 4 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} 2a - 2b & 4a - 2b \\ 2c - 2d & 4c - 2d \end{pmatrix}$$

$2a - 2b = 2$
$4a - 2b = 4$
$-2a = -2$
$a = 1$
$2(1) - 2b = 2$
$2 - 2b = 2$
$-2b = 2 - 2$
$-2b = 0$

$b = 0$

$2c - 2d = 2$
$4c - 2d = 6$
$-2c = -4$
$c = 2$
$2(2) - 2d = 2$
$4 - 2d = 2$
$-2d = 2 - 4$
$-2d = -2$
$d = 1$

$$M = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

$\therefore$The shear factor is k = 2 and the invariant line is the $x - axis$ or the line $y = 0$

CALCULUS

Objectives

1 Define and explain the concept of limits and their importance in calculus.
2 Apply limit properties and theorems to solve problems involving limits.
3 Define and explain the concept of derivatives and their applications in physics, engineering, and economics.
4 Apply differentiation rules, including the power rule, product rule, and quotient rule, to solve problems involving derivatives.
5 Define and explain the concept of integrals and their applications in physics, engineering, and economics.
6 Apply integration rules, including substitution, integration by parts, and integration by partial fractions, to solve problems involving integrals.
7 Apply calculus concepts to solve problems involving optimization, related rates, and area between curves.

Definition

- Calculus is a branch of mathematics that deals with the study of continuous change, particularly in the context of functions and limits.
- It consists of two main branches called Differential Calculus and Integral Calculus.
- Differential Calculus deals with the study of rates of change and slopes of curves.
- It helps us understand how functions change as their input changes.

- Integral Calculus deals with the study of accumulation of quantities.
- It helps us find the area under curves, volumes of solids, and other quantities.

Applications of Calculus

- Calculus has numerous applications in various fields, including:

1 Physics and Engineering: To describe the laws of motion, energy, and gravity.
2 Economics: To model economic systems, understand the behavior of markets, and make predictions.
3 Computer Science: To develop algorithms, model complex systems, and analyze data.
4 Biology: To model population growth, understand the behavior of complex systems, and analyze data.



DIFFERENTIAL CALCULUS

Differentiation from first principles

- If a curve in the $x-y$ plane (Cartesian plane) is defined as $y = f(x)$, there exists another function that describes the nature of its slope at any point on it denoted as $f'(x)$ or $\frac{dy}{dx}$.
- In mathematics at this level, the possible extent to which an expression can be deduced under a given situation is called a limit.
- A limit is denoted with the situation as $\lim_{situation}( )$.
- However, the derived function that describes the gradient of $f(x)$ at any point $(x, f(x))$ is given as $f'(x) = \lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right)$.

- If k is a constant and a is a point on the x-axis,
  (i) $\lim k = k$; that is the limit of a constant is the constant itself.
  (ii) $\lim f(a) \pm g(a) = \lim f(a) \pm g(a)$; that is, the limit of a series or a polynomial is the sum of the limits of its individual terms.

MODES OF DIFFERENTIATION

- To some extent, it is possible but hard or tiresome to differentiate $y = (x + 2)^2$ with respect to x.
- It would require us to first expand the component $(x + 2)^2$ into individual terms that we can easily differentiate.
- If $y = x^n$ then $\frac{dy}{dx} = nx^{n-1}$

The Chain Rule

- If y is a function of x, and there exists a parameter t that can be expressed as another function of x in y, then y is a function of t and gradient/derivation of y is given as $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

The Product Rule

- If y is a function of x, and a product of two functions u and v, then u and v are also functions of x and the gradient/derivative function of y with respect to x is given as $\frac{dy}{dx} = vu' + uv'$.

The Quotient rule

- ✓ Quotient rule is a rule applied to find the gradient function or the derivative function of functions that involves fractions.
- ✓ If y is a function of x and y is a ratio of functions u and v, where u and v are also functions of x, the gradient or derivative function of y with respect to x is given as:
  $\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$.

Differentiating Exponential Functions

- ➢ If $y = e^x$ is a function defined with respect to x, then its gradient function is given as $\frac{d(e^x)}{dx} = e^x$.
- ➢ If the function of x is given as $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$.

Derivatives of Trigonometric Functions

- ➢ If $y = sinx$, $\frac{dy}{dx} = \cos x$
- ➢ If $y = \cos x$, $\frac{dy}{dx} = -\sin x$
- ➢ If $y = \tan x$, $\frac{dy}{dx} = \sec^2 x$

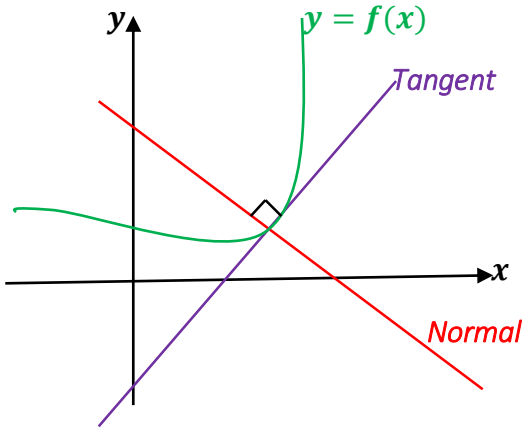
Second Derivatives

- ✓ Differentiation is also important in detecting the nature of gradients that a gradient function holds.
- ✓ Differentiating a derivative function is detecting the rate at which the gradient changes.

- ✓ The derivative of a derivative function is known as the Second Derivative Function.
- ✓ If we are differentiating function $y = f(x)$ in terms of x, the first derivative is written as $\frac{dy}{dx} = f'(x) = \frac{d(f(x))}{dx}$.
- ✓ The second derivative is written as $\frac{d^2y}{dx^2} = f''(x) = \frac{d}{dx}\left(f'(x)\right)$, pronounced as
  "d two y by dx squared".

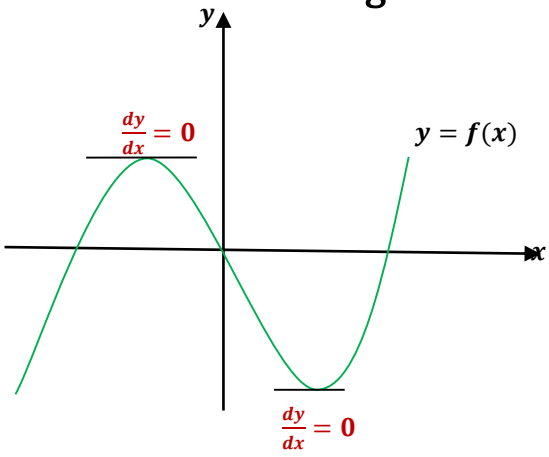
Tangents and Normals

- ➢ The gradient m_1 of a tangent that touches the curve at a given point is the gradient of that curve at that point and there exists the gradient m_2 of a line that is perpendicular to that tangent at the same point such that $m_1 \times m_2 = -1$, that perpendicular is called a normal to the curve at that point.



Stationary points

✓ A stationary point on a curve also known as a turning point is that point where the curve has zero gradient.



Integration

- Integration is the reverse of differentiation.
- Integration is a mathematical process that finds the area under curves, accumulates quantities, and solves problems involving rates of change.
- It is a fundamental concept in calculus and has numerous applications in physics, engineering, economics, and computer science.

Definite and Indefinite Integrals

- The symbol $\int (\dots)\, dx$ denotes 'integration with respect to x'.
- $\int_a^b y\, dx$ is a definite integral while $\int y\, dx$ is an indefinite integral.

Integration by Parts

- Integration by Parts is a technique used to integrate products of functions.
- It is based on the product rule of differentiation, which states that the derivative of a product of two functions is equal to the derivative of the first function times the second function, plus the first function times the derivative of the second function.

Formula for Integration by Parts

- The formula for Integration by Parts is given by $\int (u\,dv) = uv - \int (v\,du)$ where u is the first function (usually the one that becomes simpler when differentiated), dv is the differential of the second function (usually the one that becomes simpler when integrated), du is the differential of the first function and v is the integral of the second function.

How to Apply Integration by Parts

1. Choose the functions u and dv.
2. Differentiate u to get du.
3. Integrate dv to get v.
4. Plug the values into the formula and simplify.

2018.P1.Nov

Given that $y = 2x^3 - 5x^2 + 4x + 2$, find $\frac{dy}{dx}$.

Solution

$$y = 2x^3 - 5x^2 + 4x + 2$$
$$\frac{dy}{dx} = 3 \times 2x^{3-1} - 2 \times 5x^{2-1} + 1 \times 4x^{1-1} + 0$$
$$= 6x^2 - 10x + 4$$

2019.P1.Nov

Find $\int (3x^2 + 8x - 5)\, dx$

Solution

$$\int (3x^2 + 8x - 5) dx$$
$$\frac{3x^{2+1}}{2+1} + \frac{8x^{1+1}}{1+1} - 5x + c$$
$$\frac{3x^3}{3} + \frac{8x^2}{2} - 5x + c$$
$$x^3 + 4x^2 - 5x + c$$

2020.P1.Nov

Integrate $y = x^{-3} - 3x^4 + 5$.

Solution

$$y = x^{-3} - 3x^4 + 5$$
$$= \frac{x^{-3+1}}{-3+1} - \frac{3x^{4+1}}{4+1} + 5x + c$$
$$= \frac{x^{-2}}{-2} - \frac{3x^5}{5} + 5x + c$$
$$= -\frac{2}{x^2} - \frac{3}{5}x^5 + 5x + c$$
$$= \frac{3}{5}x^5 - \frac{2}{x^2} + 5x + c$$

2021.P1.Nov

Find $\int (6x^2 - 2x + 7)\, dx$

Solution

$$\int (6x^2 - 2x + 7)\, dx$$

$$\frac{6x^{2+1}}{2+1} - \frac{2x^{1+1}}{1+1} + 7x + c$$

$$\frac{6x^3}{3} - \frac{2x^2}{2} + 7x + c$$

$$2x^3 - x^2 + 7x + c$$

2022.P1.Nov

Given that $y = 2x^2 - 4x + 3$, find $\frac{dy}{dx}$.

Solution

$$y = 2x^2 - 4x + 3$$

$$\frac{dy}{dx} = 4x - 4$$

2023.P1.Nov

Find $\int (x^3 + 2x + 3)\, dx$.

Solution

$$\int (x^3 + 2x + 3) dx$$

$$= \frac{x^{3+1}}{3+1} - \frac{2x^{1+1}}{1+1} + 3x + c$$

$$= \frac{x^4}{4} + \frac{2x^2}{2} + 3x + c$$

$$= \frac{1}{4}x^4 + x^2 + 3x + c$$

2024.P1.GCE

Given that $y = 4x^3 - 2x^2 + 1$, find $\frac{dy}{dx}$.

Solution

$$y = 4x^3 - 2x^2 + 1$$

$$\frac{dy}{dx} = 12x^2 - 4x$$

2018.P2.GCE

(i) Determine the equation of the normal to the curve $y = 2x^2 - 3x - 2$ that passes through the point (3, 7). [3]

(ii) Evaluate $\int_0^1 (x^2 - 2x - 3)\,dx$. [3]

Solution

(i) $y = 2x^2 - 3x - 2$

$\frac{dy}{dx} = 4x - 3$

$m_1 = 4(3) - 3$

$= 9$

$m_1 \times m_2 = -1$

$m_2 = -\frac{1}{9}$

$y - y_1 = m(x - x_1)$

$y - 7 = -\frac{1}{9}(x - 3)$

$9y - 63 = -1(x - 3)$

$9y - 63 = -x + 3$

$9y + x = 3 + 63$

$9y + x = 66$

(ii) $\int_0^1 (x^2 - 2x - 3)dx$

$\left[\frac{x^{2+1}}{2+1} - \frac{2x^{1+1}}{1+1} - 3x\right]_0^1$

$\left[\frac{x^3}{3} - \frac{2x^2}{2} - 3x\right]_0^1$

$\left[\frac{1}{3}x^3 - x^2 - 3x\right]_0^1$

$\frac{1}{3}(1)^3 - (1)^2 - 3(1) - \left[\frac{1}{3}(0) - (0)^2 - 3(0)\right]$

$-\frac{11}{3} - 0$

$-3\frac{2}{3}$

2019.P2.Nov

(i) Find the equation of the normal to the curve $y = 5x^3 - 6x^2 + 2x + 5$ at the point (1, 2). [3]

(ii) Evaluate $\int_1^3 (3x^2 + 4x)dx$. [3]

Solution

(i) $y = 5x^3 - 6x^2 + 2x + 5$

$\frac{dy}{dx} = 15x^2 - 12x + 2$

$m_1 = 15(1)^2 - 12(1) + 2$

$= 5$

$m_1 \times m_2 = -1$

$m_2 = -\frac{1}{5}$

$y - y_1 = m(x - x_1)$

$y - 2 = -\frac{1}{5}(x - 1)$

$5y - 10 = -1(x - 1)$

$5y - 10 = -x + 1$

$5y + x = 1 + 10$

$5y + x = 11$

(ii) $\int_1^3 (3x^2 + 4x)dx$

$[3x^2 + 4x]_1^3$

$\left[\frac{3x^{2+1}}{2+1} + \frac{4x^{1+1}}{1+1}\right]_1^3$

$\left[\frac{3x^3}{3} + \frac{4x^2}{2}\right]_1^3$

$[x^3 + 2x^2]_1^3$

$(3)^3 + 2(3)^2 - [(1)^3 + 2(1)^2]$

$27 + 18 - (1 + 2)$

$45 - 3$

$42\ units^2$

2020.P2.Nov

(i) A curve has gradient $x^2 - 4x + 3$. Find the equation of the curve if it passes through the point $(3, -1)$. [3]

(ii) Find the equation of the normal to the curve $y = x^3 - 2x^2 - 4x + 1$ at the point $(-1, 2)$. [3]

Solution

(i) $\frac{dy}{dx} = x^2 - 4x + 3$

$y = \frac{x^{2+1}}{2+1} - \frac{4x^{1+1}}{1+1} + 3x + c$

$y = \frac{1}{3}x^3 - 2x^2 + 3x + c$

$-1 = \frac{1}{3}(3)^3 - 2(3)^2 + 3(3) + c$

$-1 = 9 - 18 + 9 + c$

$c = -1$

$\therefore y = \frac{1}{3}x^3 - 2x^2 + 3x - 1$

(ii) $y = x^3 - 2x^2 - 4x + 1$

$\frac{dy}{dx} = 3x^2 - 4x - 4$

$m_1 = 3(-1)^2 - 4(-1) - 4$

$= 3$

$m_1 \times m_2 = -1$

$m_2 = -\frac{1}{3}$

$y - y_1 = m(x - x_1)$

$y - 2 = -\frac{1}{3}(x + 1)$

$3y - 6 = -1(x + 1)$

$3y - 6 = -x - 1$

$3y + x = -1 + 6$

$3y + x = 5$

2021.P2.Nov

(i) Evaluate $\int_{-1}^{1}(1-4x+3x^2)\,dx$. [3]

(ii) Find the equation of the normal to the curve $y = x^2 - x + 4$ at the point $(-1, 6)$. [3]

Solution

(i) $\int_{-1}^{1}(1-4x+3x^2)dx$

$[1-4x+3x^2]_{-1}^{1}$

$\left[x-\frac{4x^{1+1}}{1+1}+\frac{3x^{2+1}}{2+1}\right]_{-1}^{1}$

$\left[x-\frac{4x^2}{2}+\frac{3x^3}{3}\right]_{-1}^{1}$

$[x-2x^2+x^3]_{-1}^{1}$

$1-2(1)^2+1^3-[-1-2(-1)^2+(-1)^3]$

$1-2+1-(-1-2-1)$

$0-(-4)$

$4\ units^2$

(ii) $y = x^2 - x + 4$

$\frac{dy}{dx} = 2x - 1$

$m_1 = 2(-1) - 1$

$= -3$

$m_2 = \frac{1}{3}$

$y - y_1 = m(x - x_1)$

$y - 6 = \frac{1}{3}(x + 1)$

$3y - 18 = -(x + 1)$

$3y - 18 = -x - 1$

$3y + x = -1 + 18$

$3y + x = 17$

2022.P2.Nov

(a) Evaluate $\int_{-1}^{4}(2+2x+6x^2)dx$. [3]

(b) Find the equation of the normal to the curve $y = x^2 - 3x - 4$ at the point (2, -6). [3]

Solution

(a) $\int_{-1}^{4}(2+2x+6x^2)dx$

$[2+2x+6x^2]_{-1}^{4}$

$\left[2x+\frac{2x^{1+1}}{1+1}+\frac{6x^{2+1}}{2+1}\right]_{-1}^{4}$

$\left[2x+\frac{2x^2}{2}+\frac{6x^3}{3}\right]_{-1}^{4}$

$[2x+x^2+2x^3]_{-1}^{4}$

$2(4)+(4)^2+2(4)^3-[2(-1)+(-1)^2+2(-1)^3]$

$8+16+128-(-2+1-2)$

$152-(-3)$

$152+3$

$155\ units^2$

(b) $y = x^2 - 3x - 4$

$\frac{dy}{dx} = 2x - 3$

$m_1 = 2(2) - 3$

$= 1$

$m_1 \times m_2 = -1$

$m_2 = -1$

$y - y_1 = m(x - x_1)$

$y + 6 = -1(x - 2)$

$y + 6 = -x + 2$

$y + x = 2 - 6$

$y + x = -4$

2023.P2.Nov

(a) Find the equation of the curve for which $\frac{dy}{dx} = 12x^3 - 4x + 5$ at the point (0, -2). [3]

(b) Evaluate $\int_{-1}^{2}(3x^2 - 2x + 1)dx$. [3]

Solution

(a) $\frac{dy}{dx} = 12x^3 - 4x + 5$

$y = \frac{12x^{3+1}}{3+1} - \frac{4x^{1+1}}{1+1} + 5x + c$

$y = 3x^4 - 2x^2 + 5x + c$

$-2 = 3(0)^4 - 2(0)^2 + 5(0) + c$

$-= 0 - 0 + 0 + c$

$c = -2$

$\therefore y = 3x^4 - 2x^2 + 5x - 2$

(b) $\int_{-1}^{2}(3x^2 - 2x + 1)dx$

$[3x^2 - 2x + 1]_{-1}^{2}$

$\left[\frac{3x^{2+1}}{2+1} - \frac{2x^{1+1}}{1+1} + x\right]_{-1}^{2}$

$\left[\frac{3x^3}{3} - \frac{2x^2}{2} + x\right]_{-1}^{2}$

$[x^3 - x^2 + x]_{-1}^{2}$

$2^3 - 2^2 + 2 - [(-1)^3 - (-1)^2 + (-1)]$

$8 - 4 + 2 - (-1 - 1 - 1)$

$6 - (-3)$

$6 + 3$

$9\ units^2$

2024.P2.GCE

(a) Evaluate $\int_{-1}^{2}(6x^5 - x^4 + 3x^2 - 1)dx$. [3]

(b) Find the equation of the normal to the curve $y = x^2 + x + 2$ at the point (1, 4). [3]

Solution

(a) $\int_{-1}^{2}(6x^5 - x^4 + 3x^2 - 1)dx$

$\left[6x^5 - x^4 + 3x^2 - 1\right]_{-1}^{2}$

$\left[\frac{6x^{5+1}}{5+1} - \frac{x^{4+1}}{4+1} + \frac{3x^{2+1}}{2+1} - x\right]_{-1}^{2}$

$\left[\frac{6x^6}{6} - \frac{x^5}{5} + \frac{3x^3}{3} - x\right]_{-1}^{2}$

$\left[x^6 - \frac{1}{5}x^5 + x^3 - x\right]_{-1}^{2}$

$(2)^6 - \frac{1}{5}(2)^5 + (2)^3 - 2 - \left[(-1)^6 - \frac{1}{5}(-1)^5 + (-1)^3 - (-1)\right]$

$\frac{318}{5} - \frac{6}{5}$

$\frac{312}{5}$

$62\frac{2}{5}\ units^2$

(b) $y = x^2 + x + 2$

$\frac{dy}{dx} = 2x + 1$

$m_1 = 2(1) + 1$

$= 3$

$m_2 = -\frac{1}{3}$

$y - y_1 = m(x - x_1)$

$y - 4 = -\frac{1}{3}(x - 1)$

$3y - 12 = -(x - 1)$

$3y - 12 = -x + 1$

$3y + x = 1 + 12$

$3y + x = 13$